

DRAFT

1132 Gunhill Aldershot

NBS Architectural Specification

13 October 2017

Revision P1

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E
In situ concrete/Large precast concrete

DRAFT

E60

Precast/ composite concrete floors/ roof decks

E60 Precast/ composite concrete floors/ roof decks

To be read with Preliminaries/ General conditions.

PROPRIETARY FLOORS/ ROOF DECKS

110A PRECAST BEAM AND BLOCKSUSPENDEED GROUND FLOORS TO HOUSES

Apartment Block D

Semi-detached houses 79 - 92

Terraces Houses 102 - 107

- Beams: T225mm Beam and Block Flooring - Agreement certified.
 - Manufacturer: Milbank Concrete Products Ltd
 Earls Colne Business Park,
 Earls Colne,
 Colchester,
 Essex,
 CO6 2NS.
 United Kingdom
 Tel: 01787 223931
 Fax: 01787 220535
 Product reference: To specialist design.
 - Type: Prestressed concrete 'T', depth 225 mm.
- Infill blocks: Autoclaved aerated.
- Structural concrete topping: Not required.
- Other requirements: Refer to Engineers drawings for service holes and load criteria.

Ground Floor to achieve thermal performance of 0.25 W/m²k

Floor Finish: **Refer to Finishes Schedule**

Floor Screed: Refer to M10/100A

Separating Layer: Suitable polythene vapour control layer (VCL) e.g Visqueen 500 gauge or equivalent.

Insulation: Refer to specification clause M10/290A

Damp Proof Membrane: 500 microns black polythene (2000 gauge) Visqueen DPM with all joints lapped and sealed. DPM to be lapped and sealed with all adjacent walls DPC's.

Block D to to have a gas protection membrane.

Beam and block flooring - as Above

Ventilated floor void: Refer to clause E60/316

Minimum Void: 250mm to be maintained between underside of the floor and the ground, in accordance with NHBC guidelines section 2.1,
 made up with a suitable well-compacted material or in accordance with the Engineers design specification.

111A PRECAST BEAM AND BLOCKGROUND FLOORS TO GARAGES

Garages to sem-detached houses plot 79 - 92

- Beams: T150mm Beam and Block Flooring - Agreement certified.
 - Manufacturer: Milbank Concrete Products Ltd
Earls Colne Business Park,
Earls Colne,
Colchester,
Essex,
CO6 2NS.
United Kingdom
Tel: 01787 223931
Fax: 01787 220535
Product reference: T225.
 - Type: Prestressed concrete 'T', depth 150 mm.
- Infill blocks: Autoclaved aerated.
- Structural concrete topping: Not required.
- Other requirements: Refer to Engineers drawings for service holes and load criteria.

Ground Floor to achieve thermal performance of n/a W/m²k

Floor Screed: Refer to M10/11A

Separating Layer: Suitable polythene vapour control layer (VCL) e.g Visqueen 500 gauge or equivalent.

Damp Proof Membrane: 500 microns black polythene (2000 gauge) Visqueen DPM with all joints lapped and sealed. DPM to be lapped and sealed with all adjacent walls DPC's.

Beam and block flooring - as Above

Ventilated floor void: Refer to clause E60/316

Minimum Void: 250mm to be maintained between underside of the floor and the ground in accordance with NHBC guidelines section 2.1, made up with a suitable well-compacted material or in accordance with the Engineers design specification.

130A PRECAST SLAB TO UPPER FLOORS FOR BLOCK D ONLY

- Manufacturer: Contractor's choice.
 - Product reference: Contractor's choice.
- Slab type: Hollow.
- Slab depth: 150 mm.
- Finish:
 - Soffit: Manufacturer's standard.
 - Top surface: Prepared to receive levelling screed as section M10.
- In situ continuity reinforcement: Not required.
- Other requirements: Drainage holes to hollow cores - refer to Engineers drawings for service holes and loading criteria.

Party Floor Construction

Floor Finish: Refer to

Floor Screed: Refer to M10/130B

Acoustic Layer: Refer to clause M10/295

Hollow Core Slab As above

Ceiling Void 150mm Void

Insulation 50mm Mineral Wool (10kg/m²)

Plasterboard Ceiling Refer to K10/

180A PRECAST STAIRS TO APARTMENTS BLOCKS A, B, C and D

- Manufacturer: Creagh Concrete Products Ltd.
 - Web: www.creaghconcrete.com.
 - Email: info@creaghconcrete.com.
 - Product reference: Precast Stairs
- Design: To specialist design - refer to Structural engineers drawings for setting out and load criteria.
Refer to Architects drawing for general configuration. Flights to be Part M compliant.
- Finish: Type B.

PERFORMANCE SPECIFIED FLOORS/ ROOF DECKS

GENERAL/ PERFORMANCE

240 DETAILING OF PROPRIETARY SYSTEM

- Installation details: Submit location and assembly drawings showing every aspect of the construction, incorporated components and features, trimming for voids, holes for services, and related work by others.
 - Purpose: To allow checking of compatibility with surrounding structure and coordination of services.
- Method statement and risk assessment for installation: Submit.
- Programme: Submit well in advance of construction.

295 AUTOCLAVED AERATED CONCRETE INFILL BLOCKS TO BEAM AND BLOCK FLOORS

- Type: Regular block to BS EN 771-4.
 - Size: 440 x 215 x 100 mm.
- Density: Maximum 1500 kg/m³.
- Compressive strength (minimum): 3.5 N/mm².
- Transverse load capacity (minimum): 3.5 kN/m² measured on a 420 mm span.
- Other requirements: Refer to Engineer's drawings for service penetrations and loading criteria.

INSTALLATION

315 INFILLING BEAM BEARINGS

- Type: Beam manufacturer's proprietary infill blocks.
- Installation: Infill gaps in walling below built in standard flooring blocks.

316A UNDERFLOOR VENTILATION

- Manufacturer: PUFV Rytons Periscope@undefloor Ventilator with Multifix Air Brick
All external visible vents and extracts are to be coloured to minimise their visual impact, subject to the material they are within then the following Multi-fix Air Brick are to be used.

Brick: Terracotta MFABTC

• **Suspended Concrete Floors:**

- Ventilate: the underfloor space by providing ventilation in two opposing external walls so that the ventilating air will have a free path between opposite sides and to all parts.
- Ventilate voids by openings providing not less than 1500mm²/m run of external wall or 500mm²/m² of floor area, whichever gives the greater opening area. Provide ventilation in internal sleeper walls or similar obstructions to maintain the path of air.
- Ventilators must NOT be located directly under floor beams or door openings.
"The NHBC recommends ventilators for suspended timber floors be spaced at not more than 2m centres and within 450mm of each end of any wall.
"The Building Regulations state any pipes needed to carry ventilating air should have a diameter of at least 100mm.
"Also refer to: The Building Regulations 2000, Approved Document C,
"NHBC Standards,
"British Standard BS 5250.

320 CONCRETE INFILL

- Designated concrete: To BS 8500-2.
 - Designation: RC20/25.
 - Aggregate size (maximum): 10 mm generally.
- Preparation: Thoroughly clean and wet surfaces of precast units.
- Placing: Avoid segregation and compact thoroughly to eliminate voids.
 - Extent: Fill troughs and other holes.
 - Finish: Flush with top of precast units.
- Protection: Prevent movement of units until concrete has gained sufficient strength.

325 GROUTING

- Mix: 1:6 cement:sharp sand (1:4 if soft sand or in wet weather).
- Preparation: Thoroughly clean and wet surfaces.
- Fill all joints and surface irregularities.

330 EXPANDED POLYSTYRENE VOID FORMERS

- Expanded polystyrene: To BS EN 13163.
- Manufacturer: Contractor's choice.
 - Product reference: Contractor's choice.
- Compressive strength (minimum): Refer to structural engineers.
- Size: Contractor's choice.

350 LEVELS OF STRUCTURAL CONCRETE FLOORS

- Tolerances: See Preliminaries section A33.

DRAFT

F
Masonry

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F10
Brick/ block walling

F10 Brick/ block walling

To be read with Preliminaries/ General conditions.

BLOCKWORK VARIATIONS IDENTIFIED

Engineering Brickwork called, we want to use the facing brickwork below ground for at least 5 courses below dpc level

Apartment Block A, B, C

355B - Inner leaf of external cavity wall: 100mm blockwork (Medium Density) 7.3 N/mm²
Party walls: N/A all lightweight

Apartment Block D

355 - Sleeper walls: 140mm Concrete Block 10.4N/mm² 140/75/140
355A - Inner leaf of external cavity wall GF: 100mm Blockwork (Medium Density) 10.4 N/mm² with wind posts
355B - Inner leaf of external cavity wall to upper floors: 100mm Blockwork 7.3 N/mm² with wind posts
356A - Party wall at GF: 140mm Blockwork (Medium Density) 10.4 N/mm² 140/75/140
356B - Party wall at Upper floor: 100mm Blockwork (Medium Density) 7.3 N/mm²
100/100/100

Semi Detached Houses

Sleeper walls/Dwarf Walls
355A & B - Inner leaf of external cavity walls
Party Walls
Garage Walls

Houses Plots 102 - 104

355 - Sleeper Walls/Dwarf Walls: 140mm Concrete Block 10.4 N/mm² 140/100/140
355B - Inner leaf of external cavity wall to GF: 100mm Blockwork 7.3 N/mm² with wind posts
355C - Inner leaf of external cavity wall to FF: 100mm Blockwork 3.6 N/mm² with wind posts
Party wall at GF: 140mm Blockwork (Medium Density) 10.4 N/mm² 140/75/140

Houses Plots 105 - 107

355 - Sleeper Walls/Dwarf walls: 140mm Concrete Block 10.4 N/mm² 140/100/140
355B - Inner leaf of external cavity wall to GF: 100mm Blockwork 7.3 N/mm² with wind posts
355 C - Inner leaf of external cavity wall to FF: 100mm Blockwork 3.6 N/mm² with wind posts
Party wall at GF: 100mm Blockwork (Medium Density) 7.3 N/mm² 100/100/100
Party wall at FF: 100mm Blockwork (Medium Density) 3.6 N/mm² 100/100/100

TYPES OF WALLING

110A CLAY FACING BRICKWORK TO EXTERNAL WALLS

Refer to DLA General Arrangement and detail drawings 1132-40, 45 and 46

Typical Wall Construction to Provide a U Value of 0.22W/m²k

O/A wall thickness = 327.5

102.5mm	Facing brickwork
50mm	Clear Cavity
50mm	Kingspan insulation as clause F30/155A
100mm	Blockwork Inner Leaf as clause 355A/355B
12.5mm	Plaster dabs
12.5mm	Plasterboard/Moisture Resistant in wet areas
3mm	Skim coat plaster

- Bricks: To BS EN 771-1.
 - Manufacturer: IBSTOCK BRICK LIMITED Leicester Road, Leicestershire, LE67 6HS
Tel +44 (0)1530 261999 www.ibstock.com .
 - Product reference: Parham Red Stock .
 - Recycled content: None permitted .
 - Special shapes: As shown on drawings .
- Mortar: As section Z21.
 - Standard: To BS EN 998-2 .
 - Mix: 1:1:6 cement:lime:sand .
 - Additional requirements: Light Grey coloured mortar eg 'natural grey' to Architects Approval .
- Bond: Half lap stretcher .
- Joints: Bucket handle .
- Features: Plinth Brick as detailed, Brick Headers over windows .
Brickwork to extend a min of 3 courses below ground level.

355 CONCRETE COMMON BLOCKWORK BELOW GROUND, LOADBEARING TO DWARF/SLEEPER WALLS

In accordance with NHBC section 5.1, D6

- Blocks: To BS EN 771-3.
 - Manufacturer: Contractor's choice.
 - Product reference: Contractor's choice.
 - Configuration: Group 1.
 - Compressive strength:
Mean value: 10.4 N/mm².
Characteristic value: 10.4 N/mm².
Category: I.
 - Freeze/ Thaw resistance: Frost resistant.
 - Thermal properties: Not applicable.
 - Recycled content: 50% (minimum) to BS EN ISO 14021.
 - Work sizes (length x width x height): 440 x 140 x 215 mm.
Tolerance category: D1.
 - Special shapes: Coursing blocks.
 - Additional requirements: None.
- Mortar: As section Z21.
 - Standard: To BS EN 998-2.
 - Mix: 1:1:6 cement:lime:sand.
 - Additional requirements: None.
- Bond: Half lap stretcher.

355A CONCRETE COMMON BLOCKWORKLOADBEARING TO EXTERNAL CAVITY WALLS

Block D - Ground Floors

- Blocks: To BS EN 771-3.
 - Manufacturer: Contractor's choice.
 - Product reference: 100/140mm Blocks Submit proposals.
 - Configuration: Group 1.
 - Compressive strength:
 - Mean value: 10.4 N/mm².
 - Characteristic value: 10.4 N/mm².
 - Category: II.
 - Freeze/ Thaw resistance: Not applicable.
 - Thermal properties: Blocks for use in External cavity wall to be 0.11m²K/W or better.
 - Recycled content: Submit proposals.
 - Work sizes (length x width x height): 440 x 100 x 215 mm & 440 x 140 x 215mm.
 - Tolerance category: D2.
 - Special shapes: None.
 - Additional requirements: Sulphate resisting where ground conditions dictate.
- Mortar: As section Z21.
 - Standard: To BS EN 998-2.
 - Mix: 1:1:6 cement:lime:sand.
 - Additional requirements: None.
- Bond: Half lap stretcher.

355B CONCRETE COMMON BLOCKWORKLOADBEARING TO EXTERNAL CAVITY WALLS

Block A, B and C - All floors

Block D - Upper Floors

- Blocks: To BS EN 771-3.
 - Manufacturer: Submit proposals.
 - Product reference: 100mm Blocks Submit proposals.
 - Configuration: Group 1.
 - Compressive strength:
 - Mean value: 7.3 N/mm² refer to structural drawings.
 - Characteristic value: 7.3 N/mm² refer to structural drawings.
 - Category: II.
 - Freeze/ Thaw resistance: Not applicable.
 - Thermal properties: Blocks for use in External cavity wall to be 0.11m²K/W or better.
 - Recycled content: Submit proposals.
 - Work sizes (length x width x height): 440 x 100 x 215 mm.
 - Tolerance category: D2.
 - Special shapes: None.
 - Additional requirements: Sulphate resisting where ground conditions dictate.
- Mortar: As section Z21.
 - Standard: To BS EN 998-2.
 - Mix: 1:1:6 cement:lime:sand.
 - Additional requirements: None.
- Bond: Half lap stretcher.

355C CONCRETE COMMON BLOCKWORK TO EXTERNAL CAVITY WALL

Houses Plots 102-104 - Upper Floors

Houses Plots 105 -107 - Upper Floors

- Blocks: To BS EN 771-3.
 - Manufacturer: Submit proposals.
 - Product reference: 100mm Blocks Submit proposals.
 - Configuration: Group 1.
 - Compressive strength:
 - Mean value: 3.6 N/mm².
 - Characteristic value: 3.6 N/mm².
 - Category: II.
 - Freeze/ Thaw resistance: Not applicable.
 - Thermal properties: Blocks for use in External cavity wall to be 0.11m²K/W or better.
 - Recycled content: Submit proposals.
 - Work sizes (length x width x height): 440 x 100 x 215 mm .
 - Tolerance category: D2.
 - Special shapes: None.
 - Additional requirements: None.
- Mortar: As section Z21.
 - Standard: To BS EN 998-2.
 - Mix: 1:1:6 cement:lime:sand.
 - Additional requirements: None.
- Bond: Half lap stretcher.

356A CONCRETE COMMON BLOCKWORK TO PARTY WALLS BLOCK D GROUND FLOOR ONLY, HOUSE PLOTS 102-104 GF ONLY

Party wall to achieve Acoustic performance of Building Regulations + 5dB

Two leaves of 140mm blockwork

Outer faces to receive 8mm scratch coat render as clause M20/295A

75mm cavity filled with 75mm Acoustic insulation as clause P10/

One layer of 12.5mm wallboard on dot dabs to both sides as clause K10/

- Blocks: To BS EN 771-3.
 - Manufacturer: Submit proposals.
 - Product reference: 140mm Dense Blocks (1900kgm) Submit proposals.
 - Configuration: Group 1.
 - Compressive strength:
 - Mean value: 10.4 N/mm².
 - Characteristic value: 10.4 N/mm².
 - Category: II.
 - Freeze/ Thaw resistance: Not applicable.
 - Thermal properties: As required to meet thermal performance of wall.
 - Recycled content: Submit proposals.
 - Work sizes (length x width x height): 440 x 140 x 215 mm .
 - Tolerance category: D2.
 - Special shapes: None.
 - Additional requirements: None.
- Mortar: As section Z21.
 - Standard: To BS EN 998-2.
 - Mix: 1:1:6 cement:lime:sand.
 - Additional requirements: None.
- Bond: Half lap stretcher.

356B CONCRETE COMMON BLOCKWORK TO PARTY WALLS TO HOUSE PLOTS GROUND FLOOR & UPPER FLOORS

Party wall to achieve Acoustic performance of Building Regulations + 5dB

Two leaves of 100mm blockwork

Outer faces to receive 8mm scratch coat render as clause M20/295A

100mm cavity filled with 100mm Acoustic insulation as clause P10/

One layer of 12.5mm wallboard on dot dabs to both sides as clause K10/

- Blocks: To BS EN 771-3.
 - Manufacturer: Submit proposals.
 - Product reference: 100mm Dense Blocks (1900kgm) Submit proposals.
 - Configuration: Group 1.
 - Compressive strength:
 - Mean value: 7.3 N/mm² OR 3.6 N/mm² where noted on structural drawings .
 - Characteristic value: 7.3 N/mm² OR 3.6 N/mm² where noted on structural drawings .
 - Category: II.
 - Freeze/ Thaw resistance: Not applicable.
 - Thermal properties: As required to meet thermal performance of wall.
 - Recycled content: Submit proposals.
 - Work sizes (length x width x height): 440 x 100 x 215 mm .
 - Tolerance category: D2.
 - Special shapes: None.
 - Additional requirements: None.
- Mortar: As section Z21.
 - Standard: To BS EN 998-2.
 - Mix: 1:1:6 cement:lime:sand.
 - Additional requirements: None.
- Bond: Half lap stretcher.

360 AUTOCLAVED AERATED CONCRETE (AAC) COMMON BLOCKWORK IN PARTY WALLS

- Blocks: To BS EN 771-4.
 - Manufacturer: Contractor's choice.
 - Product reference: Contractor's choice.
 - Configuration: Group 1.
 - Mean compressive strength (minimum): As per engineers drawings.
 - Category: I.
 - Thermal conductivity (maximum): tbc.
 - Recycled content: Contractor's choice.
 - Work sizes (length x width x height): 440 x 100 x 215 mm.
 - Tolerance category: GPLM.
 - Special shapes: Coursing blocks.
 - Additional requirements: Density: not to exceed 1500 kg/m³.
- Mortar: As section Z21.
 - Standard: To BS EN 998-2.
 - Mix: 1:1:6 cement:lime:sand.
 - Additional requirements: None.
- Bond: Half lap stretcher.

385 ENGINEERING BRICKWORK IN MANHOLES

- Bricks: To BS EN 771-1.
 - Manufacturer: Contractor's choice.
 - Product reference: Contractor's choice.
 - Type: HD.
 - Mean compressive strength: Greater than or equal to 75 N/mm².
 - Category: I.
 - Water absorption: Equal to or less than. 7.5%.
 - Freeze/ Thaw category: F2.
 - Active soluble salts content category: S2.
 - Additional requirements: None.
- Mortar: As section Z21.
 - Standard: To BS EN 998-2.
 - Mix: 1:0.25:3 cement:lime:sand.
 - Additional requirements: None.
- Bond: English.
- Joints: Flush.

395 DAMP PROOF COURSE BRICKWORK

- Bricks to BS EN 771-1.
 - Manufacturer: As clause 110A .
 - Product reference: As Clause 110A.
 - Type: HD.
 - Water absorption: Equal to or less than 4.5%.
 - Net dry density: Equal to or greater than 2100 kg/m³.
 - Freeze/ Thaw category: F2.
 - Active soluble salts content category: S2.
 - Additional requirements: None.
- Mortar: As section Z21.
 - Standard: To BS EN 998-2.
 - Mix: 1:0.25:3 cement:lime:sand.
 - Additional requirements: Light Grey coloured mortar to match mortar being used for facing brickwork.
- Bond: Half lap stretcher.
- Joints: Bucket handle.

TESTING

400 HARD LANDSCAPING MATERIALS SPECIFICATION

- Minimum BRE 'Green Guide to Specification' online rating: A+.

410 COMPRESSIVE STRENGTH OF MORTAR FOR EACH WALLING TYPE

- Testing authority: An approved laboratory.
- Test method: To BS EN 1015-11.
- Preliminary tests procedure: As follows:
 - Specimens:
 - Number of specimens: 6.
 - Type: 40 x 40 x 160 mm prism.
 - Preparation: At least six weeks before walling commences.
 - Specimen testing: Half of specimens at 7 days. Remainder at 28 days.
 - Retarded mixes: Extend curing periods to include retardation period.
 - Response to result: If mean compressive strength at 28 days is not within the range given below repeat tests with more suitable sand or next higher designation of mortar.
- Site tests procedure: As follows.
 - Number of specimens: Six per 150m² of walling or per storey whichever the more frequent.
 - Specimen types: As preliminary test, but prepared during construction.
 - Specimen testing: Half of specimens at 7 days. Remainder at 28 days.
 - Retarded mixes: Extend curing periods to include retardation period.
- Required test mean compressive strength at 28 days (N/mm²): To be within the following range:
 - Walling type: F10/110A.
 - Preliminary tests minimum (N/mm²): 4.0.
 - Preliminary tests maximum (N/mm²): 6.0.
 - Site tests minimum (N/mm²): 4.0.
 - Site tests maximum (N/mm²): 6.5.
- Results: Submit.

WORKMANSHIP GENERALLY

430 CONDITIONING OF CLAY AND CALCIUM SILICATE BRICKS AND CLAY BLOCKS

- Bricks and blocks delivered warm from manufacturing process: Do not use until cold.
- Absorbent bricks in warm weather: Wet to reduce suction. Do not soak.

440 CONDITIONING OF CONCRETE BRICKS/ BLOCKS

- Autoclaved concrete bricks/ blocks delivered warm from manufacturing process: Do not use.
- Age of nonautoclaved concrete bricks/ blocks: Do not use until at least four weeks old.
- Avoidance of suction in concrete bricks/ blocks: Do not wet.
 - Use of water retaining mortar admixture: Submit details.

460 MORTAR GROUPS

- Mix proportions: For a specified group select a mix design from the following:
 - Group 1:
 - 1:0–0.25:3 (Portland cement:lime:sand with or without air entraining additive).
 - 1:3 (Portland cement:sand and air entraining additive).
 - Group 2:
 - 1:0.5:4–5 (Portland cement:lime:sand with or without air entraining additive).
 - 1:3 (masonry cement:sand containing Portland cement and lime in approximate ratio 1:1, and an air entraining additive).
 - 1:2.5–3.5 (masonry cement:sand containing Portland cement and inorganic materials other than lime and air entraining additive).
 - 1:3–4 (Portland cement:sand and air entraining additive.)
 - Group 3:
 - 1:1:5–6 (Portland cement:lime:sand with or without air entraining additive).
 - 1:3.5–4 (masonry cement:sand containing Portland cement and lime in approximate ratio 1:1, and an air entraining additive).
 - 1:4–5 (masonry cement:sand containing Portland cement and inorganic materials other than lime and air entraining additive).
 - 1:5–6 (Portland cement:sand and air entraining additive).
 - Group 4:
 - 1:2:8–9 (Portland cement:lime:sand with or without air entraining additive).
 - 1:4.5 (masonry cement:sand containing Portland cement and lime in approximate ratio 1:1, and an air entraining additive).
 - 1:5.5–6.5 (masonry cement:sand containing Portland cement and inorganic materials other than lime and air entraining additive).
 - 1:7–8 (Portland cement:sand and air entraining additive).
- Batching: Mix proportions by volume.
- Mortar type: Continuous throughout any one type of masonry work.

500 LAYING GENERALLY

- Mortar joints: Fill vertical joints. Lay bricks, solid and cellular blocks on a full bed.
- AAC block thin mortar adhesive and gypsum block adhesive joints: Fill vertical joints. Lay blocks on a full bed.
- Clay block joints:
 - Thin layer mortar: Lay blocks on a full bed.
 - Interlocking perpend: Butted.
- Bond where not specified: Half lap stretcher.
- Vertical joints in brick and concrete block facework: Even widths. Plumb at every fifth cross joint.

520 ACCURACY

- Courses: Level and true to line.
- Faces, angles and features: Plumb.
- Permissible deviations:

- Position in plan of any point in relation to the specified building reference line and/ or point at the same level - - - - - Level of bed joints up to 5 m (brick masonry) - Level of bed joints up to 5 m	± 10 mm. - Straightness in any 5 m length - Verticality up to 3 m height - Verticality up to 7 m height - Overall thickness of walls ± 11 mm. ± 13 mm.	± 5 ± 10 mm. ± 14 mm. ± 10 mm.
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mm.

- 535 HEIGHT OF LIFTS IN WALLING USING CEMENT GAUGED OR HYDRAULIC LIME MORTAR
 - Quoins and advance work: Rack back.
 - Lift height (maximum): 1.2 m above any other part of work at any time.
 - Daily lift height (maximum): 1.5 m for any one leaf.

- 540 HEIGHT OF LIFTS IN WALLING USING THIN JOINT MORTAR GLUE
 - Quoins and advance work: Rack back.
 - Lift height (maximum): 1.3 m above any other part of work at any time.

- 545 LEVELLING OF SEPARATE LEAVES
 - Locations for equal levelling of cavity wall leaves: As follows:
 - Every course containing vertical twist type ties or other rigid ties.
 - Every third tie course for double triangle/ butterfly ties.
 - Courses in which lintels are to be bedded.

- 560 COURSING BRICKWORK
 - Gauge: Four brick courses including bed joints to 300 mm.

- 580 LAYING FROGGED BRICKS
 - Single frogged bricks: Frog uppermost.
 - Double frogged bricks: Larger frog uppermost.
 - Frog cavity: Fill with mortar.

- 585 LAYING CELLULAR BRICKS
 - Orientation: Cavities downward.

- 595 LINTELS
 - Bearing: Ensure full length masonry units occur immediately under lintel ends.

- 610 SUPPORT OF EXISTING WORK
 - Joint above inserted lintel or masonry: Fully consolidated with semidry mortar to support existing structure.

- 635 JOINTING
 - Profile: Consistent in appearance.

- 645 ACCESSIBLE JOINTS NOT EXPOSED TO VIEW
 - Jointing: Struck flush as work proceeds.

- 671 FIRE STOPPING
 - Avoidance of fire and smoke penetration: Fit tightly between cavity barriers and masonry. Leave no gaps.

- 690 ADVERSE WEATHER
 - General: Do not use frozen materials or lay on frozen surfaces.
 - Air temperature requirements: Do not lay bricks/ blocks:
 - In cement gauged mortars when at or below 3°C and falling or unless it is at least 1°C and rising.
 - In hydraulic lime:sand mortars when at or below 5°C and falling or below 3°C and rising.
 - In thin joint mortar glue when outside the limits set by the mortar manufacturer.
 - Temperature of walling during curing: Above freezing until hardened.
 - Newly erected walling: Protect at all times from:
 - Rain and snow.
 - Drying out too rapidly in hot conditions and in drying winds.

ADDITIONAL REQUIREMENTS FOR FACEWORK

- 710 THE TERM FACEWORK
- Definition: Applicable in this specification to brick/ block walling finished fair.
 - Painted facework: The only requirement to be waived is that relating to colour.
- 730 BRICK/ CONCRETE BLOCK SAMPLES
- General: Before placing orders with suppliers submit for approval of appearance labelled samples of the following: Bricks as in clause F10/110A .
 - Selection of samples: Representative of the range in variation of appearance.
- 740 FINISHED MASONRY WORK REFERENCE PANELS
- General: Before proceeding to construct the following walling types, construct panels as specified. Give notice when panels are dry.
 - Selection of masonry units: Reasonably representative of the average quality of the whole order to be delivered .
 - Panel types:
 - Walling type: All .
 - Location: The panels should be located in good natural light and, if possible, so that they can be seen in conjunction with the finished work .
 - Size: 1.5 x 1.5 m .
 - Other requirements: None .
- 745 MASONRY SAMPLE PANELS
- Sampling frequency: A panel for each type and delivery of masonry unit.
 - Selection of masonry units: Reasonably representative of the average quality of the whole order to be delivered .
 - Panel types: As clause 740.
- 750 COLOUR CONSISTENCY OF MASONRY UNITS
- Colour range: Submit proposals of methods taken to ensure that units are of consistent and even appearance within deliveries.
 - Conformity: Check each delivery for consistency of appearance with previous deliveries and with approved reference panels; do not use if variation is excessive.
 - Finished work: Free from patches, horizontal stripes and racking back marks.
- 760 APPEARANCE
- Brick/ block selection: Do not use units with damaged faces or arrises.
 - Cut masonry units: Where cut faces or edges are exposed cut with table masonry saw.
 - Quality control: Lay masonry units to match relevant reference panels.
 - Setting out: To produce satisfactory junctions and joints with built-in elements and components.
 - Coursing: Evenly spaced using gauge rods.
 - Lifts: Complete in one operation.
 - Methods of protecting facework: Submit proposals.
- 780 GROUND LEVEL
- Commencement of facework: Not less than 150 mm below finished level of adjoining ground or external works level.
- 790 PUTLOG SCAFFOLDING
- Use: Not permitted in facework.

800 TOOTHED BOND

- New and existing facework in same plane: Bond together at every course to achieve continuity.

830 CLEANLINESS

- Facework: Keep clean.
- Mortar on facework: Allow to dry before removing with stiff bristled brush.
- Removal of marks and stains: Rubbing not permitted.

DRAFT

F22

Cast stone ashlar walling/dressings

F22 Cast stone ashlar walling/dressings

To be read with Preliminaries/General conditions.

TYPES OF DRESSINGS

Refer to DLA Drawing 1132-??? for details of cills

WINDOW CILLS - Option 1 Clause 110A
Option 2 Clause 110B
Option 3 Clause 110C

PARAPET COPINGS & CORNICE - Option 1 Clause 110A
Option 2 Clause 110B
Option 3 Refer to H41/130B

110A CAST STONE OPTION 1 - WINDOW CILLS

- Cast stone units:
 - Manufacturer: KEY STONework LTD
Architectural Stonework Solutions,
28 The Leys - Long Buckby, Northamptonshire NN6 7YD
Tel: 01327 842 585
www.keystonework.co.uk.
Product reference: Wet or Dry Cast reconstituted stone, Glass Reinforced Stone (GRS).
 - Absorption: To BS 1217 as manufacturers details.
 - Compressive strength:
Average cube strength (minimum): 35MPa (35 N/mm²).
Single cube strength (minimum): manufacturer to advise.
 - Finish: Smooth.
 - Colour: Portland.
- Mortar: As section Z21.
- Mortar used for the installation of Cast Stone Dressings should not be stronger than Designation 111 (1:1'5/6) for standard elevations.
Refer to Forticrete for other applications.
Mix: 1:1:5-6 cement:lime:sand .
 - Sand: To BS EN 13139; crushed stone with grading to approval. .
- Bond: As drawings .
- Joints: Flush.
 - Width: 5 mm .
 - Pointing: As clause 535 .
- Other requirements: All cast stone units are decorative and not designed for load bearing.
The structural stability of all units in their finished state should be approved by the structural engineers with special attention to larger surrounds and lintols.
Stone surrounds to project a min of 25mm from face of wall and to incorporate a drip to the underside - Refer to Architects Details .

110B CAST STONE OPTION 2 - WINDOW CILLS

- Cast stone units:
 - Manufacturer: FORTICRETE
Cast Stone, Kiveton Park Statio, Kiveton Park, Sheffield S26 6NP
Tel: 01909 775000
Email: caststone@forticrete.com
www.forticrete.co.uk.
Product reference: Forticrete Stone.
 - Absorption: To BS 1217 as manufacturers details.
 - Compressive strength:
Average cube strength (minimum): 35MPa (35 N/mm²).
Single cube strength (minimum): manufacturer to advise.
 - Finish: Smooth.
 - Colour: Portland stone.
- Mortar: As section Z21.
- Mortar used for the installation of Cast Stone Dressings should not be stronger than Designation 111 (1:1'5/6) for standard elevations.
Refer to Forticrete for other applications.
Mix: 1:1:5-6 cement:lime:sand .
 - Sand: To BS EN 13139; crushed stone with grading to approval. .
- Bond: As drawings .
- Joints: Flush.
 - Width: 5 mm .
 - Pointing: As clause 535 .
- Other requirements: All cast stone units are decorative and not designed for load bearing.
The structural stability of all units in their finished state should be approved by the structural engineers with special attention to larger surrounds and lintols.
Stone surrounds to project a min of 25mm from face of wall and to incorporate a drip to the underside - Refer to Architects Details .

110C CAST STONE OPTION 3 - WINDOW CILLS

- Cast stone units:
 - Manufacturer: PROCTER CAST STONE
Isabella Road, Garforth, West Yorkshire, LS25 2 DY
Tel: 0113 286 3329
Email: websales@proctergarforth.co.uk
www.caststoneuk.co.uk.
Product reference: Procter Cast Stone - Semi Dry.
 - Absorption: To BS 1217 as manufacturers details.
 - Compressive strength:
Average cube strength (minimum): as manufacturers standard).
Single cube strength (minimum): manufacturer to advise.
 - Finish: Smooth.
 - Colour: Natural.
- Mortar: As section Z21.
- Mortar used for the installation of Cast Stone Dressings should not be stronger than Designation 111 (1:1'5/6) for standard elevations.
Refer to Forticrete for other applications.
Mix: 1:1:5-6 cement:lime:sand .
 - Sand: To BS EN 13139; crushed stone with grading to approval. .
- Bond: As drawings .
- Joints: Flush.
 - Width: 5 mm .
 - Pointing: As clause 535 .
- Other requirements: All cast stone units are decorative and not designed for load bearing.
The structural stability of all units in their finished state should be approved by the structural engineers with special attention to larger surrounds and lintols.
Stone surrounds to project a min of 25mm from face of wall and to incorporate a drip to the underside - Refer to Architects Details .

115 CAST STONE GABLE VENTTO PLOTS 79, 80, 83, 84, 87, 88, 90 and 91

- Cast stone units:
 - Manufacturer: ALTERNATIVES AS OPTIONS 1, 2 and 3.
Product reference: To match cills and copings.
 - Absorption: To BS 1217 as manufacturers details.
 - Compressive strength:
Average cube strength (minimum): as manufacturers standard).
Single cube strength (minimum): manufacturer to advise.
 - Finish: Smooth.
 - Colour: Natural.
- Mortar: As section Z21.
- Mortar used for the installation of Cast Stone Dressings should not be stronger than Designation 111 (1:1'5/6) for standard elevations.
Refer to Forticrete for other applications.
Mix: 1:1:5-6 cement:lime:sand .
 - Sand: To BS EN 13139; crushed stone with grading to approval. .

GENERAL REQUIREMENTS

220 CONTROL SAMPLES

- Required samples: After finalization of design, one each of the following components: Stone cills.
- Approval of appearance: Obtain before manufacture of remaining units.
- Identification and storage location: Clearly label and retain at factory for comparison with production units.

230 QUALITY ASSURANCE

- Records: Maintain for each type of component, including:
 - Correlation with records of mixes, including batch numbers and test samples.
 - Type of reinforcement.
 - Test reports (including cube identification numbers) and testing authority (cast stone manufacturer and/ or laboratory accredited by United Kingdom Accreditation Service (UKAS) for the tests specified).
 - Dates for casting and delivery to site.
 - Any other pertinent data, e.g. identification of approved control samples.
- Submit copies of records: On request.

255 COMPRESSIVE STRENGTH

- Cube strength: Sampling to BS 1217. Testing to BS EN 12390-1, 2 and 3 as appropriate, with compaction and curing representative of the methods used in production of the cast stone components.

300 REINFORCEMENT

- Non structural reinforcement: Include to resist shrinkage and handling stresses.
- Carbon steel reinforcement: To BS 4449 and BS 4483 as appropriate.
 - Cutting and bending: To BS 8666.
- Galvanized reinforcement: Galvanizing to BS EN ISO 1461 after cutting.
- Stainless steel reinforcement: To BS 6744, designation 1.4301.
 - Cutting and bending: To BS 8666.
- Condition at time of placement: Clean, free of corrosive pitting, loose materials and substances which adversely affect the reinforcement, mix, or bond between the two.
- Fixing: Accurate and secure. Prevent intrusion of wire or clips into the cover.

305 BED JOINT REINFORCEMENT

- Standard: To BS EN 845-3.

312 COVER TO REINFORCEMENT

- General: Minimum cover to reinforcement:
 - Carbon steel:
 - 40 mm: External faces (exposed to weather).
 - 30 mm: Internal faces (protected from weather).
 - Galvanized steel:
 - 30 mm: External faces (exposed to weather).
 - 20 mm: Internal faces (protected from weather).
 - Austenitic stainless steel:
 - 10 mm or 2 x bar diameter: All faces (select the greater of the two).
 - Other materials/ coatings: Submit proposals including minimum cover.

335 MANUFACTURING ACCURACY

- Standard: To BS 1217, section 8.
- Additional tolerances: none.

340 MIXES GENERALLY

- Constituent materials and mix design: Achieve constant colour and texture for each finish type.
- Aggregates for facing mixes: Free from particles which may cause 'popouts', or unsightly marking or staining.
- Total chloride ion content (maximum): Cl 0.40.
- Admixtures containing calcium chloride: Do not use.

350 SEPARATE FACING AND BACKING MIXES

- Thickness of facing mix at any point (minimum): 20 mm.
- Distance of reinforcement from the junction of the two mixes (minimum): 10 mm.
- Joining facing and backing mixes: Bond so that they are effectively monolithic.

360 CASTING AND CURING

- Compaction: Thorough.
- Protection: Prevent premature drying out during curing period.
- Immature components: Prevent distortion from movement, vibration, overloading, physical shock, rapid cooling and thermal shock.
- Delivery to site: Not until at least 14 days after casting.

370 QUALITY OF FINISHES

- Appearance standard: As established by samples.

380 INSPECTION

- Completed components: Make available for inspection before delivery to site. Give notice when ready for inspection.

LAYING AND JOINTING

400 SAND SAMPLES

- Submit: Before placing orders, provide representative samples of Sands for bedding and pointing .

430 ADVERSE WEATHER

- General: Do not use frozen materials and do not lay on frozen surfaces.
- Temperature: Do not lay blocks/ dressings:
 - In cement gauged mortars when the air temperature is at or below 3°C and falling or below 1°C and rising (unless mortar has a temperature of not less than 4°C when laid and walling is thoroughly protected).
 - In hydraulic lime:sand mortars when the air temperature is at or below 5°C and falling or below 3°C and rising.
- Freezing conditions: Maintain temperature of the work above freezing until mortar has fully set.
- Precipitation: Protect newly erected walling against rain and snow by covering when precipitation occurs and at all times when work is not proceeding.
- Hot and dry conditions: Prevent newly erected masonry from drying out too rapidly.
- Remedial work: Rake out and replace mortar damaged by frost and where instructed, rebuild damaged work.

440 LAYING GENERALLY

- Selection: Do not use units with damaged faces or arrises.
- Accuracy:
 - Courses: Level and true to line.
 - Faces, angles and features: Plumb.
 - Setting out: Achieve satisfactory junctions and joints with adjoining or built-in elements and components.
- Absorbent units: Dampen in warm weather to reduce suction.
- Mortar joints:
 - Laying: Full bed of mortar with all joints and voids filled.
 - Temporary distance pieces: Lead or stainless steel. Remove when mortar is sufficiently strong.
 - Appearance: Neat and consistent.
- Cleanliness: Keep facework clean. Rubbing and other abrasive or chemical cleaning methods to remove marks and stains, not permitted.
- Cutting of reinforced units: Not permitted.

450 WALLING/ DRESSINGS BELOW GROUND LEVEL

- Extent of facework below finished level of adjoining ground or external works level (minimum): 150 mm.

490 PUTLOG SCAFFOLDING

- Use: Not permitted.

500 ONE PIECE SILLS/ THRESHOLDS

- Bed joints: Leave open except under:
 - End bearings.
 - Masonry mullions.
- Pointing on completion: Mortar to match adjacent work.

510 OPENINGS

- Method of forming: Rigid templates, accurately fabricated to the required size.

520 JOGGLE JOINTS

- General: Fill with bedding mortar. Tamp to expel air.

535 POINTING

- Joint preparation: Rake out to depth of 7–10 mm as work proceeds. Remove debris. Dampen surface.
- Mortar application: Neat and consistent.

540 SUPPORT OF EXISTING WORK

- Joint above inserted lintel or masonry: Fully consolidated with semidry mortar to support existing structure.

DRAFT

F30
Accessories/ sundry items for brick/ block/ stone
walling

F30 Accessories/ sundry items for brick/ block/ stone walling

To be read with Preliminaries/ General conditions.

CAVITIES

- 110 CONCRETE FILL TO BASE OF CAVITY
 - Concrete generally: To BS EN 206 and BS 8500-2.
 - Concrete type: Designated GEN1.
 - Workability: High.
 - Extent: Maintain 75 mm between top of fill and external ground level and a minimum of 225 mm between top of fill and ground level dpc.
 - Placement: Compact to eliminate voids.
- 120 CLEANLINESS
 - Cavity base and faces, ties, insulation and exposed dpcs: Free from mortar and debris.
- 130 PERPEND JOINT WEEP HOLES
 - Form: Open perpend joint.
 - Locations: Through outer leaf immediately above base of cavity, at cavity trays, stepped dpcs and external openings. 75 mm above top of cavity fill at base of cavity.
 - Provision: At not greater than 1000 mm centres and not less than two over each opening.
- 132A PERPEND JOINT PLASTICS WEEP HOLES
 - Manufacturer: Manthorpe Building Products Ltd.
 - Web: www.manthorpebuildingproducts.co.uk .
 - Product reference: G951 Peep Weep .
 - Colour: Terracotta
 - Locations: Through outer leaf immediately above base of cavity, at cavity trays, stepped dpcs and external openings. 75 mm above top of cavity fill at base of cavity.
 - Provision: At not greater than 1000 mm centres and not less than two over each opening.
- 155A PARTIAL FILL CAVITY INSULATION GENERALL TO EXTERNAL CAVITY WALLS
 - Insulation: Phenolic foam boards to BS EN 13166.
 - Product certification: British Board of Agreement (BBA) Certificate number 16/5299.
 - Manufacturer: Kingspan Insulation.
www.kingspaninsulation.co.uk
info@kingspaninsulation.co.uk.
 - Product reference: Kooltherm® K108 Cavity Board .
 - Recycled content: Not applicable.
 - Face size (length x width): 1200 x 450 mm.
 - Thickness (nominal): 50 mm.
 - Thermal conductivity: 0.018 W/(m.K).
 - Reaction to fire class: A1.
 - Additional requirements: To be installed in accordance with instructions issued by Kingspan..
 - Placement: Secure against face of inner leaf.
 - Residual cavity: Clear and unobstructed.
 - Joints between boards, at closures and penetrations: No gaps and free from mortar and debris.

171A VENTILATION DUCTS IN EXTERNAL WALLING GENERALLY

- All externally visible vents and extracts to be coloured to minimise their visual impact. Depending on material they are set within the col;ours should be as follows:
 - Brick - Red, provide sample for approval
 - Manufacturer: Contractor's choice .
 - Product reference: Submit proposals .
- Placement: Across cavity, sloping away from inner leaf. Full mortar joints to seal cavity.
- Protection from water penetration to inner leaf: Where barrier is not integral to duct, form stepped dpc cavity tray with stop ends above duct, extending 150 mm on each side.

180A CAVITY CLOSERSGENERALLY TO EXTERNAL WALLS

- Manufacturer: Kingspan Insulation.
 - Web: www.kingspaninsulation.co.uk.
 - Email: info@kingspaninsulation.co.uk .
 - Product reference: Kingspan Thermabate®100 to suit final details .
- Accessories: To include integral dpc .

REINFORCING/ FIXING ACCESSORIES

210 CAVITY WALL TIESFOR ALL CAVITY WALLS

- Standard: To BS 1243.
 - Type: 1 (Masonry heavy duty) .
- Material/ finish: Stainless steel .
- Sizes: Dependant on wall thickness - see clause 215A .

215A CAVITY WALL TIES USED WITH PARTIAL FILL INSULATION

- Manufacturer: Ancon Building Products.
 - Web: www.ancon.co.uk.
 - Email: info@ancon.co.uk.
 - Product reference: ST1 Wall tie
- Length: 225 mm.
- Material: Stainless steel grade 1.4301 (304)

225 FIXING TIES IN MASONRY CAVITY WALLS

- Embedment in mortar beds (minimum): 50 mm.
- Placement: Sloping slightly downwards towards outer leaf, without bending. Drip centred in the cavity and pointing downwards.
- Spacing: Staggered in alternate courses.
 - Horizontal centres: 900 mm.
 - Vertical centres: 450 mm.
- Additional ties: Provide within 225 mm of reveals of unbonded openings and at the vertical reveals of unsupported masonry.
 - Spacing: In every course.

233 FIXING TIES IN MASONRY CAVITY WALLS WITH PARTIAL FILL CAVITY INSULATION

- Embedment in mortar beds (minimum): 50 mm.
- Placement: Sloping slightly downwards towards outer leaf, without bending. Drip centred in the cavity and pointing downwards.
- Spacing: Evenly space in non staggered horizontal and vertical rows.
 - Horizontal centres: 600 mm.
 - Vertical centres: 450 mm.
- Spacing centres of top (eaves) row of ties: Not more than 450 mm.
- Provision of additional ties: Within 225 mm of reveals of unbonded openings and at the vertical reveals of unsupported masonry.
 - Spacing: At not more than 300 mm centres vertically.

255A WIND POSTS TO BLOCK D

- Manufacturer: Contractor's choice.
 - Product reference: Submit proposals.
- Material: Austenitic stainless steel grade 1.4301 (304) .
- Sizes: Refer to Structural engineers, as design requires.
- Fixings (complete with washers and shims): As manufacturers details.

260 ANGLE SUPPORTSGENERALLY

- Manufacturer: Ancon or similar approved.
 - Product reference: Ancon MDC as clause 265A.
- Material: Austenitic stainless steel.
- Size: To suit wall construction.
- Fixings (complete with washers and shims): To manufacturers design.

265A SUPPORT SYSTEM TO APARTMENT BLOCKS A, B and C.

- Manufacturer: Ancon Building Products.
 - Web: www.ancon.co.uk.
 - Email: info@ancon.co.uk.
 - Tel: +44 (0)114 275 5224
 - Product reference: MDC/120/8.2/P50 system
- MDC Bracket Angle Support System fixed to Ancon 30/20 Cast-In Channel.
- Accessories: Thermal break, BK03-135.
- Refer to Structural engineers for loading requirements.

FLEXIBLE DAMP PROOF COURSES/ CAVITY TRAYS

330A DAMP PROOF COURSE GENERALLY

- Manufacturer: IKO Plc
 Appley Lane North, Appley Bridge, Wigan, Lancashire WN6 9AB
 Tel: 01257 255771 E-mail: technical.uk@iko.com Web: www.ikogroup.co.uk.
Product reference: [Hyload Original High Performance DPC system].
Installation Reference: Agrément Certificate No. 95/3133
Properties
 - Material: Polymeric
 - Nominal Roll Length: 20m
 - Nominal thickness: 1.27mm
 - Nominal Weight: 1.5 kg/m2
 - Roll Width: As required

330B DAMP PROOF COURSE LINEAR BELOW COPING STONES

- Manufacturer: IKO Plc
 Appley Lane North, Appley Bridge, Wigan, Lancashire WN6 9AB
 Tel: 01257 255771 E-mail: technical.uk@iko.com Web: www.ikogroup.co.uk .
Product reference: Hyload Permabit High Performance DPC system.
Installation Reference: Agrément Certificate No. 88/1966
Properties
 - Material: Bitumen Polymer
 - Nominal Roll Length: 20m
 - Nominal thickness: 1.25mm
 - Nominal Weight: 1.6 kg/m2
 - Roll Width: As required

345A SITE FORMED FLEXIBLE SHEET CAVITY TRAYS ABOVE LINTELS, CAVITY BARRIERS

- Material: Polymeric.
- Manufacturer: Manufacturer: IKO PLC.
Appley Lane North, Appley Bridge, Wigan, Lancashire WN6 9AB
Tel: 01257 255771 E-mail: technical.uk@iko.com Web: www.ikogroup.co.uk.
 - Product reference: Hyload Original High Performance DPC system.
 - Installation reference: Agrément Certificate No. 95/3133.
- Properties
 - Nominal Roll Length: 20m
 - Nominal thickness: 1.27mm
 - Nominal Weight: 1.5 kg/m2
- Roll Width: As required

345B SITE FORMED FLEXIBLE SHEET CAVITY TRAYS WITHIN PARAPETS

- Material: Bitumen polymer.
- Manufacturer: Manufacturer: IKO PLC.
Appley Lane North, Appley Bridge, Wigan, Lancashire WN6 9AB
Tel: 01257 255771 E-mail: technical.uk@iko.com Web: www.ikogroup.co.uk.
 - Product reference: Hyload Permabit High Performance DPC system.
 - Installation reference: Agrément Certificate No. 88/1966.
- Properties
 - Nominal Roll Length: 20m
 - Nominal thickness: 1.25mm
 - Nominal Weight: 1.6 kg/m2
- Roll Width: As required

380A PREFORMED CAVITY TRAY JUNCTION CLOAKS/STOP ENDS

- Manufacturer: IKO PLC.
Appley Lane North, Appley Bridge, Wigan, Lancashire WN6 9AB
Tel: 01257 255771 E-mail: technical.uk@iko.com Web: www.ikogroup.co.uk
 - Product reference: Hyload Preformed Cloaks
- Material: Polymeric
Colour: Black.
- Accessories: Hyload DPC Jointing Tape, Product Code: 29510000.

INSTALLATION OF DPCS/ CAVITY TRAYS

400A COLD WEATHER WORKING

- In cold weather, warm DPC rolls before unrolling and Hyload DPC Jointing Tape prior to use.
-

415A HORIZONTAL DPCS

- Placement: In continuous lengths sandwich between even beds of fresh mortar, with 100 mm laps at joints and full laps at angles.
- All lap joints in the DPC must have 100mm overlap and be completely sealed using Hyload DPC Jointing Tape.
Width of Hyload Original/Permabit DPC to be at least full width of masonry leaf unless otherwise specified. Do not cover edges of DPC with mortar.
Where there are separate DPCs in each leaf of a cavity wall, ensure that edges do not project into the cavity.
Hyload Original/Permabit DPC must be sandwiched between even beds of wet mortar and perforations in adjacent courses of brickwork must be completely filled with mortar.
Complete all work strictly in accordance with Agrément Certificate and with good practice guidance within BS8215:1991

425 GROUND LEVEL DPCS

- Joint with damp proof membrane: Continuous and effectively sealed.

435A STEPPED DPCS IN EXTERNAL WALLS

- Install DPC in external walls on sloping ground, ensure that the DPC is a minimum of 150 mm above finished ground level and all changes in level are created using lapped joints with Hyload Preformed Cloak Units.

445 SILL DPCS

- Form and placement: In one piece and turned up at back when sill is in contact with inner leaf.

455A COPING/ CAPPING DPCS

- Placement: Bed Hyload Permabit DPC in one operation to ensure maximum bond between masonry units, mortar and DPC.
All lap joints in the DPC must have 100mm overlap and be completely sealed using Hyload DPC Jointing Tape..
- Dpcs crossing cavity: Provide Hyload Copeclose rigid support to prevent sagging.

465 SEALING DPCSGENERALLY

- Overlaps and junctions: Seal with Hyload DPC Jointing Tape or Hyload DPC Lap Adhesive (used with Hyload Original DPC System) .

475A SITE FORMED CAVITY TRAYS

- Requirements to prevent downward ingress of water:
 - Accurately formed to the profiles shown on drawings, and firmly secured.
 - Carefully cleaned to remove debris and mortar droppings before they set.
 - Carefully protected from perforation and other damage.
 - Laid in continuous lengths where practicable. All cavity trays must be sandwiched between even beds of wet mortar in the external walls and either built in or surface-fixed to the internal leaf.
 - Surface fixing is achieved by bonding the DPC to the inner leaf with Hyload DPC Jointing Tape on IKOpro SA Bitumen Primer; the joint is secured with Hyload DPC Fixing Strip and fixings suitable for the substrate.
 - All lap joints in the cavity tray must have 100mm overlap and be completely sealed using Hyload DPC Jointing Tape.
 - End lap joints supported using Hyload DPC Joint Support Boards.
 - All awkward junctions e.g. changes in directions, levels, junctions with columns or where the cavity tray stops, require the application of Hyload Preformed Cloak Units. Ensure all joints between Preformed Cloak Units and Hyload Original/Permabit DPC are sealed as above.
 - Complete all work strictly in accordance with Agrément Certificate and with good practice guidance within BS8215:1991

485A CAVITY TRAYS OVER OPENINGS AND OTHER CAVITY BRIDGINGS

- Length: To extend not less than 150 mm beyond ends of lintels/ bridgings ensuring Hyload Preformed Cloak Stop Ends are incorporated.

515 DPC/ CAVITY TRAY LEADING EDGE IN FACEWORK - FLUSH

- Treatment at face of masonry: Finish flush and clear of mortar at the following locations: to be agreed .

535 DPC/ CAVITY TRAY LEADING EDGE IN FACEWORK - PROJECTING

- Treatment at face of masonry: Projecting 5 mm from face of wall at the following locations: At window and door heads .

560A VERTICAL DPCS GENERALLY

- Form: In one piece wherever possible.
 - Joints: Upper part overlapping lower not less than 100 mm and fully seal using Hyload DPC Jointing Tape or Hyload DPC Lap Adhesive

570 JAMB DPCS AT OPENINGS

- Joint with cavity tray/ lintel at head: Full underlap.
- Joint with sill/ horizontal dpc at base: Full overlap.
- Projection into cavity: Not less than 25 mm.
- Relationship with frame: In full contact.

JOINTS

610 MOVEMENT JOINTS WITH SEALANT TO EXTERNAL FACING BRICKWORK

- Joint preparation and sealant application: As section Z22.
- Filler: Contractors Proposals.
 - Thickness: To match design width of joint.
 - Manufacturer: Contractor's choice.
 - Product reference: Submit proposals.
 - Placement: Build in as work proceeds with no projections into cavities and to correct depth to receive sealant system.
- Sealant:
 - Designation: TBC.
 - Manufacturer: Contractor's choice.
 - Product reference: Submit proposals.
 - Colour: To match adjoining brickwork.

615 MOVEMENT JOINTS WITHOUT SEALANT TO EXTERNAL WALL STARTER LOCATIONS

- Filler: Closed cell polyethylene foam.
 - Thickness: To match design width of joint.
 - Manufacturer: Contractor's choice.
 - Product reference: Contractor's choice.
 - Placement: Build in as work proceeds filling the joint but without projecting into cavities.

630 UNEXPOSED CONTRACTION JOINTS

- Formation: Close butt as work proceeds.

650 POINTING IN FLASHINGS

- Joint preparation: Free of debris and lightly wetted.
- Pointing mortar: As for adjacent walling.
- Placement: Fill joint and finish flush.

660 PINNING UP TO SOFFITS

- Top joint of loadbearing walls: Fill and consolidate with mortar.

670 TOPS OF NONLOADBEARING WALLS

- Restraints: 75 x 100 mm channel of galvanized steel sheet to BS EN 10346 fixed to metal deck with self-tapping screws at 600 mm centres .
 - Fixing: Secure to soffit.
- Joint filler: Contractors proposals .
 - Placement: Full, no gaps.

PROPRIETARY SILLS/ LINTELS/ COPINGS/ DRESSINGS

725A CAST STONE SILLS LAID IN CEMENT GAUGED MORTAR GENERALLY TO BLOCK D, HOUSES PLOTS 102 - 107

Refer to DLA Window Types ????

- Standard: To BS 1217.
- Manufacturer: TBC.
 - Product reference: TBC.
- Dimensions: As shown on drawings.
- Finish/ colour: TBC.
- Mortar for bedding/ jointing: Cement gauged as section Z21.
 - Standard: To BS EN 998-2.
 - Mix: 1:1:6 cement:lime:sand.
 - Additional requirements: To match adjoining masonry.
- Joints: Flush.
- Bedding one piece sills: Leave bed joints open except under end bearings and masonry mullions. On completion, point to match adjacent work.

745 PRESTRESSED CONCRETE LINTELS

- Standard: To BS EN 845-2.
- Manufacturer: Contractor's choice.
 - Product reference: Contractor's choice.
- Types: Single.
- Sizes: As Required.
- Additional requirements: Resistance to fire: 1hour in compartmentation walls ..
- Placement: Bed on mortar used for adjacent work. Prop at not more than 1.2 m centres to prevent displacement during construction. Retain props in position for not less than 14 days or until mortar has matured, whichever is longer.
 - Bearing length (minimum): 150 mm.

755 PREFABRICATED STEEL LINTELS

- Standard: To BS EN 845-2.
- Manufacturer: Contractor's choice.
 - Product reference: Contractor's choice.
- Types: Single.
- Material/ finish: To manufactures details depending on location and exposure.
- Sizes: As required.
- Additional requirements: None.
- Placement: Bed on mortar used for adjacent work.
 - Bearing length (minimum): 150 mm.

755A PREFABRICATED STEEL LINTELS INSULATED TYPE TO EXTERNAL CAVITY WALL

- Standard: To BS EN 845-2.
- Manufacturer: Catnic.
 - Product reference: CG90/100.
- Types: Combined.
- Material/ finish: Manufacturers standard - Galvanised steel, powder coated.
- Sizes: as required to suit openings.
- Additional requirements: Refer to structural engineers.
- Placement: Bed on mortar used for adjacent work.
 - Bearing length (minimum): 150 mm.
- Full lintol schedule to be prepared and submitted for approval by Structural Engineers to ensure design loadings adhered to.

765A CAST STONE COPING UNITS LAID IN CEMENT GAUGED MORTAR
APARTMENTS BLOCK A, B & C COPINGS @ HIGH LEVEL
HOUSE PLOT 79-92 FRONT ELEVATION

- Standard: To BS 1217.
- Manufacturer: To be confirmed.
 - Product reference: TBC.
- Dimensions: As shown on drawings.
- Finish/ colour: To match sample.
- Mortar for bedding/ jointing: Cement gauged as section Z21.
 - Standard: To BS EN 998-2.
 - Mix: 1:1:6 cement:lime:sand.
 - Additional requirements: Coloured mortar to match brick mortar colour.
- Joints: Full and finished flush.
- Placement: Lay on a full bed of mortar to line and level.

MISCELLANEOUS ITEMS

830 BUILDING IN FRAMES

- Preparation: Remove horns and provide support.
- Fixing cramps: Fully bed in mortar.

840 OPENINGS FOR FRAMES

- Formation: Use accurate, rigid templates to required size.

850 WALL PLATES

- Placement: On full bed of mortar to correct horizontal level.

DRAFT

F31

Precast concrete sills/ lintels/ copings/ features

F31 Precast concrete sills/ lintels/ copings/ features

To be read with Preliminaries/ General conditions.

TYPES OF COMPONENT

140 CONCEALED PRECAST LINTELS

- Concrete: Designated to BS 8500-2: Minimum RC30.
 - Aggregate size (maximum): 20 mm.
- Configuration:
 - Clear span up to 900 mm:
 - Section: 140 mm deep x width of wall.
 - Bearing: 150 mm at both ends.
 - Reinforcement: One 12 mm carbon steel bar for each 105 mm of wall thickness.
 - Clear span 900 mm to 1800 mm:
 - Section: 215 mm deep x width of wall.
 - Bearing: 225 mm at both ends.
 - Reinforcement: One 16 mm carbon steel bar for each 105 mm of wall thickness.
- Cover to reinforcement (nominal): 20 mm minimum.

150 CONCEALED PRECAST PADSTONES .

- Concrete: Designated to BS 8500-2: Minimum RC30.
 - Aggregate size (maximum): 20 mm.
- Other requirements: None .

GENERAL REQUIREMENTS

210 MOULDS

- Permissible fabrication and operating tolerances: Length 0 to +6 mm, other dimensions ± 3 mm.

220 CONCRETE GENERALLY

- Specification: To BS 8500-2 and BS EN 206-1.
- Producer: Currently certified by a body accredited by UKAS to BS EN 45011 or BS EN ISO/IEC 17065 for product conformity certification of ready-mixed concrete.
- Chloride class of concrete:
 - Excluding SRPC: C1 0.40.
 - Using SRPC: C1 0.20.
 - Reinforced and heat cured: C1 0.10.
 - Prestressed: C1 0.10.
- Admixtures containing calcium chloride: Not allowed.

250 REINFORCEMENT

- Carbon steel reinforcement: As appropriate to BS 4449, BS 4482 and BS 4483.
 - Cutting and bending: To BS 8666.
- Galvanized reinforcement: Galvanized to BS EN ISO 1461 after cutting. Chromate treated.
- Stainless steel reinforcement: To BS 6744.
 - Designation 1.4301.
 - Cutting and bending: To BS 8666.
- Non structural reinforcement: Include to resist shrinkage and handling stresses.
- Bimetallic corrosion and staining: Prevent by appropriate selection and use of materials.
- Condition at time of placement: Clean, free of corrosive pitting, loose materials and substances that adversely affect reinforcement, concrete, or bond between the two.
- Fixing: Accurate and secure.
 - Method: Wire tying, approved steel clips or tack welding if permitted.
 - Concrete cover: Maintain free of all tying wire or clips.

255 QUALITY ASSURANCE OF REINFORCEMENT

Reinforcement to BS 4449, BS 4483, BS 6744: Obtain valid certificates of approval for product conformity issued by the UK Certification Authority for Reinforcing Steels.

260 CASTING AND CURING

- Placing of concrete: Thoroughly compact.
- Protection against drying out: Methods and duration to BS EN 13369 clause 4.2.1.3.
- Immature components: Avoid movement, vibration, overloading, physical shock, rapid cooling and thermal shock.
- Delivery to site: Minimum 14 days after casting.

261 CUTTING

- Cutting of precast concrete components: Not permitted.

262 RECORDS

- Records for each type of component: Maintain details including:
 - Unique identification number.
 - Identification of the producer.
 - Identification of the place of production.
 - Correlation with records of mixes, including batch numbers.
 - Date of each stage of manufacture.
 - Dates and results of all tests, checks and inspections.
 - Dimensions related to specified levels of accuracy.
 - Specific location in the finished work.
 - Weight of the unit.
 - Damage and making good.
 - Any other pertinent data, e.g. if unit is a production control unit.
- Availability of records for inspection: On request.

FAIR FACED COMPONENTS

320 DETAILS OF SAMPLES

- Submittals after approval of appearance and before manufacture of production units:
 - Aggregates: Confirm type, maximum size, grading and source.
- Conformity of designed concrete: Evidence of compliance for compressive strength class and limiting values of composition.

330 MIXES FOR VISIBLE FACED COMPONENTS

- Constituent materials and mix design for each finish type: To remain constant.
- Colour and appearance of each finish type: To remain constant.
- Aggregates: To BS EN 12620.
 - Origin: Single source for each finish type, having sufficient quantity for whole contract.

341 CONDITIONS FOR SEPARATE FACING AND BACKING MIXES

- Difference in cement content: Not greater than 80 kg/m³.
- Thickness of facing mix: 10 mm greater than maximum aggregate size, minimum 25 mm.
- Location of reinforcement: Minimum 20 mm away from the interface between mixes.
- Compaction of facing and backing mix: Carry out to create monolithic construction.

350 QUALITY OF FINISHES

- Appearance standard: As established by samples.

365 COVER ON EXPOSED AGGREGATE FACES

- Nominal cover: Exclusive of aggregate projection.

370 COVER ON VISIBLE FACES

- Spacers: Not permitted.
- Proposed method statement: Submit.

380 CONSISTENCY OF PRODUCTION METHODS

- Production methods: To remain consistent for each matching type of finish.
 - Finish appearance: To remain within the range of variation indicated by the samples submitted.
- Changes to production methods: If variations are proposed for components of the same finish, submit evidence that there will be no difference in appearance.

390 INSPECTION

- Completed components: Give notice when ready to be inspected at factory.

INSTALLATION

420 LAYING

- Mortar for bedding and jointing: As section Z21.
 - Type: Site batched and mixed.
 - Mix: As used for adjacent work.
 - Packing: If required use slate.
- Bedding components: On full bed of mortar.
- Removal of marks, stains and extraneous mortar on visible faces: Rubbing not permitted.

430 SUPPORT OF EXISTING WORK OVER NEW LINTELS

- Joint above lintels: Fully fill and compact with semidry mortar.

440 ONE PIECE SILLS/ THRESHOLDS

- Bed joints: Leave clear of mortar except at end bearings and beneath masonry mullions.
 - On completion: Point with mortar to match adjacent work.

DRAFT

G
Structural/Carcassing metal/timber

DRAFT

G10
Structural steel framing

G10 Structural steel framing

GENERAL REQUIREMENTS/ INFORMATION

- 110A GENERAL INFORMATION ALL STRUCTURAL ELEMENTS
 - TO BE READ IN CONJUNCTION WITH ROLTON'S SPECIFICATION, FOR STRUCTURAL REQUIREMENTS ROLTONS SPECIFICATION TO TAKE PRECEDENCE
- 111A DESIGN OF ALL STEELWORK GENERALLY
 - Design concept: As Design Brief for Civil and Structural Elements..
 - Design standard: BS EN 1993-1-1.
 - Supplementary requirements: None.
 - Completion of design: Design and detail members to suit conceptual layout detailed in contract drawings. Design and detail joints to BS 5950.
 - Loading requirements: To BS EN 1991 and Design Brief for Civil and Structural Elements.
 - Design parameters: Refer to Design Brief for Civil and Structural Elements.
 - Building class to Building Regulations: Block A - Class 2A, Block B1 - Class 2A, Block B2 - Class 2A, Block C - Class 2B.
 - Design working life category: 4.
 - Fixings to foundations/ walls: Contractor to propose details
- 114 EUROCODES
 - National Annex: Reference to a Eurocode shall be deemed to include the appropriate United Kingdom National Annex, and nationally determined parameters shall apply.
 - Substitution of alternative design rules for Application Rules: Not permitted.
- 115 DESIGN CONSTRAINTS - GENERAL
 - Members forming bracing systems or girders of lattice construction: Unless detailed or instructed otherwise, position so that their lines of action intersect at a point.
 - Bolts:
 - Diameter (minimum): 16 mm.
 - Number per connection (minimum): Two, unless otherwise indicated.
 - Other requirements: None.
 - Punching of bolt holes: Not permitted.
 - Welds: At least 6 mm fillet.
 - Other constraints: None.
- 125 SPECIFICATION STANDARD
 - Standard: Comply with latest edition of National Structural Steelwork Specification (NSSS CE Marking version).
 - Additional requirements: None.
 - Document availability: For the duration of the work, at fabrication shop and on site.
 - References to Engineer in NSSS CE Marking version: For the purpose of this contract, interpret such references as being to the person named in section A10 as administering the contract on behalf of the Employer.
 - Exceptions: None.

130 GENERAL STEEL SECTIONS AND PLATES GENERALLY

- Standard: To BS EN 10025-2.
- Grade: S355JR.
 - Options: None.
- Source: Obtain steel from a source accredited to a national or internationally accepted quality standard.
- Other requirements: None.

135 HOLLOW STEEL SECTIONS GENERALLY

- Standard: To BS EN 10210-1.
- Grade: S355J0H.
 - Options: None.
- Source: Obtain steel from a source accredited to a national or internationally accepted quality standard.
- Other requirements: None.

COLD FORMED MATERIALS

FABRICATION

180 NOTIFICATION OF COMMENCEMENT

- Notice: Give notice before fabrication is due to start.
 - Period of notice (minimum): Five working days .

190 MARKING

- Identifying and recording materials and components: Submit details of proposed methods.
- Location of marks:
 - Generally: Visible for checking after erection.
 - Weathering steel: On surfaces not exposed to open view in the completed work.
- Steel to be blast cleaned, pickled, metal sprayed or galvanized: Marked so that subsequent treatment cannot obliterate the marking.

195 HARD STAMPING

- Usage: Not permitted except as indicated on drawings.

210 END CONNECTIONS

- Angle cleats: Project 10 mm beyond ends of simply supported members.

215 HOLLOW SECTIONS

- Insides of sections: Debris and moisture removed before sealing ends and openings.

220 ACCESS/ VENTILATION HOLES IN BASE PLATES

- Base plates larger than 1 m²: Make 25 mm diameter holes as necessary for pressure grouting, escape of entrapped air or direct compaction of filling/ bedding material.

225 STEELWORK TO BE GALVANIZED

- Cutting, drilling and shop welding: Complete before galvanizing.
- Vent and drain holes: Provide as necessary.
 - Locations: Submit proposals.
- Sealing: Not required .

235 SHOP INSPECTION

- Give notice: Before fabricating Roof Structure - Refer to Structural drawings.
 - Period of notice (minimum): Five working days.

WELDING

255 SITE WELDING

- Usage: Permitted only where indicated on drawings.
- Working conditions: Suitable and safe. Do not weld when surfaces are wet or when ambient temperature is below 0°C.

270 ADDITIONAL WELDS

- Welds (including tack welds) not indicated on drawings: Not permitted without approval.

315 HOLDING DOWN SYSTEMS

- Bolts:
 - S tandard: To BS 7419.
 - T ype: Hexagonal headed.
 - G rade: 8.8.
 - D iameter: To be determined by Design.
 - C oating: None.
- Bolt boxes: Submit proposals.
- Anchorage: As required for design.
- Other requirements: None.

335 SPRING WASHERS

- Standard: To BS 4464.

370 GALVANIZED COATING TO BOLT ASSEMBLIES

- Standard: To BS 7371-6.
- Galvanizing: Applied by fastener manufacturer. Passivated and lubricated if no additional coatings are specified. Nuts tapped after galvanizing.
- Use/location: All shop and site connections in shop painted steelwork.

390 SEALED HOLLOW SECTIONS

- Holes: Sealed to prevent access of moisture.
 - Method of sealing: Submit proposals.

ERECTION

410 PRE-ERECTION CHECKS

- Scope: At least 7 days before proposed erection start date, check the following:
 - Foundations and other structures to which steelwork will be attached: Accuracy of setting out.
 - Holding down bolts: Position, protruding length, slackness and condition.
- Inaccuracies and defects: Report without delay.
- Permission to commence erection: Obtain.

432 TEMPORARY SUPPORT

- Permanent bracing system:
 - Vertical: Concrete shear walls around stair wells/ lift shafts.
 - Horizontal: Diaphragms formed by precast concrete floor units acting compositely with the in situ concrete topping..
- Temporary bracing/ restraints: Provide as necessary until permanent bracing system is complete and sufficiently mature to carry loads and all connections have been made to the permanent system.
- Elements to be supported: Frames.
- Bracing/ Restraints: Provide as necessary until permanent connection can be made to floor.
- Forces and moments in temporary supports: As calculations.

440 COLUMN BASES

- Levels: Adjust using steel shims or folding wedges no larger than necessary.
- Location of shims/ wedges: Position symmetrically around perimeter of base plate. Do not use a single central pack.
- Give notice: If space beneath any column base is outside specified limits for bedding thickness.
- Accuracy of erection: Check, and correct errors before filling and bedding beneath bases and carrying out other adjacent work.

443 PROPRIETARY FILLING/ BEDDING OF COLUMN BASES

Bedding thickness range: 10 - 100mm.

- Preparation: Concrete surfaces scarified to provide a good mechanical key.
- Bolt pockets and spaces beneath base plates: Completely filled with proprietary non shrink grout.

TESTING

470 SITE TESTING OF ANCHORS TO MASONRY

- Standard: To BS 5080.
- Preliminary tests: 5 No. tests for shear loading to failure in locations that will be hidden in completed structure.
- Proof tests: Test 10% of working fixings to 1.5 times the working load.
- Test results: Report failures and seek instructions.

475 PRODUCTS

- Steel: Submit test certificates.

PROTECTIVE COATINGS

535 INSPECTION OF COATING WORK

- Work in progress: Permit coating manufacturer to inspect and take samples of products.
 - Notice: Give notice of dates for:
 - Start of surface preparation and coating.
 - Coated members or components leaving the works.
- Period of notice (minimum): 5 working days.

PROTECTIVE COATING SYSTEMS

620 GALVANIZING TO BLAST CLEANED STEEL

- Use/ location: All hidden steelwork..
- Preparation: Blast cleaning to BS EN ISO 8501-1, preparation grade Sa2½ using chilled angular iron grit grade G24 to give a coarse surface profile, followed by chemical cleaning.
- Galvanizing: To BS EN ISO 1461.
 - Minimum mean coating thickness: 140 micrometres.

630 GALVANIZING PLUS SITE APPLIED EPOXY PRIMER AND POLYURETHANE FINISH

- Use/ location: All visible steelwork.
- Shop preparation: Blast cleaning to BS EN 8501-1, preparation grade Sa2½ using chilled angular iron grit grade 24 to give a coarse surface profile, followed by chemical cleaning.
- Galvanizing: To BS EN ISO 1461.
 - Minimum mean coating thickness: 70 micrometres.
- Paint manufacturer: Contractors Choice.
- Pretreatment primer: Mordant wash ('T' wash if required).
- Site intermediate coat: Not required.
 - Dry film thickness: As recommended by paint manufacturer.
- Site top coat: As recommended by paint manufacturer.
 - Dry film thickness: As recommended by paint manufacturer.
 - Colour: TBC.
- Special requirements: Stripe intermediate coat to external angles.

PREPARATION FOR PAINTING

710 OFFSITE PREPARATION AND PAINTING

- Working area: Covered and properly lit, heated and ventilated.
- Sequence of working: Select from the following and submit proposals:
 - Fabricate, blast clean, prime.
 - Blast clean, fabricate, remove flash rust with a light overall sweep blast, prime.
 - Blast clean, apply weldable prefabrication primer, fabricate, prime.
- Prefabrication primer (option 3): Type recommended by manufacturer of post fabrication primer.
 - Thickness of post fabrication primer coat: May be reduced if and as recommended by manufacturer.
- Surfaces inaccessible after assembly: Apply full treatment and coating system including, if necessary, local application of site coatings.

725 MANUAL CLEANING OF NEW STEELWORK

- Preparation: Remove fins, burrs, sharp edges, weld spatter, loose rust and loose scale.
- Surface finish: Clean but unpolished to BS EN ISO 8501-1, grade St 2.
- Finishing: Thoroughly degrease and clean down. Remove any consequent rusting back to grade St 2. Prime without delay.

736 TREATMENT OF SITE WELDED JOINTS IN GALVANIZED STEELWORK

- Preparation: After welding, and without delay, remove scale and weld spatter from weld areas. Remove traces of rust. Wash with clean water and allow to dry.
- Coating: Reinstate using one of the methods given in BS EN ISO 1461, clause 6.3.

740 BOLTED JOINTS (OTHER THAN PRELOADED JOINTS)

- Steelwork to be shop painted: Apply full shop specification to joint faces.
- Steelwork to be erected with mill finish then site painted: Before erection, prepare and prime joint faces and allow to dry.
- Bolted joints in externally exposed steelwork:
 - Immediately before assembling, apply a further coat of primer and bring surfaces together while still wet.
 - After assembling and before applying site coatings, seal crevices to bolts and joint perimeters with a compatible sealant.

760 GALVANIZED FASTENERS

- Treatment: After steelwork erection and before applying site coatings, thoroughly degrease and clean. Etch prime.

770 SITE PREPARATION OF GALVANIZED SURFACES FOR PAINTING

- Preparation: Thoroughly degrease. Remove white corrosion products. Wash off and allow to dry before applying etching wash or primer.

PAINTING

810 ENVIRONMENTAL CONDITIONS

- General requirements prior to starting coating work:
 - Surfaces: Unaffected by moisture or frost.
 - Steel temperature: At least 3°C above dew point, with conditions stable or improving, and not high enough to cause blistering or wrinkling of the coating.
 - Relative humidity: Below 85%.

815 COATINGS

- Surfaces to be coated: Clean, dust free and suitably dry. Previous coats to be adequately cured.
- Multiple coats of same material: Use different tints to assist checking of complete coverage.
- Penultimate coat: Colour recommended by paint manufacturer to suit top coat colour.
- Finish required: Smooth and even, of uniform thickness and colour, free from defects.

820 FILM THICKNESS

- Wet film thickness: During application, check thickness of each coat with a wheel or comb gauge used in accordance with BS EN ISO 2808.
- Accumulated dry film thickness: After each coat has dried, check total accumulated film thickness.
 - Method: Magnetic or electromagnetic meter.
 - Number and position of measurements: As directed.
 - Validation: Measurements to be independently witnessed.
 - Meter calibration: Check against standard shims and recalibrate regularly against a smooth steel reference plate.
- Average dry film thickness:
 - At least specified thickness over any square metre.
 - No reading to be less than 75% of specified thickness.
- Top coat dry film thickness: Sufficient to give an even, solid, opaque appearance.

825 STRIPE COAT

- External angles, nuts, bolt heads, rough weld seams, and areas difficult to coat: Apply an additional stripe coat of primer .

850 JUNCTIONS WITH CONCRETE

- Exposed steelwork partially embedded or encased in concrete: Apply two coats of bituminous coating locally to the steel/concrete junction.
- Bituminous coating: To BS 6949, type 1, class A.

DRAFT

G20

Carpentry/ timber framing/ first fixing

G20 Carpentry/ timber framing/ first fixing

To be read with Preliminaries/ General conditions.

GENERAL

READ IN CONJUNCTION WITH ROLTON'S SPECIFICATION, FOR STRUCTURAL ITEMS ROLTON'S SPECIFICATION TO TAKE PRECEDENCE

105 TIMBER PROCUREMENT

- Timber (including timber for wood based products): Obtained from well managed forests/ plantations in accordance with:
 - The laws governing forest management in the producer country or countries.
 - International agreements such as the Convention on International Trade in Endangered Species of wild fauna and flora (CITES).
- Documentation: Provide either:
 - Documentary evidence (which has been or can be independently verified) regarding the provenance of all timber supplied, or
 - Evidence that suppliers have adopted and are implementing a formal environmental purchasing policy for timber and wood based products.

115 CONTRACTOR'S STRUCTURAL DESIGN

- Design responsibility: Design and detail trussed rafters to main roof including all associated items.
- Requirement:
 - Generally: As section B50/B51.
 - Modifications: None.
 - Design: Complete the design in accordance with the designated code of practice to satisfy specified performance criteria.
 - Service class: Trussed rafters Class 2, all other timber Class 1.
 - Additional requirements: All other steel components interacting with roof system.
- Design and production information:
 - Drawings: Showing Showing Showing general arrangements, locations of members and geometry of structure. To include (but not limited to) grades, strength classes and finished sizes.. .
 - Other requirements: Drawing showing line and point loads transmitted to foundations, full structural calculations for the stability of the structure and element design calculations. .
- Timing of submissions: As agreed with Main Contractor to tie in with programme.

150 STRENGTH GRADING OF TIMBER

- Grader: A company currently registered under a third party quality assurance scheme operated by a certification body approved by the UK Timber Grading Committee.

160 GRADING AND MARKING OF SOFTWOOD

- Timber of a target/ finished thickness less than 100 mm and not specified for wet exposure: Graded at an average moisture content not exceeding 20% with no reading being in excess of 24% and clearly marked as 'DRY' or 'KD' (kiln dried).
- Timber graded undried (green) and specified for installation at higher moisture contents: Clearly marked as 'WET' or 'GRN'.
- Structural timber members cut from large graded sections: Regraded to approval and marked accordingly.

PRODUCTS

- 210 STRUCTURAL SOFTWOOD (GRADED DIRECT TO STRENGTH CLASS) FOR TIMBER TRUSSES
- Grading standard: To BS 4978, BS EN 14081-1, or other national equivalent and so marked.
 - Strength class to BS EN 338: tTo Contractors Design.
 - Treatment:
 - Preservative treatment: Contractors recommendations in kline with NHBC.
 - Design service life: 30 years.
 - Fire retardant treatment: None required.
- 211 STRUCTURAL SOFTWOOD (GRADED DIRECT TO STRENGTH CLASS) FOR UPPER FLOORS
- Grading standard: To BS 4978, BS EN 14081-1, or other national equivalent and so marked.
 - Strength class to BS EN 338: tTo Contractors Design.
 - Treatment:
 - Preservative treatment: Contractors recommendations in kline with NHBC.
 - Design service life: 30 years.
 - Fire retardant treatment: None required.
- 270 UNGRADED SOFTWOODFOR INTERNAL NONSTRUCTURAL USE
- Quality of timber: Free from decay, insect attack (except pinhole borers) and with no knots wider than half the width of the section.
 - Surface finish: Regularized.
 - Treatment:
 - Preservative treatment: Organic solvent impregnation to NBS section Z12 and Wood Protection Association Commodity Specification C8.
 - Design service life: 30 years.
 - Fire retardant treatment: None required.
- 350 TRUSSED RAFTERS
- Design and fabrication standard: To BS EN 14250.
 - Manufacturer: A firm currently registered under a third party quality assurance scheme.
 - Truss system: Agrément certified.
 - Ancillary components to be supplied by the truss fabricator: All components including fixings and loose timbers.
 - Treatment: Organic solvent type as section Z12 and BWPDA Commodity Specification C8.

WORKMANSHIP GENERALLY

- 401 CROSS SECTION DIMENSIONS OF STRUCTURAL SOFTWOOD AND HARDWOOD
- Dimensions: Dimensions in this specification and shown on drawings are target sizes as defined in BS EN 336.
 - Tolerances: The tolerance indicators (T1) and (T2) specify the maximum permitted deviations from target sizes as stated in BS EN 336, clause 4.3:
 - Tolerance class 1 (T1) for sawn surfaces.
 - Tolerance class 2 (T2) for further processed surfaces.
- 402 CROSS SECTION DIMENSIONS OF NONSTRUCTURAL SOFTWOOD
- Dimensions: Dimensions in this specification and shown on drawings are finished sizes.
 - Maximum permitted deviations from finished sizes: As stated in BS EN 1313-1, clause 6 for sawn sections.

- 403 CROSS SECTION DIMENSIONS OF NONSTRUCTURAL HARDWOOD
- Dimensions: Dimensions in this specification and shown on drawings are finished sizes.
 - Maximum permitted deviations from finished sizes: As stated in BS EN 1313-2:
 - Clause 6 for sawn sections.
 - Clause NA.3 for further processed sections.
- 420 WARPING OF TIMBER
- Bow, spring, twist and cup: Not greater than the limits set down in BS 4978 or BS EN 14081-1 for softwood, or BS 5756 for hardwood.
- 430 SELECTION AND USE OF TIMBER
- Timber members damaged, crushed or split beyond the limits permitted by their grading: Do not use.
- 435 NOTCHES, HOLES AND JOINTS IN TIMBER
- Notches and holes:
 - General: Avoid if possible.
 - Sizes: Minimum needed to accommodate services.
 - Position: Do not locate near knots or other defects.
 - In same joist: Minimum 100 mm apart horizontally.
 - Notches in joists:
 - Position: Locate at top. Form by sawing down to a drilled hole.
 - Depth (maximum): 0.15 x joist depth.
 - Distance from supports: Between 0.1 and 0.2 x span.
 - Holes in joists:
 - Position: Locate on neutral axis.
 - Diameter (maximum): 0.25 x joist depth.
 - Centres (minimum): 3 x diameter of largest hole.
 - Distance from supports: Between 0.25 and 0.4 of span.
 - Notches in roof rafters, struts and truss members: Not permitted.
 - Holes in struts and columns: Locate on neutral axis.
 - Diameter (maximum): 0.25 x minimum width of member.
 - Centres (minimum): 3 x diameter of largest hole.
 - Distance from ends: Between 0.25 and 0.4 of span.
 - Scarf joints, finger joints and splice plates: Do not use without approval.
- 440 PROCESSING TREATED TIMBER
- Cutting and machining: Carry out as much as possible before treatment.
 - Extensively processed timber: Retreat timber sawn lengthways, thickened, planed, ploughed, etc.
 - Surfaces exposed by minor cutting/ drilling: Treat with two flood coats of a solution recommended by main treatment solution manufacturer.
- 450 MOISTURE CONTENT
- Moisture content of wood and wood based products at time of installation: Not more than:
 - Covered in generally unheated spaces: 24%.
 - Covered in generally heated spaces: 20%.

451 MOISTURE CONTENT TESTING

- Procedure: When instructed, test timber sections with an approved electrical moisture meter.
- Test sample: Test 5% but not less than 10 lengths of each cross-section in the centre of the length.
- Test results: 90% of values obtained to be within the specified range. Provide records of all tests.

480 TRUSSED RAFTER TESTING

- Standard: To BS EN 14250.
- Number/ timing of tests: Five at random during the contract.
- Purpose of testing: General quality control.
- Test reports: Submit as soon as available.

510 PROTECTION

- Generally: Keep timber dry and do not overstress, distort or disfigure sections or components during transit, storage, lifting, erection or fixing.
- Timber and components: Store under cover, clear of the ground and with good ventilation. Support on regularly spaced, level bearers on a dry, firm base. Open pile to ensure free movement of air through the stack.
- Trussed rafters: Keep vertical during handling and storage.

520 EXPOSED END GRAIN PROTECTION

- Components: Seal exposed end grain of the following before delivery to site:
As required.
- Sealer: Clear end grain sealer.

530 PAINTED FINISHES

- Structural timber to be painted: Primed as specified before delivery to site.

540 CLEAR FINISHES

- Structural timber to be clear finished: Keep clean and apply first coat of specified finish before delivery to site.

550 EXPOSED TIMBER

- Planed structural timber exposed to view in completed work: Prevent damage to and marking of surfaces and arrises.

JOINTING TIMBER

590 METAL PLATE FASTENERS/GUSSETS

- Manufacturer: Contractor's choice.
 - Product reference: Contractor's choice.

630 BOLTED JOINTS

- Bolt spacings (minimum): To BS EN 1995-1-1, section 8.5.
- Holes for bolts: Located accurately and drilled to diameters as close as practical to the nominal bolt diameter and not more than 2 mm larger.
- Washers: Placed under bolt heads and nuts that would otherwise bear directly on timber. Use spring washers in locations which will be hidden or inaccessible in the completed building.
- Bolt tightening: So that washers just bite the surface of the timber. Ensure that at least one complete thread protrudes from the nut.
 - Checking: At agreed regular intervals up to Completion. Tighten as necessary.

670 ANTI-CORROSION FINISHES FOR FASTENERS

- Galvanizing: To BS 7371-6, with internal threads tapped and lightly oiled following treatment.
- Sherardizing: To BS 7371-8, Class 1.
- Zinc plating: To BS EN ISO 4042 and passivated.

ERECTION AND INSTALLATION

710 PROPOSALS FOR ERECTING STRUCTURAL TIMBER

- Proposals: Submit details of:
 - Method and sequence of erection.
 - Type of craneage.
 - Temporary guys and bracing proposed for use during erection.
- Latest date for submission: 10 days before erection.

740 PRE-ERECTION CHECKING

- Timing: Not less than 10 days before proposed erection start date.
- Checklist:
 - Foundations and other structures to which timber structure will be attached: Check for accuracy of setting out.
 - Holding down bolts: Check for position, protruding length, condition and slackness.
- Inaccuracies and defects: Report without delay.
- Erection: Obtain permission to commence.

750 MODIFICATIONS/REPAIRS

- Defects due to detailing or fabrication errors: Report without delay.
- Methods of rectification: Obtain approval of proposals before starting modification or remedial work.
- Defective/damaged components: Timber members/ components may be rejected if the nature and/or number of defects would result in an excessive amount of site repair.

760 TEMPORARY BRACING

- Provision: As necessary to maintain structural timber components in position and to ensure complete stability during construction.

770 ADDITIONAL SUPPORTS

- Provision: Position and fix additional studs, noggings and/ or battens to support edges of sheets materials, and wall/ floor/ ceiling mounted appliances, fixtures, etc. shown on drawings
- Material properties: Additional studs, noggings and battens to be of adequate size and have the same treatment, if any, as adjacent timber supports.

775 BEARINGS

- Timber surfaces which are to transmit loads: Finished to ensure close contact over the whole of the designed bearing area.
- Packings: Where provided, to cover the whole of the designed bearing area.
 - Crushing strength: Not less than timber being supported.
- In external or inaccessible locations: Rot and corrosion proof.

780 WALL PLATES

- Position and alignment: To give the correct span and level for trusses, joists, etc.
- Bedding: Fully in fresh mortar.
- Joints: At corners and elsewhere where joints are unavoidable use nailed half lap joints. Do not use short lengths of timber.

784 JOISTS GENERALLY

- Centres: Equal, and not exceeding designed spacing.
- Bowed joists: Installed with positive camber.
- End joists: Positioned approximately 50 mm from masonry walls.

786 JOISTS ON HANGERS

- Hangers: Bedded directly on and hard against supporting construction. Do not use packs or bed on mortar.
- Joists: Cut to leave not more than 6 mm gap between ends of joists and back of hanger. Rebated to lie flush with underside of hangers.
- Fixing to hangers: A nail in every hole.

795 TRIMMING OPENINGS

- Trimmers and trimming joists: When not specified otherwise, not less than 25 mm wider than general joists.

800 TRUSSED RAFTER INSTALLATION

- Erection: To Trussed Rafter Association (TRA) Technical handbook. Site installation guide and TRA Product data sheet PD3.
- Trusses generally: Do not modify without approval.
- Damaged trusses: Do not use.
- Fixing: With truss clips. Bottom chords of standard trusses and rafters of raised tie trusses bearing fully on wall plates.
 - Bottom chords of standard trusses: Do not fix to internal walls until roofing is complete and cisterns are installed and filled.
- Tolerances: To NA to BS EN 1995-1-1.

805 TRUSS CLIPS

- Manufacturer: Contractor's choice.
 - Product reference: Submit proposals.
- Material/ finish: Galvanized steel.
- Fasteners: 32 x 3.5 mm galvanized or sherardized square twisted nails in every hole.

810 PERMANENT BRACING OF TRUSSED RAFTERS

- Bracing and binders:
 - Size: minimum 25x100mm and free from major strength reducing defects and distortion. .
 - Method of fixing: To every rafter, strut or tie with not less than two fasteners. Fasteners: 75 x 3.35 mm galvanized round wire nails.
- Lap joints: Extended over and nailed to at least two truss members.

840 STRUTTING TO FLOOR JOISTS

- Type: One of the following:
 - Herringbone strutting: At least 38 x 38 mm softwood.
 - Solid strutting: At least 38 mm thick softwood and at least three quarters of joist depth.
 - Proprietary metal strutting: of approved type in accordance with NHBC.
- Fixing: Between joists as follows:
 - Joist spans of 2.5 to 4.5 m: One row at centre span.
 - Joist spans over 4.5 m: Two rows equally spaced.
 - Strutting must not project beyond top and bottom edges of joists.
- Outer joists: Blocked solidly to perimeter walls.

850 INSPECTION GENERALLY

- Structural timber-work: Give reasonable notice before covering up.

860 BOLTED JOINTS INSPECTION

- Timing: Inspect all accessible bolts at the end of the Defects Liability Period and tighten if necessary.

910 EAVES SOFFIT VENTILATION

- Soffit boards: Fixed to leave a continuous ventilation opening not less than 10 mm wide for full length of eaves.
-

DRAFT

H
Cladding/Covering

DRAFT

H11
Curtain walling

H11 Curtain walling

To be read with Preliminaries/ General conditions.

TENDERING

10 INFORMATION TO BE PROVIDED WITH TENDER

- Submit the following curtain walling particulars:
 - Evidence of compliance: All curtain walling to be CE marked to BS EN 13830:2003. Submit Declaration of Performance (DoP). CE marking and Declaration of Performance (DoP) requirements for structural sealant glazing to be submitted on the basis of a manufacturer's European Technical assessment (ETA) prepared in accordance with EOTA technical guide ETAG 002 (Parts 1, 2 and 3).
 - Typical plan, section and elevation drawings at suitable scales.
 - Typical detailed drawings at large scales, including AS SCHEDULE.
 - Technical information and certification demonstrating compliance with specification of proposed incorporated products and finishes, including TBC.
 - Certification, reports and calculations demonstrating compliance with specification of proposed curtain walling.
 - Proposals for connections to and support from the building structure and building components.
 - Proposals for amendments to primary supporting structure and for secondary supporting structure additional to that shown on preliminary design drawings.
 - Schedule of builder's work, special provisions and special attendance by others.
 - Examples of standard documentation from which project quality plan will be prepared.
 - Preliminary fabrication and installation method statements and programme.
 - Schedule of products and finishes with a design life expectancy less than that specified in clause 440, with proposals for frequencies and methods of replacement.
 - Proposals for replacing damaged or failed products.
 - Areas of non-compliance with the specification.

TYPES OF CURTAIN WALLING

110A CURTAIN WALLINGCAPPED ALUMINIUM - TO STAIRCASES

Refer to DLA Drawing 1132-50, Window Types C, H and M

- Supporting structure: Cavity brick construction.
- Curtain walling system:
 - Manufacturer: Manufacturer: AluK (GB) Ltd.
 - Address: Newhouse Farm Ind Est, Chepstow, NP16 6UD
 - Telephone: 01291 639 739
 - Web: www.aluk.com
 - Contact: Nigel Gilbert -AluK Project Consultant (London & SE) m: 07976 180 333
- nigel.gilbert@aluk.com .
- Product reference: SL52 Capped Curtain Wall System .
- Type: 52mm wide stick system, pressure equalised, drained and ventilated. .
- Internal framing member:
 - Material: Aluminium - Generally P-52-402 mullion cover cap and P-52-400 transom cover cap. .
 - Finish: Polyester powder coating to BS EN 12206-1:2004 .
 - Colour/ texture: AluK standard colour range, RAL7016 - Matt finish. .
 - Minimum film thickness: A minimum 60 micrometres .
- External cover cap:
 - Material: Aluminium - Generally P-52-402 mullion cover cap and P-52-400 transom cover cap. .
 - Finish: Polyester powder coating to BS EN 12206-1:2004. .
 - Colour/ texture: AluK standard colour range, RAL7016 - Matt finish. .
 - Minimum film thickness: A minimum 60 micrometres. .
- Glazing: Hermetically sealed 28mm insulating glass units to achieve a maximum 1.0W/m²k centre pane. Glazing thermal safety checks are recommended to ensure project suitability, these are to be carried out by the glazing manufacturer. .
- Inner pane: 6mm Clear Toughened / 6,4mm Laminated (SBD) - thickness TBC. .
- Outer pane: 6mm Clear Toughened - thickness TBC. .
- Glazing system: Gaskets, cover plate fixed .
- Accessories: All aluminium flashings, cills and other perimeter trims necessary as indicated within the drawings. It is the subcontractor's responsibility to ensure that these are designed and installed in accordance with AluK's instructions and guidelines. .
- Incorporated components: Opening lights as section L10/330, doors as section L10. AOV note to be added .
- Other requirements: EPDM to be provided to allow system to be sealed to VCL for weather and air tightness. Max. air permeability 5m³/m².h The AluK SL52 curtain wall system shall be fabricated and installed by AluK authorised dealers only in complete accordance with the information and guidance released by AluK (GB) Ltd. .

150 MATERIALS SPECIFICATION

- Minimum BRE 'Green Guide to Specification' online rating: C.
- Specification of glazing materials to be in accordance with guidance in CIRIA document 'Guidance on glazing at height'.

GENERAL REQUIREMENTS/ PREPARATORY WORK

210 DESIGN

- Curtain walling and associated features: Complete the detailed design. Submit before commencement of fabrication.
- Related works: Coordinate in the detailed design.

215 DESIGN PROPOSALS

- Submission of alternative proposals: Preliminary design drawings indicate intent. Other reasonable proposals will be considered.

220 SPECIFICATION

- Compliance standards: BS EN 13830:2003 and The Centre for Window and Cladding Technology (CWCT) 'Standard for systemised building envelopes'.
- Reference information: For the duration of the contract, keep available at the design office, workshop and on site copies of:
 - The CWCT 'Standard for systemised building envelopes'.
 - Publications invoked by the CWCT 'Standard for systemised building envelopes'.

230 INFORMATION TO BE PROVIDED DURING DETAILED DESIGN STAGE

- Submit the following curtain walling particulars:
 - A schedule of detailed drawings and dates for submission for comment.
 - A schedule of loads that will be transmitted from the curtain walling to the structure.
 - Proposed fixing anchor details relevant to structural design and construction.
 - A detailed testing programme in compliance with the Main Contract master programme.
 - A detailed fabrication and installation programme in compliance with the Main Contract master programme.
 - Proposals to support outstanding applications for Building Regulation consents or relaxations.

232 QUALITY PLAN

- Requirement: Submit during detailed design.
- Content: In accordance with BS EN ISO 9001 and including the following:
 - Name of the quality manager.
 - Quality assessment procedures.
 - Inspection procedures to be adopted in checking the work.
 - Stages at which check lists will be used and samples of the lists.
 - List of work procedures on the correct use of materials or components, both off site and on site.
 - List of product information with latest revisions.
 - Subcontractors involved in the work.
 - Subcontractors' quality plans.
 - Storage, handling, transport and protection procedures.
 - Procedure for registering and reporting non compliances.
 - Maintenance procedures and calibration records.
 - Certification that completed work complies with specification.
 - Check list register to ensure all items have been inspected and non compliances discharged.

235 INFORMATION TO BE PROVIDED BEFORE COMMENCEMENT TESTING OR FABRICATION OF CURTAIN WALLING

- Submit the following curtain walling particulars:
 - Detailed drawings to fully describe fabrication and installation.
 - Detailed calculations to prove compliance with design/ performance requirements.
 - Project specific fabrication, handling and installation method statements.
 - Certification for incorporated components manufactured by others confirming their suitability for proposed locations in the curtain walling.
 - Recommendations for spare parts for future repairs or replacements.
- Recommendations for safe dismantling and recycling or disposal of products.

235A INFORMATION TO BE PROVIDED BEFORE COMMENCEMENT OF TESTING OR FABRICATION OF STRUCTURAL SEALANT CURTAIN WALLING

- Submit AluK's structural bonding sealant application for project specific approval for:
 - Compatibility and adhesion of all products and finishes.
 - Full details of structural sealant glazing design
 - Structural sealant dimensioning
 - A project specific sealant application method statement.-

DESIGN/ PERFORMANCE REQUIREMENTS

305 CWCT 'STANDARD FOR SYSTEMISED BUILDING ENVELOPES'

- General: Unless specified or agreed otherwise comply with:
 - Part 2 - Loads, fixings and movement.
 - Part 3 - Air, water and wind resistance.
 - Part 4 - Operable components, additional elements and means of access.
 - Part 5 - Thermal, moisture and acoustic performance.
 - Part 6 - Fire performance
 - Part 7 - Robustness, durability, tolerances and workmanship.
- Project performance requirements specified in this subsection: Read in conjunction with CWCT performance criteria.

313A INTEGRITY

- Requirement: The curtain walling must resist wind loads, dead loads and design live loads, and accommodate deflections and movements without damage.
- Design wind pressure: Calculate in accordance with: BS 6399-2 (the specialist contractor shall provide calculations to demonstrate this if requested by the contract administrator).
Hard body impact loads: To CWCT TN75:
 - Location and category: In accordance with CWCT requirements.
 Soft body impact loads - curtain walling to BS EN 14019:
 - Location and classification: In accordance with CWCT requirements.
 Soft body impact loads - glass to BS EN 12600:
 - Location and classification: In accordance with CWCT requirements.
- Permanent imposed loads: None .
- Temporary imposed loads: CDM requirements for maintenance, occupant impact and temporary construction loads .

320 DEFLECTION UNDER DEAD LOADS

- Requirement: Framing members parallel to the curtain walling plane must not:
 - Reduce glass bite to less than 75% of design dimension.
 - Reduce edge clearance to less than 3 mm between members and immediately adjacent glazing units, panel/ facing units or other fixed units.
 - Reduce clearance to less than 2 mm between members and movable components such as doors and windows.

330 GENERAL MOVEMENT

- Requirement: Curtain walling must accommodate anticipated building movements as follows: To be confirmed within the structural engineers report. .

340 AIR PERMEABILITY

- Requirement: Permissible air leakage rates of 1.5m³/hr/m² for fixed lights and 2.0 m³/hr/lin.m for opening lights must not be exceeded when the curtain walling is subjected to the peak test pressure.
- Permeability class to BS EN 12152: A4.
Peak test pressure: 600 Pa.

345 AIR PERMEABILITY EXFILTRATION

- Requirement: The maximum permissible air exfiltration rate through the curtain walling system must not exceed: 5.0m³/ h.m² at a test pressure of 50 Pa. .

350 WATER PENETRATION

- Watertightness class to BS EN 12154: R7.
 - Peak test pressure: 600 Pa.
- Additional requirements: Underside of any transom not to be wetted at peak test pressure.

370 THERMAL PROPERTIES

- Method of calculating the thermal transmittance (U-value) of curtain walling/ each zone of curtain walling: Weighted U-value.
- Average U-value of curtain walling: 1.6 W/m²k.
- Curtain wall zone interfaces: Co-ordinate to achieve required average U-value.
- Method for assessing thermal transmittance (U-value) of assemblies: By calculation.

380 SOLAR AND LIGHT CONTROL

- Total solar energy transmission:
 - Maximum g-value - glazing only: 0.4.
 - Maximum effective g-value - glazing with shading devices: TBC.
- Visible light transmission:
 - Minimum light transmission - glazing only: TBC.
 - Minimum effective light transmission - glazing with shading devices: TBC.

385 THERMAL STRESS IN GLAZING

- Glass panes/ units: Must have adequate resistance to thermal stress generated by orientation, shading, solar control and construction.

390 AVOIDANCE OF CONDENSATION

- Requirement: Notional psychrometric conditions under which condensation must not form on building interior surfaces of framing members or any part of infill panels/ facings are:
 - Notional outdoor psychrometric conditions as BS 6229, table A1.
 - Notional indoor psychrometric conditions:
 - Temperature: 20°C.
 - Relative humidity: 50%.
 - Vapour pressure: 1.17 kPa.

410 SOUND TRANSMITTANCE

- Minimum weighted sound reduction index (R_w) to BS EN ISO 717-1:
 - Between internal and external surfaces of curtain walling: TBC.
- Minimum weighted standardized level difference (D_{nTw}) to BS EN ISO 717-1:
 - Between adjacent floors abutting curtain walling: TBC.
 - Between adjacent rooms on same floor abutting curtain walling: TBC.

430 FIRE STOPPING

- Locations: At junctions of curtain walling with compartment walls and floors.
- Materials and methods of fixing: To ensure fire resistance not less than that specified for compartment walls and floors.

- 435 OPENING LIGHTS (WINDOWS)
- Performance criteria: To CWCT 'Standard for systemised building envelopes' Part 3.
 - Security:
 - Applicable opening lights: As schedule.
 - Security rating: To PAS 24.
 - Opening lights restrictive catches to CWCT 'Standard for systemised building envelopes' clause 4.2.5: To all opening lights in windows to floors above ground level.
 - Ventilation requirement: tbc.
 - Windows to be cleaned from inside of the building: As schedule.
 - Fasteners: Concealed multipoint, operated by an internal handle.
 - Integral locks: Suited to pass same key.

- 445 LIGHTNING PROTECTION SYSTEM
- Curtain wall components used as part of lightning protection system: None.

- 450 SAFETY
- Finished surfaces of curtain walling: Accessible internal and external areas must not:
 - Have irregularities capable of inflicting personal injury.
 - Release irritant or staining substances.

BREEAM PERFORMANCE REQUIREMENTS

- 480 VIEW OUT
- Glazed areas/ opening sizes and position: Designed to meet BREEAM 'View out' criteria for relevant building type.
 - Submit design plan and elevation drawings showing the following:
 - All BREEAM defined 'relevant areas' dependant on building type and room depths.
 - Actual or notional workstation/ desk layouts.
 - Glazed areas/ open areas.
 - Submit site plan showing: Building location and proximity to external obstructions.

- 485 POTENTIAL FOR NATURAL VENTILATION
- Submit design plan and elevation drawings, and calculations confirming the following: Locations of openings.

TESTING

- 510 COMPARISON (TYPE) TESTING
- Requirement: To CWCT 'Standard for systemised building envelopes', Part 8.
 - Test results and reports: Before commencement of curtain walling fabrication and installation, submit proof of compliance with this specification.
- 520 PROJECT TESTING (SITE)
- Test results and reports: Before installation of general areas of curtain walling, submit proof of compliance with this specification.
- 635 SITE HOSE TEST
- Requirement: To CWCT 'Standard for systemised building envelopes' Section 8.16.2, 'Standard test methods for building envelopes' Section 9.
 - Joints to be tested: tbc.

PRODUCTS

- 710 ALUMINIUM ALLOY FRAMING SECTIONS
- Standard: To relevant parts of BS EN 515, BS EN 573, BS EN 755 and BS EN 12020.
 - Alloy, temper and thickness: Suitable for the application and specified finish.
 - Structural members: To BS EN 1999-1-1.
- 712 ALUMINIUM ALLOY SHEET
- Standards: To relevant parts of BS EN 485, BS EN 515 and BS EN 573.
 - Alloy, temper and thickness: Suitable for the application and specified finish.
- 715 CARBON STEEL FRAMING SECTIONS/ REINFORCEMENT
- Standards: To relevant parts of BS 7668, BS EN 10029, and BS EN 10210.
 - Thickness: Suitable for the application, and for galvanizing or other protective coating.
- 717 CARBON STEEL SHEET
- Standards: To relevant parts of BS 1449-1, BS EN 10048, BS EN 10051, BS EN 10111, BS EN 10131, BS EN 10132, BS EN 10139, BS EN 10140, BS EN 10149, BS EN 10209 and BS EN 10268.
 - Grade and thickness: Suitable for the application, and for galvanizing or other protective coating.
- 720 STAINLESS STEEL SHEET
- Standards: To relevant parts of BS EN 10029, BS EN 10048, BS EN 10051, BS EN 10095 and BS EN ISO 9445.
 - Grade: To BS EN 10088-2, austenitic 1.4301 (304) generally, 1.4401 (316) when used externally or in severely corrosive environments.
 - Thickness: Suitable for the application.
- 730 MECHANICAL FIXINGS
- Stainless steel: To BS EN ISO 3506, grade A2 generally, grade A4 when used in severely corrosive environments.
 - Carbon steel: To BS 4190 and suitable for galvanizing or other protective coating.
 - Aluminium brackets, rivets and shear pins: To relevant parts of BS EN 755.
- 732 ADHESIVES
- General: Not degradable by moisture or water vapour.
- 735 FIXING ANCHORS
- Type and use: Reviewed and approved by fixing manufacturers. Submit confirmatory information on request.
 - Dimensions: Not less than recommended by their manufacturers.
 - Adjustment capability: Sufficient in three dimensions to accommodate building structure and curtain walling fabrication/ installation tolerances.

737 GLASS GENERALLY

- Standards: To BS 952 and relevant parts of:
 - BS EN 572 for basic soda lime silicate glass.
 - BS EN 1096 for coated glass.
 - BS EN 1748 for borosilicate glass.
 - BS EN 1863 for heat strengthened soda lime silicate glass.
 - BS EN 12150 for thermally toughened soda lime silicate glass.
 - BS EN 13024 for thermally toughened borosilicate glass.
 - BS EN ISO 12543 for laminated glass.
- Selection of glass type and thickness in accordance with recommendations of CIRIA publication 'Guidance on glazing at height'.
- Glass quality: Clean and free from obvious scratches, bubbles, cracks, ripples, dimples and other defects.
- Glass edges: Generally undamaged. Shells and chips not more than 2 mm deep and extending not more than 5 mm across the surface are acceptable if ground out.

741 DISTORTIONAL TOLERANCES ON GLASS ALL GLAZED AREAS

- Measurement of tolerances: After any thermal toughening/ heat strengthening.
- Maximum bow: 0.2% of pane dimension.
- Maximum roller wave:
 - For 3 to 5 mm thick glass: 0.5 mm.
 - For 6 to 10 mm thick glass: 0.3 mm.
 - For 12 mm and thicker glass: 0.15 mm.
- Maximum edge dip:
 - For 3 to 5 mm thick glass: 0.8 mm.
 - For 6 to 10 mm thick glass: 0.5 mm.
 - For 12 mm and thicker glass: 0.25 mm.

742 HEAT SOAKED THERMALLY TOUGHENED GLASS

- Standard: To BS EN 14179.
 - Holding period: 2 hours.
- Locations of heat soaked glass: As schedule.

745 INSULATING GLASS UNITS

- Standard and labels for hermetically sealed units: To BS EN 1279.
- Label: Each pane.
- Colour of aluminium perimeter spacers: Black.
- Perimeter taping: Not to be used.
- Perimeter seals:
 - Resistant to UV light degradation on exposed edges.
 - Compatible with structural, assembly and weather sealants.

750 INFILL PANELS/ FACINGS

- Tolerances:
 - Deviation in size (maximum): ± 1 mm.
 - Deviation in flatness from plane per 2 m length (maximum): ± 1 mm.
- Rigidity: Adequate to comply with design/ performance requirements.

760 GASKETS

- Material:
 - Noncellular rubber to BS 4255-1.
 - Cellular rubber to ASTM-C509-06.
- Continuity: Outer gaskets of single front sealed curtain walling systems and inner gaskets of drained and ventilated or pressure equalized curtain walling systems must be formed in a complete frame with sealed joints. Vulcanized rubber gaskets must have factory moulded corner joints.
- Durability: Resistant to oxidation, ozone and UV degradation.

765 WEATHERSTRIPPING OF OPENING UNITS

- Material:
 - Noncellular rubber to BS 4255-1.
 - Cellular rubber to ASTM-C509-06.
 - Polypropylene woven pile, silicone treated.
- Attachment: Fixed in undercut grooves in framing sections using preformed corners, with any joints in the length.

770 GENERAL SEALANTS

- Selection: In accordance with BS 6213 from:
 - Silicone.
 - One part polysulfide.
 - Two part polysulfide.
 - One or two part polyurethane.
- Classification and requirements: To BS EN ISO 11600.
- Reaction to contact products and finishes: Stable and compatible.

772 CURTAIN WALLING JOINT ASSEMBLY SEALANTS

- Material: One part, low modulus silicone to BS EN ISO 11600, type F or G. Neutral curing where in contact with or close proximity to other products that may be adversely affected by acetoxy curing.
- Manufacturer: As recommended by AluK (GB) Ltd..
 - Product reference: As recommended by AluK (GB) Ltd.

785 VAPOUR CONTROL LAYER

- Acceptable materials:
 - Aluminium alloy.
 - Carbon steel, galvanized or protective coated.
 - Stainless steel.
 - Reinforced membranes: Foil, plastics or rubbers, protected both sides by rigid facings/linings.
- Location: Warm side of thermal insulation.
- Integrity: Continuous, free from gaps and sealed at joints.

FINISHES

810 PROTECTIVE COATING OF CARBON STEEL FRAMING SECTIONS/ REINFORCEMENT

- Treatment: One of the following to all surfaces:
 - Hot dip galvanized to BS EN ISO 1461.
 - An appropriate equivalent coating to BS EN ISO 12944-5 or BS EN ISO 14713-1, -2 and -3.

820 PROTECTIVE COATING OF CARBON STEEL MECHANICAL FIXINGS

- Treatment: One of the following to all surfaces:
 - Hot dip galvanized to BS EN ISO 1461.
 - Sherardized to BS 7371-8, class 30 coating thickness and passivated.
 - Zinc plated to BS EN ISO 2081, coating designation Fe//Zn//C for an iridescent (yellow passivate) chromate conversion coating or Fe//Zn//D for an opaque (olive green) chromate conversion coating.

830 POWDER COATING

- Requirement: As section Z31.

840 ANODIZING

- Requirement: As section Z33.

FABRICATION AND INSTALLATION

910 GENERALLY

- Electrolytic corrosion: Prevent. Submit proposed methods.
- Fixings: Concealed unless indicated on detailed drawings. Where exposed they must match material and finish of the products fixed.
- Fabrication: Machine cut and drill products in the workshop wherever possible.
- Identification of products: Mark or tag to facilitate identification during assembly, handling, storage and installation. Do not mark surfaces visible in the completed installation.

912 METALWORK

- Requirement: As section Z11, unless specified otherwise in this section.

915 GLAZING

- Requirement: As section L40, unless specified otherwise in this section.
- Directional patterned/ wired glass: Generally fix parallel to surround and align adjacent panes where seen together at close quarters.

917 FIXINGS/ ADHESIVES APPLICATION

- Requirement: As section Z20, unless specified otherwise in this section.

920 SEALANT APPLICATION

- Requirement: As section Z22, unless specified otherwise in this section.

930 ASSEMBLY

- General: Carry out as much assembly as possible in the workshop.
- Joints (other than movement joints): Rigidly secured, reinforced where necessary and fixed with hairline abutments.
- Displacement of components in assembled units: Submit proposals for reassembly on site.

935 OPENABLE WINDOWS IN NATURALLY VENTILATED BUILDINGS

- Location: Over 10 m from sources of external pollution.

955 FIXING ANCHOR INSTALLATION

- Site drilling or cutting into structure: Submit proposals for positions other than shown on detailed drawings.
- Concrete supporting structure:
 - Cast-in inserts: Provide detailed locational information. Protect cavities in inserts from entry of concrete.
 - Edge fixing distances: Not less than recommended by fixing anchor manufacturers.
- Corrective fabrication: Minimize. Where necessary, submit proposals.

970 CURTAIN WALLING INSTALLATION

- Securing to fixing anchors: Through holes formed during fabrication only.
- Tightening mechanical fasteners: To manufacturer's recommended torque figures. Do not overtighten fasteners intended to permit differential movement.
- Protective coverings: Remove only where necessary to facilitate installation and from surfaces that will be inaccessible on completion.

975 WELDING

- In situ welding: Not permitted..

980 INTERFACES

- Flashings, closers, etc: Locate and form correctly to provide weathertight junctions with the curtain walling.

982 IRONMONGERY

- Assembly and fixing: Accurately, using fasteners with matching finish supplied by ironmongery manufacturer.
- Completion: Check, adjust and lubricate as necessary to ensure correct functioning.

985A MAINTENANCE

- Maintenance manual: Incorporate details within the Building Manual in accordance with CWCT 'Standard for systemised building envelopes' clause 7.6.1.
 - Materials certification and test reports to be included:
 - Contact details for subcontractors and suppliers.
 - Design criteria for the curtain wall system.
 - Product information for components and materials including AluK (GB) Ltd's literature, COSHH data sheets and recommendations for cleaning maintenance and repair.
 - Copies of material, components and finishes certification and test reports.
 - A full set of construction drawings, updated to include any changes made up to the time of completion.
 - Method statement for means of access for maintenance and for use of any permanent equipment.
 - Method statement covering the procedures for replacement of any parts that have a design life less than the design life of the curtain wall system.
 - Recommendations for routine maintenance and cleaning, including suitable cleaning agents and lubrication/ adjustments to working parts.

DRAFT

H41

Glass fibre reinforced plastics cladding/ features

H41 Glass fibre reinforced plastics cladding/ features

TYPES OF CLADDING/ FEATURES

- 130A GRP PURPOSE MADE FEATURES FLAT PORCH CANOPIES TO HOUSES
All to specialist design - For Location and types refer to DLA House Type Overview Sheets
- Primary support structure: Masonry cavity wall
 - GRP components:
 - Manufacturer: Stormking Plastics Ltd Amington Point Amington Industrial Estate, Tamworth B77 4ED
Telephone: 01827 311100
contact@stormking.co.uk
Contact Name: Will Charteris
 - Product
Flat Porch: Deep Fascia flat top internal fix contemporary style GRP door canopy with reproductive lead roof, flashing/fixing upstand, all internal framework, internal mild steel fixing brackets, fascia and soffit, including all fixing frames, brackets, fixing packs and colour coded caps to complete full canopy installation.
 - Code 4 lead flashing where required.
- 130B GRP PURPOSE MADE FEATURES DORMERS TO APARTMENT BLOCKS A, B & C
All to specialist design - For Location and types refer to DLA Elevations
- Primary support structure: Slate clad Trussed Rafter Roofing
 - GRP components:
 - Manufacturer: Stormking Plastics Ltd Amington Point Amington Industrial Estate, Tamworth B77 4ED
Telephone: 01827 311100
contact@stormking.co.uk
Contact Name: Will Charteris
 - Product: Full GRP dormer roof with Apex trussed ready to tile roof, dormer side cheeks, side gutter flashing and all internal framework and insulation including Roof trusses to be supplied for tile and battening on site.
Code 4 leaf flashing to suit dormer window roof, factory fitted code 4 apron flashing bonded to softwood bearer and plywood flashing support tray.

DESIGN/ PERFORMANCE REQUIREMENTS

- 205 CONTRACTOR'S DESIGN OF CANOPIES AND DORMERS
- Design responsibility: Determine sizes and thickness of units and type size, number and spacing of fixings and any further information to complete the full design.
 - Structural and fire requirements:
 - Generally: As section B50.
 - Modifications: None.
 - Design: Complete the design in accordance with the designated code of practice to satisfy specified performance criteria.
 - Functional requirements: Not applicable.
 - Additional requirements: None.
 - Design and production information: As Preliminaries section A31 or as Agreed.
 - Timing of submissions: As Preliminaries section A31 or as Agreed.
- 215 THERMAL PERFORMANCE/ BRIDGING
- Requirement: Complete thermal design to avoid excessive thermal bridging.
Assessment: To BRE Information Paper 1/06.

220 WEATHER RESISTANCE

- Requirement: Weathertight, with full allowance made for deflections and other movements.

240 HORIZONTAL DEFLECTION OF VERTICAL PANELS

- Maximum deflection under wind loads: 10 mm.

250 VERTICAL DEFLECTION OF HORIZONTAL PANELS

- Maximum deflection under wind, snow and imposed loads: 10 mm.

260 FIRE RATING CERTIFICATION

- Verification: Before commencing full manufacture, submit evidence based on laboratory testing.
- Testing authority: UKAS accredited.

270 COLOUR FASTNESS/ APPEARANCE OF GRP

- Colour fastness of pigments: Not less than standard 6 when measured to BS 1006:B01C:LFS6.
Evidence of compliance: Submit.

275 COLOUR FASTNESS/ APPEARANCE SAMPLES

- Weathered samples: If available, submit naturally weathered samples, otherwise submit artificially weathered samples.
- Naturally weathered samples:
 - Pigments and resins: As proposed GRP.
 - Age: Not less than two years.
 - Action: Submit with new unweathered control samples.
- Artificially weathered samples:
 - Pigments, resins and gel coat: As proposed GRP.
 - Test method: Accelerated weatherometer subjecting samples to moisture and ultraviolet light.
 - Duration: Not less than 1500 hours.
 - Action: Submit with new unweathered control samples.

285 COLOUR FASTNESS/ APPEARANCE OF EXISTING GRP

- Existing GRP units:
 - Type: Similar to proposed.
 - Manufacturer: As proposed GRP.
 - Pigments and resins: As proposed GRP.
 - Age: Not less than two years.
 - Action: Submit details of location and history.
- Purpose: To demonstrate typical weathering characteristics.

320 DESIGN SAMPLES

- GRP samples: Before general manufacture obtain approval of appearance of fully tested compliant design samples of: Each component type.
 - Extent: Showing proposed colour, texture and incorporating a completed section of a joint.
 - Action: Obtain approval of appearance before proceeding. Retain as production control sample.

MANUFACTURE

410 QUALITY OF WORK

- Manufacture: Compliant with design and performance requirements.
 - Materials: Appropriate and compatible.
 - Workmanship: Appropriate and in accordance with manufacturers' recommendations.
- Resins: Used as supplied and not adulterated.
- Fillers and admixtures: Submit proposals.
- Standard of finish: Appropriate to end use and position in building.
 - Prohibited blemishes: Including, but not limited to, wrinkling, spotting, striations, fibre patterning, fish eyes, blisters, crazing, cracking, dry patches and uneven or inconsistent colour.

440 MANUFACTURING ACCURACY

- Finished dimensions of completed units when erected:
 - Ambient temperature: Measurements taken at 16-18°C.
 - Maximum permissible deviations:

Overall dimension involved (m)	Up to 2	2-3	3-4.5	4.5-6
Width and height:		+0		+0
+0		+0		
	-2 mm	-3 mm	-5 mm	-6 mm
Straightness of edges: deviation from intended line, any variation to be evenly distributed with no sudden bends or irregularities.	6 mm	3 mm		4 mm 5 mm
Squareness: taking the longer of 2 sides at any corner as a base line, the deviation of shorter side from perpendicular; dimension involved is the shorter side.	6 mm	3 mm		4 mm 5 mm
Twist: deviation of any corner from the plane containing the other 3 corners; dimension involved is the shorter side.	8 mm	3 mm		5 mm 7 mm
Flatness - deviation under a 1 m straight edge placed anywhere on a flat panel surface:				
3 mm				

450 GEL COATS TO EXTERNAL SURFACES

- Mixing: Resins, pigments and fillers thoroughly mixed.
- Batching: All units of one colour from single batch of resin/ additive mix.
- Nominal overall thickness: 500 micrometres.
 - Limits for single gel coating: No reading less than 400 micrometres nor more than 600 micrometres.
 - Limits for double gel coating: Submit proposals for coat thickness limits based on resin manufacturer's recommendations.
 - Gel coat with embedded surface tissue: Submit proposals for coat thickness limits based on resin manufacturer's recommendations.
- Checking wet film thickness: All units checked in accordance with BS EN ISO 2808, four readings per m² per coat over external surface area.

460 LAMINATING

- Glass fibre content: Not less than 900 g/m² of glass fibre, in not less than two layers to each GRP skin.
- Reinforcement distribution:
 - Random reinforcement: Distributed uniformly.
 - Non random reinforcement: Correctly positioned and aligned.
- Layering of woven fabric reinforcement: Layered with chopped strand mat on both sides.
- Resin:glass fibre ratio: Not less than 2:1, higher as appropriate, with glass fully wetted out by resin.
- Layer bonding: Good overall bond between gel coats and layers of laminate.
- Consolidation of GRP: Well consolidated and free from air voids.

470 BONDING OF CONSTITUENTS

- Resin layers: Good overall bond between gel coats and layers of laminate.
- Additional components: Core materials, ties, ribs, fixings and accessories fully bonded to GRP skins over full contact surface area.

480 FIXINGS AND FASTENERS

- Loadbearing fixing type: Contractor design.
 - Material: Submit proposals.
- Restraint fixing type: Contractor design.
 - Material: Submit Proposals.
- Extent of adjustment: To accommodate support structure/ cladding fabrication/ installation tolerances and thermal movement of GRP.
- Method of fixing to primary support structure: Contractor design.
 - Avoidance of bimetallic corrosion: Select appropriate materials for fixings/ fasteners/ spacers/ washers.

490 SEALING OF UNITS

- Flow coat: Apply to all surfaces of finished units which are not gel coated.
- Cut edges and holes: Seal to avoid penetration of moisture into glass fibres.

500 CURING

- Initial curing period at 20°C: 24 hours.
- Oven curing regime: Contractor design.
- Distortion during curing: Prevent.

505 METAL POWDER FINISH

- Polishing: Remove an even film of resin to ensure natural patination of the whole exposed surface.

510 HARDNESS TESTING

- Standard: To BS EN 59.
- Timing: After curing.
- Extent: One test for each 1 m² of external surface area, not less than 2 tests per unit. The number of measurements shall be such that the average result has a standard deviation of $\pm 4\%$.
- Barcol hardness at ambient temperature: Not less than 30.

530 WEIGHT TESTING

- First unit of each type and size:
 - Compliance: In accordance with design and specification.
 - Weight: Record.
- Subsequent units:
 - Acceptable weight: Unmodified subsequent units to be within 10% of weight of first unit.

570 PRODUCTION RECORDS

- Recorded information:
 - Unique identification number.
 - Full details of composition.
 - Date of each stage of manufacture.
 - Dates and results of tests, checks and inspections.
 - Dimensions related to specified levels of accuracy.
 - Specific location in the finished work.
 - Other pertinent data, e.g. if the unit is a production control unit.
- Availability of records: For inspection on request.

580 RETENTION OF MOULDS AFTER PRODUCTION

- Purpose: Retain moulds and store in a reusable condition to allow manufacturing to recommence if required.
- Retention period: Destroy only with approval.

INSTALLATION

605 PREPARATION

- Prefabrication: Complete products and attach fixings in workshop wherever possible.
- Identification: Mark or tag products. Do not mark surfaces visible in the complete installation.
- Electrolytic corrosion: Isolate dissimilar metals.

615 SUITABILITY OF STRUCTURE

- Contractor's survey:
 - Scope: Geometric survey of supporting structure, checking line, level and fixing points.
 - Coordinate: With surveys for adjacent cladding.
 - Give notice: If structure will not allow required accuracy or security of erection.
- Setting out: Establish erection datum points, lines and levels for a complete elevation at a time unless otherwise agreed.

625 INSTALLATION OF INTERFACES

- General: Locate flashings, closers etc. correctly with neat overlaps to form weathertight junctions.

630 ACCURACY OF ERECTION OF CLADDING

- Elevation joint widths: Within joint lengths, including in-line continuations across transverse joints, as follows:
 - Tolerance: Greatest width not to exceed least width by more than 15%.
 - Variations: Evenly distribute, with no sudden changes.
- Offset in elevation: Between nominally in-line edges across transverse joints not to exceed 10% width of joint.
- Offset in plan or section: Between flat faces or adjacent panels across joints not to exceed 15% width of joint.
- Sealant joints width limitations: To recommendations of sealant manufacturer.
- Finished work: Square, regular, true to line and plane with satisfactory fit at junctions.

635 ACCURACY OF ERECTION OF GRP COMPONENTS

- Joint widths: Within joint lengths, including in-line continuations across transverse joints, as follows:
 - Tolerance: Greatest width not to exceed least width by more than 20%.
 - Variations: Evenly distribute, with no sudden changes.
- Offset in elevation: Between component edges across transverse joints not to exceed 10% width of joint.
- Offset in plan or section: Between flat faces of adjacent units across joints not to exceed 10% width of joint.
- Sealant joints width limitations: To recommendations of sealant manufacturer.
- Finished work: True to line and plane with satisfactory fit at junctions.

640 FINAL FIXING

- Torque figures: Tighten threaded fastenings to figures recommended by manufacturer. Do not overtighten restraint fixings intended to permit lateral movement.

645 METALWORK

- Material standards and fabrication: As section Z11.

655 SEALANT JOINTING PROPRIETARY SYSTEM

- Sealant: Silicone.
 - Manufacturer: Contractor's choice.
 - Product reference: Submit proposals.
 - Colour: To be agreed.
- Joint back up/ filler: As recommended by sealant manufacturer.
- Primer: As recommended by sealant manufacturer.
- Application: As section Z22.

660 OPEN DRAINED JOINTS

- Rear seals: Airtight.
- Baffles and flashings: Installed securely and accurately to function as intended.

680 CLEANING DOWN

- General: Return to site at completion or when instructed and thoroughly clean down the entire area of the GRP work. Cleaning agents for the purpose must be approved by the GRP manufacturer.

DRAFT

H62
Natural slating

H62 Natural slating

To be read with Preliminaries/ General conditions.

TYPES OF SLATING

105A ROOF SLATING OPTION 1 - GENERALLY

- Substrate: Cold Vented Roof Type R1 - Trussed rafters at 600 mm centres.
- Pitch: Pitch varies between 30° and 45°.
- Underlay: Reinforced bitumen membrane to BS 8747, type 1F.
 - Recycled content: None permitted.
 - Direction: Parallel to eaves.
 - Head-lap (minimum): 100 mm.
- Battens:
 - Size: 50 x 25 mm.
 - Fixing: 65 x 3.35 mm galvanized annular ring shank nails.
- Slates:
 - Supplier: Cembrit Ltd www.cembrit.co.uk
57 Kellner Road, London SE28 0AX
Tel: 020 8301 8900 .
 - Product reference: Spanish .
 - Type: Duquesa .
 - Size: 500 x 250 mm .
 - Head-lap (minimum): 98 mm .
- Fixing: Two nails each slate. 35 x 3.35mm x 10mm dia, head copper nails.
- Accessories: None .

105B ROOF SLATING OPTION 2 - GENERALLY

- Substrate: Cold Vented Roof Type R1 - Trussed rafters at 450 or 600 mm centres.
- Pitch: Pitch varies between 30° and 45°.
- Underlay: Reinforced bitumen membrane to BS 8747, type 1F.
 - Recycled content: None permitted.
 - Direction: Parallel to eaves.
 - Head-lap (minimum): 100 mm.
- Battens:
 - Size: 50 x 25 mm.
 - Fixing: 65 x 3.35 mm galvanized annular ring shank nails.
- Slates:
 - Supplier: Cembrit Ltd www.cembrit.co.uk
57 Kellner Road, London SE28 0AX
Tel: 020 8301 8900 .
 - Product reference: Spanish .
 - Type: Montleon .
 - Size: 500 x 250 mm .
 - Head-lap (minimum): 98 mm .
- Fixing: Two nails each slate. 35 x 3.35mm x 10mm dia, head copper nails.
- Accessories: None .

105C ROOF SLATINGGENERALLY - OPTION 3

- Substrate: Cold Vented Roof Type R1 - Trussed rafters at 450 or 600 mm centres.
- Pitch: Pitch varies between 30° and 45°.
- Underlay: Reinforced bitumen membrane to BS 8747, type 1F.
 - Recycled content: None permitted.
 - Direction: Parallel to eaves.
 - Head-lap (minimum): 150 mm.
- Battens:
 - Size: 50 x 25 mm.
 - Fixing: 65 x 3.35 mm galvanized annular ring shank nails.
- Slates:
 - Supplier: SSQ Ltd, 301 Elveden Road, Park Royal, London, NW10 7SS.
Contact: Dalgeet Desouza: 07957 143423; dalgeet@ssq.co.uk
OR alternative approved.
Product reference: Spanish.
 - Type: Del Carmen - Ultra.
 - Size: 500 x 250 mm.
 - Head-lap (minimum): 90 mm.
- Fixing: Two nails each slate. 35 x 3.35mm x 10mm dia, head copper nails.
- Accessories: None.

105D ROOF SLATINGGENERALLY - OPTION 4

- Substrate: Cold Vented Roof Type R1 - Trussed rafters at 450 or 600 mm centres.
- Pitch: Pitch varies between 30° and 45°.
- Underlay: Reinforced bitumen membrane to BS 8747, type 1F.
 - Recycled content: None permitted.
 - Direction: Parallel to eaves.
 - Head-lap (minimum): 150 mm.
- Battens:
 - Size: 50 x 25 mm.
 - Fixing: 65 x 3.35 mm galvanized annular ring shank nails.
- Slates:
 - Supplier: SSQ Ltd, 301 Elveden Road, Park Royal, London, NW10 7SS.
Contact: Dalgeet Desouza: 07957 143423; dalgeet@ssq.co.uk
OR alternative approved.
Product reference: Spanish.
 - Type: Domiz - First.
 - Size: 500 x 250 mm.
 - Head-lap (minimum): 90 mm.
- Fixing: Two nails each slate. 35 x 3.35mm x 10mm dia, head copper nails.
- Accessories: None.

SLATING GENERALLY

210 BASIC WORKMANSHIP

- General: Fix slating and accessories to make the whole sound and weathertight at earliest opportunity.
- Setting out: To true lines and regular appearance, with neat fit at edges, junctions and features.
- Fixings for slating accessories: As recommended by manufacturer.
- Gutters and pipes: Keep free of debris. Clean out at completion.

240A UNDERLAY

- Handling: Do not tear or puncture.
- Manufacturer: Permavent
 - Type; Apex
 - Contact: Martyn Halliwell; 07795 160567, martyn@permavent.co.uk
- Laying: Maintain consistent tautness.
- Vertical laps (minimum): 100 mm wide, coinciding with supports and securely fixed.
- Fixing: Galvanized steel, copper or aluminium 20 x 3 mm extra large clout head nails.
- Eaves: Where exposed, use an external grade (UV resistant) underlay or a proprietary eaves support product.
- Penetrations: Use proprietary underlay seals or cut underlay to give a watertight fit around pipes and components.
- Ventilation paths: Do not obstruct.

245A BATTENS/ COUNTERBATTENS - TREATED

- Timber: Sawn softwood
 - Supplier; John Brash Ltd.,
 - Type; PNSY, WPCA, WPCE, or WPNE
 - Species: To BS 5534, clause 4.12.1.
 - Permissible characteristics and defects: Not to exceed limits in BS 5534, annex C.
 - Grading: mechanically graded to comply with BS5534 and treated to BS8417:2003 and carry a 60year guarantee and be FSC or PEFC certified..
 - Moisture content at time of fixing and covering (maximum): 22%.
 - Preservative treatment: As section Z12 and Wood Protection Association Commodity Specification C8.
 - Type: Red.

265 BATTEN FIXING

- Setting out: Align parallel to ridge in straight horizontal lines to gauge of slates. Align on adjacent areas.
- Batten length (minimum): Sufficient to span over three supports.
- Joints in length: Square cut. Butt centrally on supports. Joints must not occur more than once in any group of four battens on one support.
- Additional battens: Provide where unsupported laps in underlay occur between battens.
- Fixing: Each batten to each support. Splay fix at joints in length.

270 BATTENS FIXED TO MASONRY

- Setting out: In straight horizontal lines. Align on adjacent areas.
- Batten length (minimum): 3 m.
- Fixing centres (maximum): 400 mm.

272 TIMBER FOR SLATING SUBSTRATE WORK

- Timber: Sawn softwood, free from wane, pitch pockets, decay and insect attack (ambrosia beetle excepted).
 - Moisture content at time of fixing and covering (maximum): 22%.
- Preservative treatment: As section Z12 and Wood Protection Association Commodity Specification C8.
 - Type: Contractor's choice.

275 SLATE FIXING

- Setting out: Lay slates with an even overall appearance with slightly open (maximum 5 mm) butt joints. Align tails.
- Slate thickness: Consistent in any one course. Lay with thicker end as tail.
- Ends of courses: Use extra wide slates to maintain bond and to ensure that cut slates are as large as possible. Do not use slates less than 150 mm wide.
- Top course: Head-nail short course to maintain gauge.
- Fixing: Centre nail each slate twice through countersunk holes 20-25 mm from side edges.
 - Nails: Copper clout to BS 1202-2 or aluminium clout to BS 1202-3.
 - Nail dimensions: Determine in accordance with BS 5534 to suit site exposure, withdrawal resistance and slate supplier's recommendations.

280 SLATES - PERFORMANCE SPECIFICATION

- Standards:
 - Product specification: To BS EN 12326-1.
 - Methods of test: To BS EN 12326-2.
- Slate type: Smooth.
- Dimensional tolerances:
 - Deviations from declared length, width, edge straightness, rectangularity, and flatness are not to exceed values specified in BS EN 12326-1, clause 5.12.
- Thickness:
 - Nominal thickness and individual thickness variation: To BS EN 12326-1, clause 5.2.
- Strength:
 - Characteristic modulus of rupture:
 - Transverse: 46MPa..
 - Longitudinal: 82MPa.
 - Mean failure load:
 - Transverse: 54MPa..
 - Longitudinal: 82MPa..
- Water absorption: Code: A1.
- Freeze-thaw resistance: Not Required.
- Thermal cycle test: Code: T1.
- Carbonate content: Less than 20%.
- Sulfur dioxide exposure test: S1.
- Non-carbonate carbon content: Less than or equal to 2%.

290 MORTAR BEDDING/ POINTING

- Mortar: As section Z21, 1:3 cement:sand, with plasticizing admixtures permitted.
 - Bond strength providing resistance to uplift: In accordance with BS 5534.
- Weather: Do not use in wet or frosty conditions or when imminent.
- Preparation of concrete and clay tile accessories to be bedded: Wet and drain surface water before fixing.
- Appearance: Finish neatly as work proceeds and remove residue.

ROOF SLATING EDGES/ JUNCTIONS/ FEATURES

305A GENERALLY

- Flashings: Fix with or immediately after slating. Form neatly.
Ensure that related trades are provided with all relevant information relating to carpentry and other work, etc. Before starting work ensure that previous related work is complete and in accordance with the project documents.
- Form all details using the specified recommended fittings and accessories: do not improvise without prior approval.
- Please be aware that this specification complies with the minimum requirements set out in British Standards so as to conform to, Building Regulations. For certain projects such as new housing there may be additional non-regulatory technical requirements from third party insurers that have to be satisfied. For example there is now a requirement for all mortar bedded ridges and hips must be mechanically fixed to comply with the NHBC Technical Standards so as to be eligible for NHBC Buildmark Warranty Cover. It is your responsibility to check.
- Fittings and accessories: As recommended by slate supplier, to match in colour and finish unless specified otherwise. do not improvise.
 - Exposed fittings and accessories: To match slate colour and finish.
- Cut slates: Cut only where necessary, to give straight, clean edges.
 - Fix edge slates and fittings securely to give neat and true lines. Ensure that all lead flashings are fixed with or immediately after the slating and are neatly dressed down, to the LSA requirements.

325 FIRE SEPARATING WALLS

- Separating walls: Completely fill space between top of wall and underside of slates with mineral wool quilt to provide fire stopping.
- Boxed eaves: Completely seal air paths in plane of separating wall with wire reinforced mineral wool, not less than 50 mm thick, fixed to rafters and carefully cut to shape to provide fire stopping.

355 VENTILATED EAVES WITH SEPARATED GRILLES/ TRAYS

- Fascia grilles:
 - Manufacturer: Glidevale or similar.
 - Product reference: FV100.
- Ventilator trays:
 - Manufacturer: Glidevale.
 - Product reference: to suit.
 - Fix to provide free passage of air over insulation.
- Underlay support: 12 mm plywood, as section G20.
 - Continuous to prevent water retaining troughs.
- Gutter: Dress underlay or underlay support tray to form drip into gutter.
- Undercourse and first course slates: Fix with tails projecting 50 mm over gutter or to centre of gutter, whichever dimension is the lesser.

445 MORTAR BEDDED VERGES WITH BEDDED UNDERCLOAK

- Underlay: Carry 50 mm onto outer leaf of gable wall and bed on mortar.
 - Undercloak: Slates.
 - Position: Over underlay, level with underside of slating battens, sloping towards verge.
 - Projection beyond face of wall: 38-50 mm.
 - Bedding: On mortar identical to that used in gable walling.
 - Slating battens: Carry onto undercloak and finish 100 mm from verge edge.
 - Verge slates:
 - Bedding: Flush with undercloak on 75 mm wide bed of mortar.
 - Pointing: Flush profile.
- Fixing: Do not displace or crack mortar.

525 MITRED HIPS

- Underlay: Lay courses over hip.
 - Overlaps (minimum): 150 mm.
- Hip slate fixing battens: tbc.
- Mitred slates: Cut extra wide slates and fix to form a straight, close mitred junction.
- Soakers: Interleave with mitred slates. Fix by turning down over head of mitred slates.

555 MORTAR BEDDED AND MECHANICALLY FIXED TILE HIPS

- Underlay: Lay courses over hip.
 - Overlaps (minimum): 150 mm.
- Hip tile fixing battens: as required.
- Roof slates: Cut and fix closely at hip.
- Hip irons: Galvanized steel in accordance with BS 5534, clause 4.15.4.
 - Fixing: To hip rafter or hip batten with not less than two zinc coated steel screws.
- Hip tiles:
 - Manufacturer: Monier Redland or similar.
Product reference: Submit proposals.
 - Bedding: On mortar, continuous to edges and solid to joints.
 - Fixing: Secure all hip tiles to hip rafters or hip tile fixing battens with self-sealing non-ferrous through fixings.
 - Bottom hip tiles: Shape neatly to align with corner of eaves and fill ends with mortar and slips of tile finished flush.

635 MITRED VALLEYS

- Underlay: Lay strips not less than 600 mm wide centred on valleys. Overlap with general roof underlay.
- Mitred slates: Cut extra wide slates and fix to form a straight, close mitred junction.
- Soakers: Interleave with mitred slates. Fix by turning down over head of mitred slates.

660 SIDE ABUTMENTS

- Underlay: Turn up not less than 100 mm at abutments.
- Abutment slates: Cut as necessary. Fix close to abutments.
- Soakers: Interleave with abutment slates. Fix by turning down over head of abutment slates.

670 TOP EDGE ABUTMENTS

- Underlay: Turn up not less than 100 mm at abutments.
- Top slate courses: Fix close to abutments.

675A ABUTMENT VENTILATOR

- Manufacturer: Manthorpe Building Products Ltd.
 - Web: www.manthorpebuildingproducts.co.uk.
 - Email: bpsales@manthorpe.co.uk.
- Product reference: G1105

690 ROOF WINDOWS

- Underlay: Turn up not less than 100 mm at window surrounds under integral flashings/soakers.
- Roof slates: Cut as necessary and fix closely all round.

700A RIDGE VENTILATING SYSTEM

- Manufacturer: Klobber Ltd.
 - Web: www.klobber.co.uk.
 - Email: technical@klobber.co.uk.
- Product reference: Klobber Uni-Dry Ventilating Ridge System.

740 MORTAR BEDDED AND MECHANICALLY FIXED TILE RIDGES

- Underlay: Lay courses over ridge.
 - Overlap (minimum): 150 mm.
- Ridge tile fixing battens: as required to suit vent.
- Ridge tiles:
 - Manufacturer: Monier Redland.
 - Product reference: Submit proposals.
 - Bedding: On mortar, continuous to edges and solid to joints.
 - Fixing: Secure all ridge tiles to ridge boards or supplementary ridge battens with self-sealing non-ferrous fixings.
 - Gable end ridge tiles: Fill ends with mortar and slips of tiles finished flush.
- Ridge terminals:
 - Manufacturer: Contractor's choice.
 - Product reference: Submit proposals.

800 CHANGE OF ROOF PITCH

- Slating substrate work: Fix timber tilting fillet to support flashing and courses of slates above change of pitch.
- Roof slates: Make weathertight with flashing.
 - Courses above change of pitch: Fix undercourse and first course slates.
 - Courses below change of pitch: Cut top two courses as necessary to maintain gauge.
- Head-nail top course.

850 ROOF SLOPE TERMINALS

- Ventilator slates: -.
 - Manufacturer: SSQ.
 - Product reference: Eclipse In-Line Natural Slate Terminal. .
 - Connect to soil and vent pipes AND/OR Mechanical extraction.

DRAFT

H71

Lead sheet coverings/ flashings

H71 Lead sheet coverings/ flashings

TYPES OF LEADWORK

- 324 SOAKERS FOR MITRED VALLEYS TO SLATE ROOFS
- Lead:
 - Thickness: 1.25 or 1.32 mm (Code 3).
 - Dimensions:
 - Length: Slate/ length of valley mitre + 25 mm.
 - Underlaps: Not less than 150 mm.
- 325 SOAKERS FOR MITRED VALLEYS TO SLATE/ PLAIN TILE ROOFS
- Lead:
 - Thickness: 1.25 or 1.32 mm (Code 3).
 - Dimensions:
 - Length: Slate/ tile gauge + lap + 25 mm.
 - Underlaps: Not less than 150 mm.
- 326 SOAKERS FOR SWEEP VALLEYS TO PLAIN TILE ROOFS
- Lead:
 - Thickness: 1.25 or 1.32 mm (Code 3).
 - Dimensions: To suit sweep and pitch of tiles. Allow adequate length for turning over head of cut tiles and width for underlay to first standard tile on each side.
- 470 FLASHINGS TO DORMERS AND CHANGE IN ROOF LEVELS - REFER TO DRAWINGS
- Lead:
 - Thickness: 1.75 or 1.80 mm (Code 4).
 - Dimensions:
 - Lengths: Not more than 1500 mm.
 - Fixing: Fixing: Nail top edge at 150 mm centres and welt edge. Clip bottom edge at laps and 500 mm centres..

GENERAL REQUIREMENTS/ PREPARATORY WORK

- 510 WORKMANSHIP GENERALLY
- Standard: To BS 6915 and latest edition of 'Rolled lead sheet. The complete manual' published by the Lead Sheet Association.
 - Fabrication and fixing: To provide a secure, free draining and completely weathertight installation.
 - Operatives: Trained in the application of lead coverings/ flashings. Submit records of experience on request.
 - Preforming: Measure, mark, cut and form lead prior to assembly wherever possible.
 - Marking out: With pencil, chalk or crayon. Do not use scribes or other sharp instruments without approval.
 - Bossing and forming: Straight and regular bends, leaving sheets free from ripples, kinks, buckling and cracks.
 - Solder: Use only where specified.
 - Sharp metal edges: Fold under or remove as work proceeds.
 - Finished work: Fully supported, adequately fixed to resist wind uplift but also able to accommodate thermal movement without distortion or stress.
 - Protection: Prevent staining, discolouration and damage by subsequent works.

- 515 LEADWELDING
 - In situ leadwelding: Not permitted.

- 516 LEADWELDING
 - In situ leadwelding: Is permitted, subject to completion of a 'hot work permit' form and compliance with its requirements.

- 520 LEAD SHEET
 - Production method:
 - Rolled, to BS EN 12588, or
 - Machine cast and BBA certified, or
 - Sand cast, from lead free from bitumen, solder, other impurities, inclusions, laminations, cracks, air, pinholes and blowholes; to code thicknesses but with a tolerance (by weight) of $\pm 10\%$.
 - Identification: Labelled to show thickness/ code, weight and type.

- 555 LAYOUT
 - Setting out of longitudinal and cross joints: Submit proposals.

- 610 SUITABILITY OF SUBSTRATES
 - Condition: Dry and free of dust, debris, grease and other deleterious matter.

- 630 PLYWOOD UNDERLAY
 - Standard: Manufactured to an approved national standard and to BS EN 636, section 8 (plywood for use in humid conditions).
 - Sheet size: 2400 or 1200 x 1200 mm and 6 mm thick.
 - Moisture content: Not more than 22% at time of covering. Give notice if greater than 16%.
 - Laying: Cross joints staggered and a 0.5 to 1 mm gap between boards.
 - Fixing: With 25 mm annular ringed shank copper or stainless steel nails, at 300 mm grid centres over the area of each sheet and at 150 mm centres along edges, set in 10 mm from perimeter edges.
 - Nail heads: Set flush or just below the surface.

- 640 TIMBER FOR USE WITH LEADWORK
 - Quality: Planed, free from wane, pitch pockets, decay and insect attack (ambrosia beetle excepted).
 - Moisture content: Not more than 22% at time of fixing and covering. Give notice if greater than 16%.
 - Preservative treatment: Organic solvent as section Z12 and Wood Protection Association Commodity Specification C8.

- 650 LAYING UNDERLAY
 - Handling: Prevent tears and punctures.
 - Laying: Butt or overlap jointed onto a dry substrate.
 - Fixing edges: With copper or stainless steel staples or clout nails.
 - Do not lay over roof edges but do turn up at abutments.
 - Wood core rolls: Fixed over underlay.
 - Protection: Keep dry and cover with lead at the earliest opportunity.

FIXING LEAD

705 HEAD FIXING LEAD SHEET

- Top edge: Secured with two rows of fixings, 25 mm and 50 mm from top edge of sheet, at 75 mm centres in each row, evenly spaced and staggered.
- Sheets less than 500 mm deep: May be secured with one row of fixings, 25 mm from top edge of sheet and evenly spaced at 50 mm centres.

710 FIXINGS

- Nails to timber substrates: Copper clout nails to BS 1202-2, or stainless steel (austenitic) clout nails to BS 1202-1.
 - Shank type: Annular ringed, helical threaded or serrated.
 - Shank diameter: Not less than 2.65 mm for light duty or 3.35 mm for heavy duty.
 - Length: Not less than 20 mm or equal to substrate thickness.
- Screws to concrete or masonry substrates: Brass or stainless steel to BS 1210, tables 3 or 4.
 - Diameter: Not less than 3.35 mm.
 - Length: Not less than 19 mm.
 - Washers and plastic plugs: Compatible with screws and lead.
- Screws to composite metal decks: Self tapping as recommended by the deck and lead manufacturer/ supplier for clips.

770 WEDGE FIXING INTO JOINTS/ CHASES

- Joint/ chase: Rake out to a depth of not less than 25 mm.
- Lead: Dress into joint/chase.
 - Fixing: Lead wedges at not more than 450 mm centres, at every change of direction and with at least two for each piece of lead.
- Sealant: Submit proposals.
 - Application: As section Z22.

780 WEDGE FIXING INTO DAMP PROOF COURSE JOINTS

- Joint: Rake/ cut out under damp proof course to a depth of not less than 25 mm.
- Lead: Dress lead into joint.
 - Fixing: Lead wedges at not more than 450 mm centres, at every change of direction and with at least two for each piece of lead.
- Sealant: Submit proposals.
 - Application: As section Z22.

790 SCREW FIXING INTO JOINTS/ CHASES

- Joint/ chase: Rake out to a depth of not less than 25 mm.
- Lead: Dress into joint/ chase and up back face.
 - Fixing: Into back face with stainless steel screws and washers and plastics plugs at not more than 450 mm centres, at every change of direction, and with at least two fixings for each piece of lead.
- Sealant: Submit proposals.
 - Application: As section Z22.

JOINTING LEAD

810 FORMING DETAILS

- Method: Bossing or leadwelding except where bossing is specifically required.
- Leadwelded seams: Neatly and consistently formed.
 - Seams: Do not undercut or reduce sheet thickness.
 - Filler strips: Of the same composition as the sheets being joined.
 - Butt joints: Formed to a thickness one third more than the sheets being joined.
 - Lap joints: Formed with 25 mm laps and two loadings to the edge of the overlap.
- Bossing: Carried out without thinning, cutting or otherwise splitting the lead sheet.
 - Details where bossing must be used: Not applicable .

DRAFT

J
Waterproofing

DRAFT

J30

Liquid applied tanking/ damp proofing

J30 Liquid applied tanking/ damp proofing

TYPES OF TANKING/ DAMP PROOFING

110A BITUMEN SOLUTION COATING TO RETAINING WALLS AND GARAGES WHERE REQUIRED

- Manufacturer: RIW.
 - Web: www.riw.co.uk.
 - Email: technical@riw.co.uk.
 - Product reference: RIW LAC®
- Application: Two coats
- Coverage per coat (maximum): First coat: 1.7 m²/litre, second coat: 2.5 m²/litre

EXECUTION

205 SUITABILITY OF SUBSTRATE

- Substrates generally:
 - Smooth, even textured, clean, dry and frost free.
 - Within tolerances for level and surface regularity.
 - Vertical and horizontal surfaces: Correctly prepared and free from irregularities.
- Curing period for concrete substrates (minimum): 7 days.
- Moisture content and stability of substrate: Must not impair integrity of finished tanking/ damp proofing.
- Preliminary work: Complete including:
 - Chases.
 - External angles.
 - Formation of upstands and kerbs.
 - Movement joints.
 - Penetrations/ Outlets.

206 ADHESION TESTING

- Tensile bond strength: Test in accordance with BS EN ISO 4624.
- Test results: Submit.

207 PRIMERS

- Application: Uniform, continuous coverage.

210 COATING APPLICATION

- Adjacent surfaces exposed to view in finished work: Protect.
- Coatings:
 - Apply in dry atmospheric conditions when primer is dry.
 - Uniform, continuous coverage. Do not allow to pool in hollows.
 - Firmly adhered to substrate and free from imperfections.
 - Prevent damage to finished coatings.
- Penetrations: Impervious.
- Final covering: Apply as soon as possible after coating has hardened.

220 COLD APPLIED COATINGS

- Thinning: Not permitted unless recommended by manufacturer.
- Successive coats:
 - Allow to dry before applying next.
 - Apply at right angles to previous.

230 BITUMINOUS COATINGS

- Air and surface temperatures (minimum): 5°C.
- Weather conditions: Do not apply if there is a risk of rain during application and drying unless effective temporary cover is provided over working area.
 - Unavoidable wetting of construction or coatings: Minimize and make good damage.

235 MODIFIED COATINGS

- Air and surface temperatures: Do not apply if below minimum recommended by coating manufacturer.
- Curing: Keep dry until fully cured.

260 JUNCTIONS WITH DPCS

- Dpcs: Clean, all edges fully exposed.
- Application: Fully coat dpc and overlap adjacent surfaces by (minimum) 75 mm.

270 BLINDING

- Coatings: Blind whilst tacky .
- Surplus material: Remove when coatings are completely dry.

COMPLETION

310 INSPECTION

- Interim and final inspections: Submit reports.

330 PROTECTION OF EXTERNAL COATINGS

- Coated surface: Clean and free from contaminants.
- Board manufacturer: Contractor's choice.
 - Product reference: Submit proposals.
 - Thickness: As recommended.
- Placement: Bond protection board to coating with double-sided tape.
 - Edges: Butt jointed and taped.
 - Perimeter treatment: As recommended.
- Contact with coating: Secure and continuous.

340 BACKFILLING TO EXTERNAL COATINGS

- Timing: Carry out as soon as possible after tanking and protection are complete.

DRAFT

J40

Flexible sheet waterproofing/ damp proofing

J40 Flexible sheet waterproofing/ damp proofing

To be read with Preliminaries/ General conditions.

TYPES OF TANKING/ DAMP PROOFING

120A LOOSE LAID POLYETHYLENE DAMP PROOFING TO GROUND FLOORS GENERALLY

- Substrate: Beam and Block Flooring or In-situ concrete slab.
- Manufacturer: Visqueen Building Products.
 - Web: www.visqueenbuilding.co.uk.
 - Email: riba@visqueenbuilding.co.uk.
 - Product reference: Visqueen EcoMembrane®
- Thickness/ Gauge: 500 micrometres (2000 gauge) .
- Recycled content: as manufacturers standard .
- Joints:
 - Surfaces to be joined: Clean and dry beyond full width of joint.
 - Laps (minimum): End and side, 150 mm .
 - Sealing: Continuous mastic strip between overlaps; edge of top sheet sealed with jointing tape .
- Laying: In strict accordance with manufacturers recommendations.

140A LOOSE LAID POLYETHYLENE GAS RETARDANT/ DAMP PROOFING TO BLOCK D

- Manufacturer: Visqueen Building Products.
 - Web: www.visqueenbuilding.co.uk.
 - Email: riba@visqueenbuilding.co.uk.
 - Product reference: Visqueen Gas Barrier
- Membrane:
 - Size: 2.0 x 25 m.
- Accessories: Double Sided Jointing Tape.

297A TANKING MEMBRANE FOR USE WHERE EXTERNAL GROUND LEVEL ABOVE FFL

- Manufacturer: Sika Limited.
 - Web: www.sika.co.uk.
 - Email: enquiries@uk.sika.com.
 - Product reference: SikaProof® P to walls to extend in line with NHBC requirements and SikaProof A under slab..
- Width: 1 m.
- Accessories: As manufacturers recommendations.
Final detail will be subject to site conditions..

WORKMANSHIP

310 WORKMANSHIP GENERALLY

- Condition of substrate:
 - Clean and even textured, free from voids and sharp protrusions.
 - Moisture content: Compatible with damp proofing/ tanking.
- Air and surface temperature: Do not apply sheets if below minimum recommended by membrane manufacturer.
- Condition of membrane at completion:
 - Neat, smooth and fully supported, dressed well into abutments and around intrusions.
 - Completely impervious and continuous.
 - Undamaged. Prevent puncturing during following work.
- Permanent overlying construction: Cover membrane as soon as possible.

320 INSPECTION

- Give notice: Before covering any part of membrane with overlying construction.

335A PRIMERS

- Manufacturer: IKO PLC.
 - Web: www.ikogroup.co.uk.
 - Email: marketing.uk@iko.com.
 - Product reference: IKOpro SA Bitumen Primer

345 COLD APPLIED BONDING COMPOUNDS

- Type and application: As recommended for the purpose by the membrane manufacturer.

350 ANGLES IN BONDED DAMP PROOFING/ TANKING

- Preformed rot proof fillet to internal angles:
 - Size (minimum): 50 x 50 mm, splay faced.
 - Bedding: Bitumen mastic or bonding compound.
- Reinforcing strip to all angles:
 - Material: As damp proofing tanking.
 - Width (minimum): 300 mm.
 - Timing: Apply before main sheeting.
- Dressing of main sheeting onto adjacent surfaces (minimum): 100 mm.

360 JUNCTIONS WITH PROJECTING DPCS/ CAVITY TRAYS

- Adjoining surfaces: Clean and dry.
- Dpcs/ Cavity trays: Lap and fully bond/ seal with sheeting.
 - Laps (minimum): 150 mm.
 - Bonding/ Sealing: As recommended by manufacturer.

365 JUNCTIONS WITH FLUSH DPCS/ CAVITY TRAYS

- Adjoining surfaces: Clean and dry.
- Dpcs/ Cavity trays:
 - Expose edge where concealed.
 - Lap and fully bond/ seal sheeting to wall.
 - Dressing of sheeting beyond dpc/ cavity tray (minimum): 50 mm.
 - Bonding/ Sealing: As recommended by manufacturers.

DRAFT

J42

Single layer polymeric sheet roof coverings

J42 Single layer polymeric sheet roof coverings

TYPES OF ROOF COVERING

TYPES OF ROOF COVERING TO FLAT ROOF AREA OF APARTMENT BLOCKS A, B, C and D

ROOF - FLAT ROOF (Rigid insulation above joists)

To achieve U-value of 0.12W/m²K or better

Flat roof comprising Sarnafil single ply membrane or equivalent approved on a built-up system.

Roof to be designed to take into consideration mounting of roof PV units.

Substrate: Timber Joists to structural engineers specification / Specialist Design.

PERFORMANCE

202 CONTRACTOR'S DESIGN OF ROOF COVERINGS

- Design responsibility: Determine types and locations of fixings and joints in sheets.
- Structural and fire requirements:
 - Generally: As section B50.
 - Modifications: None.
 - Design: Complete the design in accordance with the designated code of practice to satisfy specified performance criteria.
- Functional requirements:
 - Performance: As specified in this section.
- Additional requirements: To accommodate roof mounted PV units.
- Design and production information: tbc.
- Timing of submissions: To meet programme agreed.

210 ROOF PERFORMANCE

- Roof covering: Secure, free draining and weathertight.

220 AVOIDANCE OF INTERSTITIAL CONDENSATION: WARM AND INVERTED ROOFS

- Determine: Interstitial condensation risk of roof construction as recommended in BS 6229.
- Basic design data:
 - Outdoor notional psychrometric conditions, winter:
 - Temperature: -5°C.
 - Relative humidity: 90%.
 - Vapour pressure: 0.36 kPa.
 - Duration: 60 days.
 - Outdoor notional psychrometric conditions, summer:
 - Temperature: 18°C.
 - Relative humidity: 65%.
 - Vapour pressure: 1.34 kPa.
 - Duration: 60 days.
 - Indoor notional psychrometric conditions:
 - Temperature: _____.
 - Relative humidity: _____.
 - Vapour pressure: _____.
- Winter interstitial condensate (warm roof):
 - Calculated amount (maximum): 0.35 kg/m².
 - Calculated annual net retention: Nil.
- Vapour control layer: If necessary, provide a suitable membrane or sealed deck so that damage and nuisance from interstitial condensation do not occur.

225 AVOIDANCE OF INTERSTITIAL CONDENSATION: WARM AND INVERTED ROOFS

- Interstitial condensation within roof construction: Determine risk as recommended in BS 5250 and BS EN ISO 13788.
- Vapour control layer: If necessary, provide a suitable membrane so that damage and nuisance from interstitial condensation do not occur.

230 INSULATION

- Requirement: Determine type and thickness of insulation and integral or separate overlay to satisfy the following criteria:
 - Thermal transmittance of the roof (maximum): 0.12 W/m²K.
 - Compressive strength of insulation (minimum) at 10% compression: To SPRA Quality Standard for insulation type.
 - Finished surface: Suitably even, stable and robust to receive roof covering.
 - Insulation compliance: To a relevant European Standard, or Agrément certified.

PRODUCTS

EXECUTION GENERALLY

510 ADVERSE WEATHER

- General: Do not lay membrane at temperatures below 5°C or in wet or damp conditions unless effective temporary cover is provided over working area.
- Unfinished areas of roof: Keep dry and protect edges of laid membrane from wind action.

520 INCOMPLETE WORK

- End of working day: Provide temporary seal to prevent water infiltration.
- On resumption of work: Cut away tail of membrane from completed area and remove from roof.

530 APPLYING PRIMERS

- Coverage per coat (minimum): As recommended by manufacturer.
- Surface coverage: Even and full.
- Coats: Fully bonded. Allow volatiles to dry off thoroughly between coats.

SUBSTRATES/ VAPOUR CONTROL LAYERS/ WARM DECK ROOF INSULATION

610 SUITABILITY OF SUBSTRATES

- Surfaces to be covered: Secure, clean, dry, smooth, free from frost, contaminants, voids and protrusions.
- Preliminary work: Complete, including
 - Grading to correct falls.
 - Formation of upstands, kerbs, box gutters, sumps, grooves, chases and expansion joints.
 - Fixing of battens, fillets and anchoring plugs/ strips.
- Moisture content and stability of substrate: Must not impair integrity of roof.

660 JOINTS IN RIGID BOARD SUBSTRATES

- Cover strip: Lay centrally over substrate joints before laying vapour control layers or coverings. Adhere to substrate with bonding compound along edges only.

670 LAYING VAPOUR CONTROL LAYER

- Laying: As recommended by manufacturer.
- Side and head laps: As recommended by manufacturer.
- Upstands, kerbs and other penetrations: Enclose edges of insulation. Fully seal at abutment by bonding or taping.

680 LAYING WARM DECK ROOF INSULATION

- Setting out:
 - Long edges: Fully supported and running at right angles to side laps in single ply membrane.
 - End edges: Adequately supported.
 - Joints: Butted together.
 - End joints: Staggered.
- Attachment: As recommended by manufacturer.
- Mechanical fixing: As recommended by manufacturer.
- Completion: Boards must be in good condition, well fitting and secure.

WATERPROOF MEMBRANES/ ACCESSORIES

710 MECHANICAL FIXING OF WATERPROOF MEMBRANE

- Setting out: Perpendicular to the deck direction.
- Laying: Loose, do not wrinkle or stretch.
- Installing fasteners:
 - Use manufacturer's/ supplier's recommended methods and equipment.
 - Insertion: Correct and consistent.
- Washers/ Pressure plates/ Bars:
 - Distance from fixed edge (minimum): 10 mm.
 - Fixing: Flush with membrane.
- Sheet overlaps: Extend beyond washers/ pressure plates by minimum 50 mm.
- Surface condition at completion: Fully sealed, smooth, weatherproof and free draining.

- 720 ADHESIVE BONDING OF WATERPROOF MEMBRANE
- Setting out: Perpendicular to the deck direction.
 - Attachment: Fully adhered on a continuous even coating of adhesive.
 - Do not wrinkle or stretch.
 - Surface condition at completion: Fully sealed, smooth, weatherproof and free draining.
- 730 WELDED JOINTING OF WATERPROOF MEMBRANE
- Side and end joints:
 - Laps (minimum): As recommended by manufacturer .
 - Preparation: Clean and dry surfaces beyond full width of joint.
 - Sealing: Weld together.
 - Seam sealant: As recommended by manufacturer.
 - Condition at completion: Fully sealed, smooth, weatherproof and free draining.
- 740 ADHESIVE JOINTING OF WATERPROOF MEMBRANE
- Side and end joints:
 - Laps (minimum): As recommended by manufacturer.
 - Preparation: Prime, clean and dry surfaces beyond full width of joint and lap.
 - Sealing: Apply continuous even coverage of adhesive to both surfaces. Mate and roll together. Do not wrinkle or stretch membrane.
 - Seam sealant: As recommended by manufacturer.
 - Condition at completion: Fully sealed, smooth, weatherproof and free draining.
- 750 TAPE JOINTING OF WATERPROOF MEMBRANE
- Side and end joints:
 - Laps (minimum): As recommended by manufacturer.
 - Preparation: Prime, clean and dry surfaces beyond full width of joint.
 - Sealing: Apply self adhesive tape. Mate and roll together. Do not wrinkle or stretch membrane.
 - Condition at completion: Fully sealed, smooth, weatherproof and free draining.
- 760 PERIMETER OF MEMBRANE
- General: Secure membrane at roof edge conditions, changes of plane, curb flashings, upstands to roof lights, etc. with mechanical fasteners.
- 765 PERIMETER DETAILS FOR THERMOPLASTIC MEMBRANES
- Upstands, edge trims, drips, kerbs, etc: Secure preformed metal sections to roof structure with mechanical fasteners.
 - Roof membrane: Dress over perimeter profile. Overlap beyond fasteners by minimum 50 mm.
 - Sealing: Weld together.
- 770 PERIMETER DETAILS FOR THERMOPLASTIC MEMBRANES
- Upstands, edge trims, drips, kerbs, etc: Form flashings from waterproof membrane material.
 - Roof membrane: Terminate in horizontal plane immediately adjacent to change in direction and secure with mechanical fasteners.
 - Flashings: Dress over perimeter profile. Overlap horizontal roof membrane beyond perimeter securement by minimum 50mm.
 - Sealing: Bond with adhesive and weld at overlap.

775 PERIMETER DETAILS FOR ELASTOMERIC MEMBRANES

- Upstands, edge trims, drips, kerbs, etc: Preformed from waterproof membrane material.
- Reinforcing strip: Lay at edge of horizontal roof plane.
 - Securing: Mechanically fasten.
- Roof membrane: Dress over perimeter profiles.
 - Sealing: Bond to substrate and to secured perimeter reinforcing strip.

780 ROOF PENETRATIONS THROUGH THERMOPLASTIC MEMBRANES

- Roof membrane: Cut around penetrations and secure to deck.
- Flanged sleeve:
 - Type: Preformed flanged sleeve.
 - Installation: Dress over and around penetration.
 - Roof membrane overlap to flange (minimum): 50 mm beyond fasteners.
 - Sealing: Weld flange to roof membrane.
- Protection to top edge of sleeve: Flashing or weathering cravat.

785 ROOF PENETRATIONS THROUGH ELASTOMERIC MEMBRANES

- Roof membrane: Cut around penetrations.
- Flanged sleeve:
 - Type: Preformed flanged sleeve.
 - Installation: Dress over and around penetration.
 - Seating: Flush to roof membrane.
 - Roof membrane overlap to flange (minimum): 75 mm beyond fasteners.
 - Sealing: As manufacturers recommendations.

SURFACING

810 LAYING INVERTED ROOF INSULATION

- Condition of substrate: Clean.
- Setting out: Loose lay with staggered joints.
 - Cutting: Minimize.
 - Small cut pieces: Avoid at perimeters and penetrations.
 - Joints: Butt together.
- Projections, upstands, rainwater outlets, etc: Cut insulation cleanly and fit closely around.
- Completion:
 - Insulation boards to be in good condition, well fitting and stable.
 - Cover to prevent wind uplift and flotation as soon as practicable.

850 LAYING MEMBRANE WALKWAYS

- Attachment: As manufacturers recommendations .

COMPLETION

910 INSPECTION

- Interim and final roof inspections: Submit reports.

930 FLOOD TEST TO FLAT ROOF

- Condition of roof prior to testing:
 - Waterproof membrane complete to a stage where integrity can be tested.
- Outlets: Externally cover and seal. Protect against damage from water pressure using temporary kerbs. Do not use plugs to seal outlets.
- Flood levels: Submit proposals. In no case higher than kerbs.
- Flood duration: Three days.
- Inspection: Regular, to detect leaks.
- Completion of test: Slowly drain roof. Do not overload or flood outlets.
- Test results and warranty: Submit on completion of testing.

935 COMMISSIONING AND TESTING OF PHOTOVOLTAIC ROOF MEMBRANE

- Commissioning and testing authority: Submit proposals.
- Timing of test: Prior to completion.
- Condition of photovoltaic system prior to testing: Complete to a stage where commissioning tests can be undertaken.
- Test results and warranty: Submit on completion of commissioning tests.

940 COMPLETION

- Roof areas: Clean.
 - Outlets: Clear.
- Work necessary to provide a weathertight finish: Complete.
- Storage of materials on finished surface: Not permitted.
- Completed membrane: Do not damage. Protect from traffic and adjacent or high level working.

DRAFT

K

Linings/Sheathing/Dry partitioning

DRAFT

K10

Gypsum board dry linings/ partitions/ ceilings

K10 Gypsum board dry linings/ partitions/ ceilings

To be read with Preliminaries/ General conditions.

TYPES OF DRY LINING

MOISTURE RESISTANT BOARD

To all wet areas such as WC'S, bathrooms and en-suites replace Gyproc Wallboard with Moisture resistant board.

120C METAL STUD PARTITION SYSTEM GENERALLY TO HOUSES & FLATS

- Manufacturer: British Gypsum.
 - Web: www.british-gypsum.com.
 - Email: bgtechnical.enquiries@bpb.com.
 - Product reference: GypWall Classic
- Studs:
 - Type: 70 mm Gypframe 70 S 50 'C' studs.
 - Centres: 600 mm
- Head condition:
 - Type: Gypframe 72 C 50 Floor and Ceiling Channels, fixed to ???.
 - Deflection allowance: TBC.
- Insulation:
 - Type: Isover APR 1200.
 - Thickness: 50 mm.
- Linings:
 - Type: 1 x 12.5 mm Gyproc WallBoard to each side or 15mm Gyproc Wallboard where 30 minute rated.
 - Edge: Tapered edge.
- Finishing:
 - Type: Skim coat plaster finish.
 - Primer/ Sealer: One coat of Gyproc Drywall Prime.
- Accessories: Rigid beads/ stops.
- Other requirements: None.

Partitions forming WC's/Bathrooms/En-suites where tiling is required to wet areas the boards to be moisture resistant.

In accordance with requirement E2(b) of Approved Document E, partitions between bedrooms,

or any room containing a WC and any other room are to have a sound insulation rating of 40dB Rw or better.

This is not a requirement where a wall contains a door or where the wall separates an en-suite from the associated bedroom
- Plywood lining: 15mm plywood to one side under plasterboard to bathrooms and ensuites

125A METAL STUD PARTITION SYSTEM PARTY WALL TO APARTMENT BLOCK A,B & C
Acoustic performance of Party wall to achieve Building Regulations + 5dB
O/All Construction - 286mm TBC

- Manufacturer: British Gypsum.
 - Web: www.british-gypsum.com.
 - Email: bgtechnical.enquiries@bpb.com.
 - Product reference: GypWall QUIET IWL
- Partition thickness: 286mm Nom .
- Studs:
 - Type: 60 mm Gypframe 60 I 50 'I' twin studs.
 - Centres: 600 mm.
- Head condition:
 - Type: Standard Gypframe 62 C 50 floor and ceiling channels, fixed to underside of floor slab.
 - Deflection allowance: 15mm TBC.
- Insulation: 100 mm Isover APR 1200 laid between studs.
- Fire Performance: Refer to Fire Strategy Drawings for Fire Ratings
 Note: Compartment walls all to be 1 hour fire rated to Apartment Blocks.
- Linings:
 - Type: 2 x 15 mm Gyproc SoundBloc to each side.
 - Edge: Tapered edge
- Finishing:
 - Type: Skim coat plaster finish.
 - Primer/ Sealer: One coat of Gyproc Drywall Prime.
- Accessories: Rigid beads/ stops.
- Other requirements: 22mm Gyproc Thermaline Basic board to one side to achieve the thermal performance.

125B METAL STUD PARTITION SYSTEM CORRIDOR & STAIRCASE WALL TO APARTMENT
BLOCK A,B & C

**Acoustic performance of Party wall to achieve Building Regulations + 5dB
O/All Construction - 308mm TBC**

- Manufacturer: British Gypsum.
 - Web: www.british-gypsum.com.
 - Email: bgtechnical.enquiries@bpb.com.
 - Product reference: GypWall QUIET IWL
- Partition thickness: 308mm Nom with Thermaline board .
- Studs:
 - Type: 60 mm Gypframe 60 I 50 'I' twin studs.
 - Centres: 600 mm.
- Head condition:
 - Type: Standard Gypframe 62 C 50 floor and ceiling channels, fixed to underside of floor slab.
 - Deflection allowance: 15mm TBC.
- Insulation: 100 mm Isover APR 1200 laid between studs.
- Linings:
 - Type: 2 x 15 mm Gyproc SoundBloc to each side.
 - Edge: Tapered edge
- Finishing:
 - Type: Skim coat plaster finish.
 - Primer/ Sealer: One coat of Gyproc Drywall Prime.
- Accessories: Rigid beads/ stops.
- Other requirements: 22mm Gyproc Thermaline Basic board to one side to achieve the thermal performance
Corridor Walls: 0.31W/m²k
Staircase Walls / Shear Walls: 0.50W/m²k.

155A WALL LINING SYSTEM TO CONCRETE SHEAR WALLS TO PARTY WALLS OF APARTMENT BLOCK A, B & C

Acoustic performance of Party wall to achieve Building Regulations + 5dB O/All Construction - 395mm TBC

- Manufacturer: British Gypsum
 - Web: www.british-gypsum.com.
 - Email: bgtechnical.enquiries@bpb.com.
- Frame: 85mm Gypliner Universal each side of 175mm concrete shea walls/columns
- Lining: Two layers of 12.5mm Soundbloc each side of wall
- Finish: Tape and jointed with a 3mm skim coat
- Acoustic Insulation: 50mm APR Isover acoustic insulation laid both sides of wall within the Gypliner frame void.
 - All to be installed in accordance with manufacturers recommendations
- Deflection: To be taken into consideration with final details - refer to structural engineer

ALTERNATIVE OPTION:

- Manufacturer: Siniat
- Frame: 60mm C or I studs each side of 175mm concrete shear walls/columns
- Lining: Two layers of 15mm GTEC dB board each side of wall
- Finish: Tape and jointed with a 3mm skim coat
- Acoustic Insulation: 50mm APR Isover acoustic insulation laid both sides of wall within the Gypliner frame void.
 - All to be installed in accordance with manufacturers recommendations
- Deflection: To be taken into consideration with final details - refer to structural engineer

FINAL DETAILS MUST BE APPROVED ACOUSTIC CONSULTANT.

205A LINING ON TIMBER TO ALL WINDOW REVEALS AND SOFFITS

- Background: Battens at 450/660 mm centres.
- Metal resilient (acoustic) bars: Not required.
- Linings: 27.5mm Celotex PL4015 (15+12.5mm) Insulated plasterboard. .
 - Fixing: Contractor's choice.
- Finishing: Skim coat plaster.
 - Primer/ Sealer: As recommended by board manufacturer for vapour control.
 - Accessories: Beads/ stops as clause 692.
- Other requirements: None.

220A PROPRIETARY SUSPENDED CEILING SYSTEM ON METAL FRAMING TO STAIRCASE AND CORRIDIOR TO APARTMENTS

- Manufacturer: British Gypsum, East Leake, Loughborough, Leicestershire LE12 6HX
Website: www.british-gypsum.com, Tel: 0115 945 6123, E-Mail: bgtechnical.enquiries@bpb.com.
British Gypsum contact: (insert name and contact details)
- Product reference: CasoLine MF.
 - Structural soffit: Concrete Soffit.
 - Performance criteria:
 - Fire resistance to BS 476: Part 21 or 22: n/a.
 - Fire protection to steels BS 476: Part 23: n/a.
- Sound insulation:
 - Sound Absorption Class: C.
 - Sound Absorption Coefficient (aw): 0.65
 - Plenum depth: 45mm (min.)
 - Suspension system:
- Hangers: Gypframe MF8 Strap hanger or Gypframe FEA1 Steel angle fixed to soffit via Gypframe MF12 Soffit Cleats.
- Primary grid: Gypframe MF7 channels at 1200mm centres, suspended from hangers at 1200mm centres.
- Secondary grid: Gypframe MF5 sections at 600mm centres, fixed to primary grid with British Gypsum Wafer Head Jack-Point Screws or alternatively Gypframe MF9 clips.
- Perimeters: Gypframe MF6 channel.
 - Lining: One layer 12.5mm Gyptone QUATTRO 41 (as clause K10/410).
Fixing: As clause 591A.
Screws: 25mm British Gypsum Drywall Screws.
 - Cavity insulation: n/a.
Recycled content: n/a.
 - Gyproc Sealant: As clause 516A.
 - Finishing: Taped seamless finish as clause 671A.
 - Services fittings: Install _____
 - Accessories: Gyptone Access Hatch – QUATTRO (510mm x 510mm x 19mm) or Gyproc Profilex access panels as clause 531A – if required.
 - Other requirements:

220B PROPRIETARY SUSPENDED CEILING SYSTEM TO APARTMENT BLOCK PARTY FLOORS A, B, C & D

- Standard: To BS EN 13964.
- Evidence of compliance: All ceilings kits to be CE marked. Submit Declaration of Performance (DoP).
- Manufacturer: British Gypsum Ltd.
 - Product reference: MF Ceiling.
- Lining board: 1 Layer of 12.5 mm plasterboard
Moisture resistant to bathrooms, kitchens, ensuite bathrooms and WC's.
 - Finishing: Skim coat plaster.
 - Accessories: Metal beads/ stops recommended by lining board manufacturer.
- Suspension system:
 - Grid centres: 400 mm.
 - Hangers: Type recommended by board manufacturer.
Length: To give ceiling soffit height above finished floor level of 2400 mm and to achieve a min ceiling void of 150mm.
Centres: As recommended by manufacturer.
Top fixing: To suit structural soffit of 150mm Bison (Soundslab) Hollow Core pre cast planks to
Structural specification OR 200mm Reinforced Concrete slab to Structural engineers specification.
- Insulation: Mineral wool to BS EN 13162; density (minimum) As recommended by Acoustic Consultant kg/m³.
 - Thickness: 50 mm.
- Accessories/ Other requirements: Isoedge flanking surround strips.
Downlighters: Approved downlighters may be installed in accordance with Robut Details to the ceiling but at not more than 1 No per 2sqm of ceiling area in each room, at centres not less than .75 and into openings not exceeding 100mm diameter or 100mm sq.

245A CEILING LINING ON TIMBER TO SEPARATING FLOOR TO HOUSES

This floor achieves Rw 40dB sound reduction for Part E see Test Report L 03-282. O/All Depth: 290mm Overall (dependant on ceiling finish)

22mm (Robust details min 18kg/m²) T&G moisture resistant P5 flooring grade chipboard or other approved decking material
to comply with NHBC Clause 6.4 D14 and to meet the structural requirement

- Background: 253mm Posi-Joists at 600mm centres to contractors design.
- Metal resilient (acoustic) bars: Not required.
- Linings: 12.5 mm plasterboard.
 - Fixings: Screws.
- Finishing: Skim coat plaster.
 - Primer/ Sealer: As recommended by board manufacturer for vapour control.
 - Accessories: Metal beads/ stops recommended by board manufacturer .
- Other requirements: Fire stopping around service penetrations as section P12.

GENERAL/ PREPARATION

305 COMPLIANCE WITH PERFORMANCE REQUIREMENTS

- Testing/ Assessment: Submit UKAS accredited laboratory reports for the following: Fire resistance: Partitions (including deflection heads and doorsets) and suspended ceilings (including access units)..
- Materials, components and details: As used in testing/ assessment reports. If discrepancies arise, give notice.

325 PREPARATION OF MASONRY TO RECEIVE WALL LININGS

- General: Suitable to receive lining system. Redundant fixtures and services removed. Cutting, chasing and making good completed.
- Holes, gaps, service penetrations, perimeter junctions and around openings: Seal.
- Adhesive fixings: Prepare substrate to achieve effective bonding.
 - Contaminants: Remove loose material, dirt, grease, oil, paper, etc.
 - Absorption: Control by dampening, priming or applying bonding agents as necessary.

335 ADDITIONAL SUPPORTS

- Framing: Accurately position and securely fix to give full support to:
 - Partition heads running parallel with, but offset from main structural supports.
 - Fixtures, fittings and service outlets. Mark framing positions clearly and accurately on linings.
 - Board edges and lining perimeters, as recommended by board manufacturer to suit type and performance of lining.

375 NEW WET LAID BASES

- Dpcs: Install under full width of partitions/ freestanding wall linings.
 - Material: Bituminous sheet or plastics.

COMPONENTS

400 GYPSUM BOARDS GENERALLY

- Standard:
 - Gypsum plasterboard to BS EN 520.
 - Gypsum fibre board to BS EN 15283-2.
 - Evidence of compliance: All sheets to be CE marked. Submit Declaration of Performance (DoP).

401 GYPSUM PLASTERBOARD

- Type: To BS EN 520, type A.
- Core density (minimum): 650 kg/m³.
- Reaction to fire: Manufacturer's standard.
- Water vapour resistance factor: Manufacturer's standard.
- Thermal conductivity: Manufacturer's standard.
- Other BS EN 520 characteristics: None.
- Recycled content: Submit proposals.
- Exposed surface and edge profiles: Suitable to receive specified finish.

430A ACCESS PANELS LOFT ACCESS HATCH TO HOUSES

- Manufacturer: Profilex Access Panel Solutions
- Product: Profilex Loft Hatch Panel
- Product code: 5200115198
- Application: First floor ceiling loft hatch within stair hallway
- Performance: Non-fire rated
- Lock: Touch catch
- Frame: Beaded frame for taping and skim finish for non-visible frame.
- Door: Flush metal door, primed white to allow on-site decoration.
- Size (mm): 540x540

432 METAL STUDS

- Manufacturer: Contractor's choice .
- Product reference: Submit proposals .

INSTALLATION

435 DRY LININGS GENERALLY

- General: Use fixing, jointing, sealing and finishing materials, components and installation methods recommended by board manufacturer.
- Cutting gypsum boards: Neatly and accurately without damaging core or tearing paper facing.
 - Cut edges: Minimize and position at internal angles wherever possible. Mask with bound edges of adjacent boards at external corners.
- Fixings boards: Securely and firmly to suitably prepared and accurately levelled backgrounds.
- Finishing: Neatly to give flush, smooth, flat surfaces free from bowing and abrupt changes of level.

445 CEILINGS

- Sequence: Fix boards to ceilings before installing dry lined walls and partitions.
- Orientation of boards: Fix with bound edges at right angles to supports and with ends staggered in adjacent rows.
- Two layer boarding: Stagger joints between layers.

455 METAL FRAMING FOR PARTITIONS/ WALL LININGS

- Setting out: Accurately aligned and plumb.
 - Frame/ Stud positions: Equal centres to suit specified linings, maintaining sequence across openings.
 - Additional studs: To support vertical edges of boards.
- Fixing centres at perimeters (maximum): 600 mm.
- Openings: Form accurately.
 - Doorsets: Use sleeved or boxed metal studs and/ or suitable timber framing to achieve strength grade requirements for framing assembly and adequately support weight of door.
 - Services penetrations: Allow for associated fire stopping.

475 METAL FURRINGS FOR WALL LININGS

- Setting out: Accurately aligned and plumb.
 - Vertical furring positions: Equal vertical centres to suit specified linings, maintaining sequence across openings. Position adjacent to angles and openings.
 - Additional vertical furrings: To support vertical edges of boards and at junctions with partitions.
 - Horizontal furring positions: To provide continuous support to edges of boards.
- Adhesive bedding to furrings:
 - Dabs: Length 200 mm (minimum). Located at ends of furrings and thereafter at 450 mm (maximum) centres.
 - Junctions with partitions: Continuous bed with no gaps across cavity.

485 SUSPENDED CEILING GRIDS

- Setting out: Accurately aligned and level.
 - Grid members and hangers: Centres to suit specified linings and imposed loads.
 - Additional grid members: Provide bracing and stiffening at upstands, partition heads, access hatches, etc.
- Fixing: Securely at perimeters, grid joints, top and bottom hanger fixings.

505 INSTALLING MINERAL WOOL INSULATION

- Fitting insulation: Closely butted joints and no gaps. Use fasteners to prevent slumping or displacement.
- Services:
 - Electrical cables overlaid by insulation: Sized accordingly.
 - Ceilings: Cut insulation around electrical fittings, etc.

510 SEALING GAPS AND AIR PATHS

- Location of sealant: To perimeter abutments and around openings.
 - Pressurized shafts and ducts: At board-to-board and board-to-metal frame junctions.
- Application: To clean, dry and dust free surfaces as a continuous bead with no gaps.
 - Gaps greater than 6 mm between floor and underside of gypsum board: After sealing, fill with jointing compound.

530 CAVITY FIRE BARRIERS WITHIN PARTITIONS/ WALL LININGS

- Metal framed systems:
 - Material: Plasterboard 12.5 mm thick.
 - Installation: Form accurately and fix securely with no gaps to provide a complete barrier to smoke and flame.
- Adhesive fixed wall lining systems:
 - Material: Adhesive compound.
 - Installation: Form in a continuous line with no gaps to provide a complete barrier to smoke and flame.

545 CAVITY FIRE BARRIERS WITHIN SUSPENDED CEILINGS

- Type: As recommended by board manufacturer to meet specified.
- Fire resistance: To BS 476-20, 30/15 minutes (Integrity/ Insulation).
- Ceiling void subdivision: Fix barriers not more than 20 m apart in any direction.
- Fixing at perimeters and joints: Secure, stable and continuous with no gaps, to provide a complete barrier to smoke and flame.
- Service penetrations: Cut and pack to maintain barrier integrity. Sleeve flexible materials. Adequately support services passing through barrier.
- Ceiling systems for fire protection: Do not impair fire resisting performance of ceiling system.

555 FIRE STOPPING AT PERIMETERS OF DRY LINING SYSTEMS

- Material: Tightly packed mineral wool or intumescent mastic/ sealant.
- Application: To perimeter abutments to provide a complete barrier to smoke and flame.

560 JOINTS BETWEEN BOARDS

- Tapered edged gypsum boards:
 - Bound edges: Lightly butted.
 - Cut/ unbound edges: 3 mm gap.
- Square edged plasterboards: 3 mm gap.
- Square edged gypsum fibre boards: 5 mm gap.

565 VERTICAL JOINTS

- Joints: Centre on studs.
 - Partitions: Stagger joints on opposite sides of studs.
- Two layer boarding: Stagger joints between layers.

570 HORIZONTAL JOINTS

- Surfaces exposed to view: Horizontal joints not permitted. Seek instructions where height of partition/ lining exceeds maximum available length of board.
- Two layer boarding: Stagger joints between layers by at least 600 mm.
- Edges of boards: Support using additional framing.
 - Two layer boarding: Support edges of outer layer.

575 PLANK PLASTERBOARD

- First layer in two layer boarding: Square edged with long edges at right angles to studs.

580 INSULATION BACKED PLASTERBOARD

- General: Do not damage or cut away insulation to accommodate services.
- Installation at corners: Carefully cut back insulation or plasterboard as appropriate along edges of boards to give a continuous plasterboard face, with no gaps in insulation.

590 FIXING GYPSUM BOARD TO METAL FRAMING/ FURRINGS

- Partitions/ Wall linings: Fix securely and firmly at the following centres (maximum):
 - Single layer boarding: To all framing at 300 mm centres. Reduce to 200 mm centres at external angles.
 - Multi-layer boarding: Face layer at 300 mm centres, and previous layers around perimeters at 300 mm centres.
- Ceilings: 230 mm. Reduce to 150 mm at board ends and at lining perimeters.
- Position of screws from edges of boards (minimum): 10 mm.
 - Screw heads: Set in a depression. Do not break paper or gypsum core.

592 FIXING INSULATION BACKED PLASTERBOARD TO METAL FURRINGS

- Fixing to furrings: In addition to screw fixings apply continuous beads of adhesive sealant to furrings.

595 DEFLECTION HEADS

- Fixing boards: Do not fix to head channels.

610 FIXING GYPSUM BOARD TO TIMBER

- Fixing to timber: Securely at the following centres (maximum):
 - Nails: 150 mm.
 - Screws to partitions/ wall linings: 300 mm. Reduce to 200 mm at external angles.
 - Screws to ceilings: 230 mm.
- Position of nails/ screws from edges of boards (minimum):
 - Bound edges: 10 mm.
 - Cut/ unbound edges: 13 mm.
- Position of nails/ screws from edges of timber supports (minimum): 6 mm.

620 FIXING GYPSUM BOARD WITH ADHESIVE DABS

- Setting out boards: Accurately aligned and plumb.
- Fixing to substrates: Securely using adhesive dabs.
- Adhesive dab spacings for each board:
 - Horizontally: One row along top edge and one continuous dab along bottom edge.
 - Vertically: One row along each edge and thereafter at intermediate spacings to suit size of board:

Thickness (mm)	Width (mm)	Dab centres (mm)
9.5	1200	400
9.5/12.5		900 450
12.5	1200	600
- Adhesive dab dimensions (width x length): At least 50-75 mm x 250 mm.
- Position of dabs from edges/ ends of boards (minimum): 25 mm.

- 625 **FIXING INSULATION BACKED PLASTERBOARD WITH ADHESIVE DABS**
- Fixing to substrates: In addition to adhesive dab fixings, secure boards with nailable plugs in locations recommended by board manufacturer.
- 630 **FIXING INSULATION BACKED PLASTERBOARD WITH ADHESIVE SPOTS**
- Setting out boards: Accurately aligned and plumb.
 - Fixing to substrates: Securely using adhesive spots and mechanical fastenings.
 - Adhesive spot spacings to each board: Four vertical rows, at 400 mm centres in each row.
 - Adhesive spot diameters (minimum): 25 mm.
 - Mechanical fasteners: Nailable plugs in locations recommended by board manufacturer.

FINISHING

- 650 **LEVEL OF DRY LINING ACROSS JOINTS**
- Sudden irregularities: Not permitted.
 - Joint deviations: Measure from faces of adjacent boards using methods and straightedges (450 mm long with feet/ pads) to BS 8212, clause 3.3.5.
 - Tapered edge joints:
Permissible deviation (maximum) across joints when measured with feet resting on boards: 3 mm.
 - External angles:
Permissible deviation (maximum) for both faces: 4 mm.
 - Internal angles:
Permissible deviation (maximum) for both faces: 5 mm.
- 670 **SEAMLESS JOINTING TO GYPSUM BOARDS**
- Cut edges of boards: Lightly sand to remove paper burrs.
 - Filling and taping: Fill joints, gaps and internal angles with jointing compound and cover with continuous lengths of paper tape, fully bedded.
 - Protection of edges/ corners: Reinforce external angles, stop ends, etc. with specified edge/ angle bead.
 - Finishing: Apply jointing compound. Feather out each application beyond previous application to give a flush, smooth, seamless surface.
 - Nail/ screw depressions: Fill with jointing compound to give a flush surface.
 - Minor imperfections: Remove by light sanding.
- 680 **SKIM COAT PLASTER FINISH**
- Plaster type As recommended by board manufacturer..
 - Thickness: 2-3 mm.
 - Joints: Fill and tape except where coincident with metal beads.
 - Finish: Tight, matt, smooth surface with no hollows, abrupt changes of level or trowel marks.
- 692 **RIGID BEADS/STOPS**
- Internal: To BS EN 13658-1.
 - External: To BS EN 13658-2.
- 695 **INSTALLING BEADS/ STOPS**
- Cutting: Neatly using mitres at return angles.
 - Fixing: Securely using longest possible lengths, plumb, square and true to line and level, ensuring full contact of wings with substrate.
 - Finishing: After joint compounds/ plasters have been applied, remove surplus material while still wet from surfaces of beads exposed to view.

725 REPAIRS TO EXISTING GYPSUM BOARD

- Filling small areas with broken cores: Cut away paper facing, remove loose core material and fill with jointing compound.
 - Finish: Flush, smooth surface suitable for redecoration.
- Large patch repairs: Cut out damaged area and form neat hole with rectangular sides. Replace with matching gypsum board.
 - Fixing: Use methods to suit type of dry lining, ensuring full support to all edges of existing and new gypsum board.
 - Finishing: Fill joints, tape and apply jointing compound to give a flush, smooth surface suitable for redecoration.

DRAFT

K11

**Rigid sheet flooring/ sheathing/ decking/ sarking/
linings/ casings**

K11 Rigid sheet flooring/ sheathing/ decking/ sarking/ linings/ casings

110 WOOD-BASED SHEETS GENERALLY

- Standard: To BS EN 13986.
 - Evidence of compliance: All sheets to be CE marked. Submit Declaration of Performance (DoP).

285 SEPARATING FLOOR SYSTEM TO HOUSESGENERALLY TO HOUSES

Refer to DLA Drawings 1132-45 & 46

Floor to achieve acoustic performance of 40dB in line with Building Regulations Part E Table 0.1b and NHBC requirements

Substrate: 253mm Posi-Joists at 600mm centres.

- Ceiling/ Soffit:
 - Lining: 12.5mm plasterboard - Refer to K10.
- Insulation/ Pugging: 100mm min mineral wool quilt insulation (10-36kg/m3) between joists - 100mm Isover Acoustic Partition Roll APR 1200. .
 - Recycled content: None permitted .
- Flooring: 22mm (Robust details min 18kg/m²) T&G moisture resistant P5 flooring grade chipboard or other approved decking material to comply with NHBC Clause 6.4 D14 and to meet the structural requirements of BS EN 1995-1-1 .
 - Recycled content: None permitted .

815A PLYWOODPATTRESSES

- Allow for Plywood pattresses to kitchen area, 400mm plywood noggins behind wall units. Allow for 1800mm plywood sheet within partitions for bathroom fittings.

WORKMANSHIP

910 INSTALLATION GENERALLY

- Timing: Building to be weathertight before fixing boards internally.
- Moisture content of timber supports (maximum): 18%.
- Joints between boards: Accurately aligned, of constant width and parallel to perimeter edges.
- Methods of fixing, and fasteners: As section Z20 where not specified otherwise.

915 DRYNESS OF CONCRETE/ SCREED SUBSTRATES FOR FLOATING FLOORS

- Relative humidity above substrate when tested with a hygrometer to BS 8201, Appendix A (maximum): 75%.
 - Test points: All corners, around perimeter, and random points over area being tested.
 - Drying aids: Turned off for not less than 4 days before testing.

920 VAPOUR CONTROL LAYER IN FLOATING FLOOR CONSTRUCTION

- Location: Immediately below floating layer.
- Installation:
 - Joints: Overlapped by minimum 150 mm and sealed with vapour resistant tape.
 - Treatment of membrane at perimeter of flooring and upstands: Turned up and sealed to top face of flooring using a method approved by the board manufacturer.
- Excess material: Trimmed off neatly after fixing skirtings/ cover beads.
- Condition of membrane before laying flooring: Clean and dry.

930 ADDITIONAL SUPPORTS

- Additional studs, noggings/ dwangs (Scot) and battens:
 - Provision: In accordance with board manufacturer's recommendations and as follows:
Tongue and groove jointed rigid board areas: To all unsupported perimeter edges.
Butt jointed rigid board areas: To all unsupported edges.
 - Size: Not less than 50 mm wide and of adequate thickness.
 - Quality of timber: As for adjacent timber supports.
 - Treatment (where required): As for adjacent timber supports.

940 BOARD MOISTURE CONTENT AND CONDITIONING

- Moisture content of boards at time of fixing: Appropriate to end use.
- Conditioning regime: Submit proposals.

950 MOISTURE CONTENT TESTING

- Test regime and equipment: Submit proposals.
- Test results: Submit record of tests and results.

960 FIXING GENERALLY

- Boards/ sheets: Fixed securely to each support without distortion and true to line and level.
- Fasteners: Evenly spaced in straight lines and, unless otherwise recommended by board manufacturer, in pairs across joints.
 - Distance from edge of board/ sheet: Sufficient to prevent damage.
- Surplus adhesive: Removed as the work proceeds.

975 METAL WALL FRAMING

- Setting out: Framing accurately aligned, vertical and securely fixed to surrounding structure at maximum 600 mm centres. All board edges supported.

976 MINERAL WOOL INSULATION TO METAL FRAMING

- Installation: Neat and secure with close butted joints and no gaps. Where insulation is not self supporting, fixed at head of frame using clips or other suitable proprietary fixings.

980 OPEN JOINTS

- Perimeter joints, expansion joints and joints between boards: Free from plaster, mortar droppings and other debris.
- Temporary wedges and packings: Removed on completion of board fixing.

990 ACCESS PANELS

- Size and position: Agree before boards are fixed.
- Additional noggings/ dwangs (Scot), battens, etc: Provide and fix as necessary.

DRAFT

K21

Wood strip/ board fine flooring/ linings

K21 Wood strip/ board fine flooring/ linings

TYPES OF FLOORING/ LINING

Refer to WYG 'Residential Demise and Fittings Schedule' and Careys 'Outline Specification'

Any variations to be highlighted.

ALL ITEMS TO BE FULLY REVIEWED BEFORE SPECIFICATION CONFIRMED

146A LAMINATE FLOORING GENERALLY

- Substrate: Screed.
- Underlayer: Acoustic Underlay 4mm thick.
 - Recycled content: Submit proposals.
- Flooring: Liverpool Oak FinFloor Fiesta AC3, Narrow board 7mm.
 - Accessories: As recommended.
- Other requirements: tbc.

GENERAL/ PREPARATION

210 WORKMANSHIP GENERALLY

- Moisture content of timber supports: 12-14%.
- Methods of fixing and fasteners: As section Z20 where not specified.
- Protection: Protect from dirt, stains and damage using suitable coverings and boards laid as the work proceeds.

220 ENVIRONMENTAL CONDITIONS

- General requirements prior to starting work specified in this section: Building weathertight, wet trades completed and affected areas dried out.
- Temperature and humidity before, during and after installing strips/ boards: Maintained at levels approximating to those which will prevail after building is occupied.

230 HEATING SYSTEM

- Operating mode: Continuous.
- Room temperatures for which the system has been designed: tbc.
- Operation up to Completion: Submit proposals.

250 FIXTURES

- Fixtures around which strip flooring is to be fixed: Installed before starting work specified in this section.

260 DRYNESS OF CONCRETE/ SCREED SUBSTRATES FOR FLOORING

- Relative humidity above substrate when tested with a hygrometer to BS 8201, Appendix A (maximum): 75%.
 - Test points: All corners, around perimeter, and random points over area being tested.
- Drying aids: Turned off for not less than four days before testing.

270 STRIP/ BOARD MOISTURE CONTENT TESTING

- Test regime and equipment: Submit proposals.
- Test results: Submit record of tests and results.

FIXING/ FINISHING

310 VAPOUR CONTROL LAYER INSTALLATION

- Location: Immediately below tbc .
- Installation:
 - Joints: Overlapped by minimum 150 mm and sealed with vapour resistant tape.
 - Treatment of membrane at perimeter of flooring and upstands: Turned up and sealed to top face of flooring using a method approved by the strip/ board manufacturer.
- Excess material: Trimmed off neatly after fixing skirtings/ cover beads.
- Condition of membrane before laying flooring: Clean and dry.

335 TREATED TIMBER

- Surfaces exposed by minor cutting and drilling: Treated with two flood coats of a solution recommended for the purpose by main treatment solution manufacturer.

340 ACCESS PANELS

- Size and position: Agree before strips/ boards are fixed.
- Additional noggings/ dwangs (Scot), battens, etc: Provide and fix as necessary.

350 FIXING STRIPS/ BOARDS

- Strips/ Boards: Fixed securely to each support with flat, true surfaces free from undulations, splits, hammer marks, scratches and protruding fastenings.
- Movement of timber: Allowed for when positioning strips/ boards and fastenings to prevent cupping, springing, opening of joints or other defects.
- Heading joints (where permitted): End matched, butted and, where applicable, positioned centrally over supports and distributed across the flooring to achieve a random effect.
- Surface fixing: tbc.

360 EXPANSION PROVISION

- Expansion gaps:
 - Edges of flooring: Parallel to lie of strips/ boards and As recommended by manufacturer mm wide.
 - Ends of flooring: 10 mm wide.
- Spacer blocks and debris: Removed before fixing skirtings/ cover fillets.
- Intermediate expansion/ movement joints: Formed as recommended by flooring manufacturer/ supplier.

DRAFT

L
Windows/Doors/Stairs

DRAFT

L10

Windows/ Rooflights/ Screens/ Louvres

L10 Windows/ Rooflights/ Screens/ Louvres

To be read with Preliminaries/ General conditions.

GENERAL

110 EVIDENCE OF PERFORMANCE

- Certification: Provide independently certified evidence that all incorporated components comply with specified performance requirements.

120 SITE DIMENSIONS

- Procedure: Before starting work on designated items take site dimensions, record on shop drawings and use to ensure accurate fabrication.
- Designated items: All.

155 VIEW OUT

- Windows/ opening sizes and position: Design to meet BREEAM 'View out' criteria for relevant building type.
- Submit design plan and elevation drawings showing the following:
 - All BREEAM defined 'relevant areas' dependent on building type and room depths.
 - Actual or notional workstation/ desk layouts.
 - Window/ open areas.
- Submit site plan showing: Building location and proximity to external obstructions.

160 POTENTIAL FOR NATURAL VENTILATION

- Submit design plan and elevation drawings, and calculations confirming the following: Locations of openings.

PRODUCTS

205 WINDOW MATERIALS SPECIFICATION

- Minimum BRE 'Green Guide to Specification Online' rating: A.

330A ALUMINIUM WINDOWS GENERALLY

Refer to DLA drawings 1132-50 indicating Window Types

Supporting structure:
As Drawings: Blocks A – D (Units 1-78; Semi's (79-92) & Terraces (102-107).

• Window System:

- Manufacturer: AluK (GB) Ltd.
- Address: Newhouse Farm Ind Est, Chepstow, NP16 6UD
- Telephone: 01291 639 739
- Web: www.aluk.com
- Contact: Nigel Gilbert -AluK Project Consultant (London & SE) m: 07976 180 333
nigel.gilbert@aluk.com
- Product reference: 58BW ST (Sash Option AW685 preferred) + 58BW Commercial (Sash Option AW620 where over ST Limits) Thermally Broken Window System.
 - Type: 58mm wide window system, pressure equalised, drained and ventilated.
 - AluK profile sections:
 - AW605 wide outer frame (AW608 outer frame at window head for trickle ventilation required) - Sub-Contractor to Clarify Specifiers Ventilation requirements, may be Achieved.
 - AW631/632 mullion/transom (dependent upon design, application and structural calculations) - FRENCH CASEMENTS TO Semi's (79-92) - using AW631/632 mullion attached to slave leaf; both leaves locked by handle - 58BW ST (Sash Option AW685 preferred) opening sash (dependent upon design, application and structural calculations) or
 - 58BW Commercial (Sash Option AW620 where over ST Limits) opening sash (dependent upon design, application and structural calculations) - AW660 square bead (dependent upon design, application and structural calculations)
 - Windows to be coupled with appropriate coupling profile.
- Standard: System and fabricator certification to secure by design to meet PAS24 and BS6375 requirements.
 - Description: Outward opening vent
 - Beading: All opening lights to be internally beaded.
All fixed lights are to be externally beaded as standard.
- Finish: Polyester powder coating to BS EN 12206-1:2004.
Colour/ texture: AluK standard colour range, RAL7016 - Matt finish.
Minimum film thickness: A minimum 60 micrometres.
- Glazing: Hermetically sealed 28mm insulating glass units to achieve a maximum 1.0W/m²k centre pane. Glazing thermal safety checks are recommended to ensure project suitability, these are to be carried out by the glazing manufacturer.
 - Inner pane: 6mm Clear Toughened / 6,4mm Laminated (SBD) - thickness TBC.
 - Outer pane: 6mm Clear Toughened - thickness TBC.
- Ironmongery/ Accessories: Open outward vent with standard Easy Clean / Egress Hinge Range hinges (58BW ST) or AluK heavy duty friction hinges 58BW Commercial (Sash Option AW620 where over ST Limits) and Aluk locking handle with appropriate locking to PAS 24 requirements.
- Accessories: All aluminium flashings, cills and other perimeter trims necessary as indicated within the drawings. It is the subcontractor's responsibility to ensure that these are designed and installed in accordance with AluK's instructions and guidelines. -
DESIGN TO BE DEVELOPED
- Other requirements: EPDM to be provided to allow system to be sealed to VCL for weather and air tightness. Max. air permeability 5m³/m².h
The AluK 58BW window system shall be fabricated and installed by AluK authorised dealers only, in complete accordance with the information and guidance released by AluK (GB) Ltd.

335A ALUMINIUM WINDOWS GENERALLY

Refer to DLA drawings 1132-50 indicating Window Types

- Standard: To BS 4873.
- Exposure category to BS 6375-1/ Design wind load: To be advised by Structural engineer.
- Finish as delivered: Polyester powder coating to BS EN 12206-1:2004.
- Thermal performance (U-value maximum): A maximum 1.5 W /m²k for overall frame and glazing. Area weighted average.
- Glazing details: Double glazed units incorporating low emissivity glass and argon gas filled cavity..
 - Beading: All opening lights to be internally beaded and all fixed lights to be externally beaded in accordance with AluK's secured by design test evidence..
- Ironmongery/ Accessories: The AluK window system shall be fabricated and installed by AluK authorised dealers only, in complete accordance with the information and guidance released by AluK (GB) Ltd..
- Fixing: Screwed to masonry reveal as clause 782.

460 ROOFLIGHTS

- Manufacturer: _____.
- Product reference: _____.
- Type: _____.
- Frame: _____.
- Finish: _____.
- Colour: _____.
- Kerb: _____.
- Thermal performance (U-value maximum): _____.
- Fire performance (minimum): _____.
- Glazing details: _____.
- Other requirements: _____.
- Fixing: _____.

480 ROOF WINDOWS

- Manufacturer: _____.
- Product reference: _____.
- Thermal performance (U-value maximum): _____.
- Fire performance (minimum): _____.
- Glazing details: _____.
- Beading: _____.
- Flashings: By window manufacturer to suit _____.
- Accessories: _____.
- Fixing: _____.

490 ROOF VENTILATORS

- Manufacturer: _____.
- Product reference: _____.
- Type: _____.
- Controls: _____.
- Materials: _____.
- Finish as delivered: _____.
- Seals: _____.
- Guards: _____.
- Typical U-value: _____.
- Fire/ smoke control performance: _____.
- Accessories/ Special features: _____.
- Fixing: _____.

EXECUTION

710 PROTECTION OF COMPONENTS

- General: Do not deliver to site components that cannot be installed immediately or placed in clean, dry floored and covered storage.
- Stored components: Stack vertical or near vertical on level bearers, separated with spacers to prevent damage by and to projecting ironmongery, beads, etc.

730 PRIMING/ SEALING

- Wood surfaces inaccessible after installation: Prime or seal as specified before fixing components.

740 CORROSION PROTECTION

- Surfaces to be protected: All exposed aluminium surfaces..
- Protective coating: Two coats of bitumen solution to BS 6949 or an approved mastic impregnated tape.
Timing of application: Before fixing components.

750 BUILDING IN

- General: Not permitted unless indicated on drawings.
 - Brace and protect components to prevent distortion and damage during construction of adjacent structure.

760 REPLACEMENT WINDOW INSTALLATION

- Standard: To BS 8213-4.

765 WINDOW INSTALLATION GENERALLY

- Installation: Into prepared openings.
- Gap between frame edge and surrounding construction:
 - Minimum: 6mm.
 - Maximum: 10mm.
- Distortion: Install windows without twist or diagonal racking.

766 LOCATION OF OPENABLE WINDOWS IN NATURALLY VENTILATED BUILDINGS

- Location: Over 10 m from sources of external pollution.

782 FIXING OF ALUMINIUM FRAMES

- Standard: As section Z20.
- Fasteners: As AluK (GB) Ltd's recommendation..
 - Spacing: When not predrilled or specified otherwise, position fasteners not more than 250 mm from ends of each jamb, adjacent to each hanging point of opening lights, and at maximum 600 mm centres.

810 SEALANT JOINTS

- Sealant:
 - Manufacturer: Contractor's choice.
 - Product reference: Contractor's choice.
 - Colour: To match window frames.
 - Application: As section Z22 to prepared joints. Finish triangular fillets to a flat or slightly convex profile.

820 IRONMONGERY

- Fixing: In accordance with any third party certification conditions applicable. Assemble and fix carefully and accurately using fasteners with matching finish supplied by ironmongery manufacturer. Do not damage ironmongery and adjacent surfaces.
- Checking/ Adjusting/ Lubricating: Carry out at Completion and ensure correct functioning.

DRAFT

L20

Doors/ shutters/ hatches

L20 Doors/ shutters/ hatches

GENERAL

Refer to WYG 'Residential Demise and Fittings Schedule' and Careys 'Outline Specification'

Any variations to be highlighted - ALL ITEMS TO BE FULLY REVIEWED BEFORE SPECIFICATION CONFIRMED

110 EVIDENCE OF PERFORMANCE

- Certification: Provide independently certified evidence that all incorporated components comply with specified performance requirements.

112 TIMBER PROCUREMENT

- Timber (including timber for wood-based products): Obtained from well-managed forests and/ or plantations in accordance with:
 - The laws governing forest management in the producer country or countries.
 - International agreements such as the Convention on International Trade in Endangered Species of wild fauna and flora (CITES).
- Documentation: Provide either:
 - Documentary evidence (which has been or can be independently verified) regarding the provenance of all timber supplied.
 - Evidence that suppliers have adopted and are implementing a formal environmental purchasing policy for timber and wood-based products.
- Certification scheme: Forest Stewardship Council (FSA).
 - Other evidence: UK Timber procurement policy Category B evidence: Completed supply chain information within attached proforma.

115 FIRE RESISTING DOORS/ DOOR ASSEMBLIES/ DOORSETS

- Door products: As defined in BS EN 12519.
- Evidence of fire performance: Provide certified evidence, in the form of a product conformity certificate, directly relevant fire test report or engineering assessment, that each door/ door assembly/ doorset supplied will comply with the specified requirements for fire or smoke resistance if tested to BS 476-22, BS EN 1634-1 or BS EN 1634-3. Such certification must cover door and frame materials, glass and glazing materials and their installation, essential and ancillary ironmongery, hinges and seals.
- Components, assemblies or sets will be marked to the relevant product standard and/ or third party certification rating.

120 NON FIRE RESISTING DOORS/ DOOR ASSEMBLIES/ DOORSETS

- Provide certified evidence, in the form of a product conformity certificate or engineering assessment, that each door/ doorset/ assembly supplied will comply with the specified requirements to BS EN 14351-1. Such certification must cover door and frame materials, glass and glazing materials and their installation, essential and ancillary ironmongery, hinges and seals.
- Components and assemblies will be marked to the relevant product standard and/ or third party certification rating.

150 SITE DIMENSIONS

- Procedure: Before starting work on designated items take site dimensions, record on shop drawings and use to ensure accurate fabrication.
- Designated items: All doors.

170 CONTROL SAMPLES

- Procedure:
 - Finalize component details.
 - Fabricate one of each of the following designated items as part of the quantity required for the project.
 - Obtain approval of appearance and quality before proceeding with manufacture of the remaining quantity.
- Designated items: Internal Timber Door.

PRODUCTS

230 WOOD FLUSH DOORS_____

- Manufacturer: _____.
- Product reference: _____.
- Facings: _____.
- Lippings: _____.
- Preservative treatment: _____.
- Finish as delivered: _____.
- Glazing/ Infill details: _____.
- Manifestation: _____.
- Beading: _____.
- Thermal performance (U-value maximum): _____.
- Other requirements: _____.

310 WOOD DOOR FRAMES_____

- Manufacturer: _____.
- Product reference: _____.
- Species: _____.
- Preservative treatment: _____.
- Finish as delivered: _____.
- Perimeter seals: _____.
- Thermal performance: _____.
- Fixing: _____.

330 WOOD DOOR FRAMES_____

- Materials: Generally to BS EN 942.
 - Species: _____.
 - Appearance class: _____.
- Assembly:
 - Adhesive: _____.
 - Joinery workmanship: As section Z10.
- Preservative treatment: _____.
- Moisture content on delivery: _____.
- Finish as delivered: _____.
- Perimeter seals: _____.
- Thermal performance (U-value maximum): _____.
- Fixing: _____.

370 DOOR FRAMES_____

- Manufacturer: _____.
- Product reference: _____.
- Perimeter seals: _____.
- Finish as delivered: _____.
- Thermal performance (U-value maximum): _____.
- Fixing: _____.

- 410 WOOD DOORSETS_____
- Manufacturer: _____.
 - Product reference: _____.
 - Door leaf:
 - Facings: _____.
 - Lippings: _____.
 - Finish as delivered: _____.
 - Frame and architraves:
 - Wood species: _____.
 - Finish as delivered: _____.
 - Preservative treatment: _____.
 - Glazing/ Infill details: _____.
 - Manifestation: _____.
 - Beading: _____.
 - Ironmongery: _____.
 - Perimeter seals: _____.
 - Thermal performance (U-value maximum): _____.
 - Other requirements: _____.
 - Fixing: _____.

480A DOORSETSALUMINIUM - ENTRANCE DOORS

- Manufacturer: AluK (GB) Ltd.
Address: Newhouse Farm Ind Est, Chepstow, NP16 6UD
Telephone: 01291 639 739
Web: www.aluk.com
Contact: Nigel Gilbert -AluK Project Consultant (London & SE) m: 07976 180 333
nigel.gilbert@aluk.com
 - Product reference: GT55TB thermally broken commercial door system. .
- Door leaf: Generally - Fully glazed within thermally broken GT622 anti finger trap (AFT) stile & GT620 standard stile, GT600 top rail & GT604 bottom rail. .
 - Finish as delivered: Polyester powdercoated .
- Standard: System and fabricator certification to secure by design to meet PAS24 and BS6375 requirements.
 - Description: Bi-Directional Door – Single / Double within Screen.
 - Manifestation: TBC .
 - Beading: Doors internally beaded.
Screen fixed lights externally beaded as standard. .
 - Frame and architraves: Generally, thermally broken - GT202 jambs, GT325 header bar and GT660 thermally broken low threshold.
 - Finish: Polyester powder coating to BS EN 12206-1:2004.
Colour/ texture: AluK standard colour range, RAL7016 - Matt finish
Minimum film thickness: A minimum 60 micrometres.
- Glazing / Panel Details: Hermetically sealed 28mm (Door Stile); Insulating glass units to achieve a maximum 1.0W/m2k centre pane.
 - Inner pane: 6mm Clear Toughened / 6,4mm Laminated (SBD) - thickness TBC.
 - Outer pane: 6mm Clear Toughened - thickness TBC.
 - Manifestation: As drawing.

Ironmongery: MANUAL OPTION: AluK XA616 back to back pull handles polyester powder coated in contrasting RAL colour to door sections with concealed Sentinel M multi-point locking. Doors to provide secured by design certification and PAS24 compliance ELECTRONIC OPTION: AluK XA616 back to back pull handles polyester powder coated in contrasting RAL colour to door sections with concealed Trimec electronic V lock to be operated via external key pad and internal push pad for day to day activity. V-lock to be linked to fire alarm with fail safe activation in the event of a fire. Doors to provide secured by design certification and PAS24 compliance. .
- Thermal performance (U-value maximum): 1.7 W/m²K .
- Other requirements: AFT stiles with pivot operation, thermally broken profiles and Dorma closer.
EPDM to be provided to allow system to be sealed to VCL for weather and air tightness. Max. air permeability 5m3/m2.h The AluK GT55TB door system shall be fabricated and installed by AluK authorised dealers only, in complete accordance with the information and guidance released by AluK (GB) Ltd. .
- Accessories: All aluminium flashings, cills and other perimeter trims necessary as indicated within the drawings. It is the subcontractor's responsibility to ensure that these are designed and installed in accordance with AluK's instructions and guidelines .

680 UP AND OVER GARAGE DOORS

- Manufacturer: Garador, Bunford Lane, Yeovil, Somerset, BA20 2YA
Tel +44 (0)1935 443722
Fax +44 (0)1935 443744
www.garador.co.uk.
- Product reference: Ascot door type, steel up and over door Size 2259mm wide X 2136mm high Standard Size..
- Type: One piece canopy up and over, insulated core.
- Performance: n/a.
- Finish as delivered: Polyester powder coated, colour to be confirmed.
- Operation: Manual.
- Other requirements: Secure Euro cylinder locks with solid locking rods
Handles: Colour Black.

EXECUTION

710 PROTECTION OF COMPONENTS

- General: Do not deliver to site components that cannot be installed immediately or placed in clean, dry, floored and covered storage.
- Stored components: Stacked on level bearers, separated with spacers to prevent damage by and to projecting ironmongery, beads, etc.

730 PRIMING/ SEALING

- Wood surfaces inaccessible after installation: Primed or sealed as specified before fixing components.

740 CORROSION PROTECTION

- Surfaces to be protected: All exposed aluminium bare edges .
- Protective coating: Two coats of bitumen solution to BS 6949 or an approved mastic impregnated tape.
- Timing of application: Before fixing components.

750 FIXING DOORSETS

- Timing: After associated rooms have been made weathertight and the work of wet trades is finished and dried out.

760 BUILDING IN

- General: Not permitted unless indicated on drawings.

780 DAMP PROOF COURSES IN PREPARED OPENINGS

- Location: Correctly positioned in relation to door frames. Do not displace during fixing operations.

790 FIXING OF WOOD FRAMES

- Spacing of fixings (frames not predrilled): Maximum 150 mm from ends of each jamb and at 600 mm maximum centres.

800 FIXING OF LOOSE THRESHOLDS

- Spacing of fixings: Maximum 150 mm from each end and at 600 mm maximum centres.

820 SEALANT JOINTS

- Sealant:
 - Manufacturer: Contractor's choice .
 - Product reference: Contractor's choice .
 - Colour: To match doors frames .
 - Application: As section Z22 to prepared joints. Triangular fillets finished to a flat or slightly convex profile.

830 FIXING IRONMONGERY GENERALLY

- Fasteners: Supplied by ironmongery manufacturer.
 - Finish/ Corrosion resistance: To match ironmongery.
- Holes for components: No larger than required for satisfactory fit/ operation.
- Adjacent surfaces: Undamaged.
- Moving parts: Adjusted, lubricated and functioning correctly at completion.

840 FIXING IRONMONGERY TO FIRE RESISTING DOOR ASSEMBLIES

- General: All items fixed in accordance with door leaf manufacturer's recommendations ensuring that integrity of the assembly, as established by testing, is not compromised.
- Holes for through fixings and components: Accurately cut.
 - Clearances: Not more than 8 mm unless protected by intumescent paste or similar.
 - Lock/ Latch cases for fire doors requiring $\geq$ 60 minutes integrity performance: Coated with intumescent paint or paste before installation.

850 LOCATION OF HINGES

- Primary hinges: Where not specified otherwise, positioned with centre lines 250 mm from top and bottom of door leaf.
- Third hinge: Where specified, positioned with centre line 250 mm below centre line of top hinge .
- Hinges for fire resisting doors: Positioned in accordance with door leaf manufacturer's recommendations.

860 INSTALLATION OF EMERGENCY EXIT DEVICES

- Standard: Unless specified otherwise, install panic bolts/ latches in accordance with BS EN 1125.

DRAFT

L30

Stairs/ ladders/ walkways/ handrails/ balustrades

L30 Stairs/ ladders/ walkways/ handrails/ balustrades

To be read with Preliminaries/ General conditions.

PRELIMINARY INFORMATION/ REQUIREMENTS

105 CONTRACTOR'S DESIGN FOR INTERNAL STAIRCASES TO HOUSES

- Design responsibility: Determine section sizes and strengths and type, sizes and numbers of fixings.
- Structural and fire requirements:
 - Generally: As section B50.
 - Modifications: None.
 - Design: Complete the design in accordance with the designated code of practice to satisfy specified performance criteria.
- Functional requirements: Straight stairs and winders to BS 5395-1.
- Additional requirements: None.
- Design and production information: To be agreed.
- Timing of submissions: To be agreed.

115 TIMBER PROCUREMENT

- Timber (including timber for wood based products): Obtained from well managed forests and/ or plantations in accordance with:
 - The laws governing forest management in the producer country or countries.
 - International agreements such as the Convention on International Trade in Endangered Species of wild fauna and flora (CITES).
- Documentation: Provide either:
 - Documentary evidence (which has been or can be independently verified) regarding the provenance of all timber supplied.
 - Evidence that suppliers have adopted and are implementing a formal environmental purchasing policy for timber and wood based products.
- Certification scheme: Forest Stewardship Council (FSC).
 - Other evidence: None.

130 SITE DIMENSIONS

- Procedure: Before starting work on designated items take site dimensions, record on shop drawings and use to ensure accurate fabrication.
 - Designated items: all staircases.

COMPONENTS

230A INTERNAL WOOD STAIRS ALL HOUSES

Refer to DLA drawings

- Components:
 - Treads: 21mm MDF.
 - Risers: 9 mm moisture resistant plywood.
 - Strings: 27 mm whitewood.
 - Newels: 90mm Square whitewood.
 - Guarding: 41mm square whitewood.
 - Handrails: Traditional white primed handrails.

Lower handrail: Not required.
- Timber: Generally to BS EN 942.
 - Appearance class: Generally J30; balusters J10.
 - Moisture content at time of installation: 9-13%.
- Adhesive: Contractor's choice.
- Finish as delivered: Prepared and sealed, as section M60.
 - Slip resistance value of integral tread – water wet (minimum): Not applicable.
 - Slip resistance value of integral nosing – water wet (minimum): Not applicable.
 - Colour of integral nosing: Not applicable.
- Workmanship: To section Z10.
- Other requirements: Allow for future stairlift to the following plots: Houses 104, 105, 106 and 107.

Staircase Flights: Floor to Floor 2775

Going: 230mm

Risers: 213.46

Pitch: 41.52 degrees

Stairs to receive carpet finish Finishes Specification

560A PROPRIETARY BALUSTRADE STO APARTMENT BLOCKS A, B & C

BALCONY TO BE DETAILED FOR APPROVAL OF PLANNING

- Manufacturer: _____.
 - Product reference: _____.
- Component material and finish as delivered:
 - Guarding: _____.
 - Handrails: _____.
 - Lower handrail: _____.
- Other requirements: _____.
- Fixing: _____.
 - Centres: _____.

590A APPLIED STAIR NOSING STO APARTMENT COMMON STAIRS

- Manufacturer: Gradus.
 - Web: www.gradus.com.
 - Email: imail@gradus.com.
 - Product reference: Traditional Aluminium G Stair Edging
- Profile: G11: Square shape, 5 mm gauge, 60 mm tread, 32 mm riser, single channel.
- Insert:
 - Type: Interior Xtra-grip.
 - Colour: to be confirmed.
- Fixing: Mechanical (screw) and adhesive fix.

INSTALLATION

610 MOISTURE CONTENT

- Temperature and humidity: Monitor and control internal conditions to achieve specified moisture content in wood components at time of installation.

620 PRIMING/SEALING/PAINTING

- Surfaces inaccessible after assembly/installation: Before fixing components, apply full protective/decorative treatment/coating system.

630 CORROSION PROTECTION OF DISSIMILAR MATERIALS

- Components/ substrates/ fasteners of dissimilar materials: Isolate using washers/ sleeves or other suitable means to separate materials to avoid corrosion and/ or staining.

640 INSTALLATION GENERALLY

- Fasteners and methods of fixing: To section Z20.
- Structural members: Do not modify, cut, notch or make holes in structural members, except as indicated on drawings.
- Temporary support: Do not use stairs, walkways or balustrades as temporary support or strutting for other work.
- Applied finishes: Substrates to be even, dry, sound and free from contaminants. Make good substrate surfaces and prepare/ prime as finish manufacturer's recommendation before application.

670 INSTALLATION OF TREAD INSERTS/ NOSINGS

- Treads: Fully cured, sound and level.
- Fixing:
 - Location/ position: In accordance with BS 8300.
 - Fixings: As manufacturer's recommendations.
 - Centres: As manufacturer's recommendations.

COMPLETION

910 INSPECTION

- Timing: Two weeks prior to date when Contractor expects work to be practically complete.
- Period of notice (minimum): 3 working days.

920 DOCUMENTATION

- Contents:
 - Copies of structural design calculations/ test reports.
 - General product information.
 - Installation information.
 - Inspection and maintenance reports.
- Number of copies: Two.
- Submission: Two weeks after request by Contract Administrator.

DRAFT

L40
General glazing

L40 General glazing

To be read with Preliminaries/ General conditions.

GENERAL REQUIREMENTS

- 111 PREGLAZING
 - Preglazing of components: Permitted.
 - Prevention of displacement: Submit details of precautions to be taken to protect glazing and compound/ seals during delivery and installation.
 - Defective/ displaced glazing/ compound/ seals: Reglaze components in situ.
- 150 WORKMANSHIP AND POSITIONING GENERALLY
 - Glazing generally: In accordance with BS 6262 series.
 - Integrity: Glazing must be wind and watertight under all conditions with full allowance made for deflections and other movements.
 - Dimensional tolerances: Panes/ sheets to be within ± 2 mm of specified dimensions.
 - Materials:
 - Compatibility: Glass/ plastics, surround materials, sealers, primers and paints/ clear finishes to be used together to be compatible. Avoid contact between glazing panes/ units and alkaline materials such as cement and lime.
 - Protection: Keep materials dry until fixed. Protect insulating glass units and plastics glazing sheets from the sun and other heat sources.
- 151 PREPARATION
 - Surrounds, rebates, grooves and beads: Cleaned and prepared by others.
- 152 PREPARATION
 - Surrounds, rebates, grooves and beads: Clean and prepare before installing glazing; ensure compliance with any certified installation requirements.
- 155 GLASS GENERALLY
 - Standards: To BS 952 and relevant parts of:
 - BS EN 572 for basic soda lime silicate glass.
 - BS EN 1096 for coated glass.
 - BS EN 1748-1 for borosilicate glass.
 - BS EN 1748-2 for ceramic glass.
 - BS EN 1863 for heat strengthened soda lime silicate glass.
 - BS EN 12150 for thermally toughened soda lime silicate safety glass.
 - BS EN 12337 for chemically strengthened soda lime silicate glass.
 - BS EN 13024 for thermally toughened borosilicate safety glass.
 - BS EN ISO 12543 for laminated glass and laminated safety glass.
 - Panes/ sheets: Clean and free from obvious scratches, bubbles, cracks, rippling, dimples and other defects.
 - Edges: Generally undamaged. Shells and chips not more than 2 mm deep and extending not more than 5 mm across the surface are acceptable if ground out.
- 165 HEAT SOAKING OF THERMALLY TOUGHENED GLASS
 - Standard: To BS EN 14179.
 - Holding period (minimum): 2 hours.
 - Mean glass temperature: $290^{\circ} \pm 10^{\circ}\text{C}$.
 - Certified evidence of treatment: Submit.
 - Designated locations: all designated toughened glass.

550 GLASS MIRRORS Generally

- Standard: To BS EN 1036.
 - BS EN 1036-2 characteristics: None relevant.
- Mirror material: Float glass, silvered to give maximum reflection, free from tarnishing, discoloration, scratches and other defects visible in the designed viewing conditions.
 - Thickness: 4 mm.
 - Backing: Polypropylene safety film.
 - Edge treatment: Polished arris.
- Background: Plastered masonry.
- Fixing method: Dome top wood screws with polyethylene sleeves and washers at 600 mm centres.
- Installation: Fixed accurately and securely without overtightening fasteners, to provide a flat surface giving a distortion free reflection.

DRAFT

M
Surface finishes

DRAFT

M10

Cement based levelling/ wearing screeds

M10 Cement based levelling/ wearing screeds

To be read with Preliminaries/General conditions.

TYPES OF SCREED

100A GROUND FLOOR BUILD-UP TO HOUSES

- Ground Floor to achieve thermal performance of 0.25 W/m²k
- Floor Finish: Refer to Finishes Specification
- Floor Screed: Refer to M10/115A
- Separating Layer: Suitable polythene vapour control layer (VCL) e.g Visqueen 500 gauge or equivalent.
- Insulation: Refer to specification clause M10/290A
- Damp Proof Membrane: 500 microns black polythene (2000 gauge) Visqueen DPM with all joints lapped and sealed. DPM to be lapped and sealed with all adjacent walls DPC's.
- Beam and block flooring: Refer to E60/110A
- Ventilated floor void: Refer to clause E60/316
- Minimum Void: 250mm to be maintained between underside of the floor and the ground, made up with a suitable well-compacted material or in accordance with the Engineers design specification. Void to be treated with approved weedkiller in accordance with NHBC Section 5.2

100B GROUND FLOOR BUILD-UP TO APARTMENT BLOCKS A, B & C

- Ground Floor to achieve thermal performance of 0.24 W/m²k
- Floor Finish: Refer to Finishes Specification
- Floor Screed: Refer to M10/115B
- Separating Layer: Suitable polythene vapour control layer (VCL) e.g Visqueen 500 gauge or equivalent.
- Insulation: Refer to specification clause M10/290B
- Damp Proof Membrane: 500 microns black polythene (2000 gauge) Visqueen DPM with all joints lapped and sealed. DPM to be lapped and sealed with all adjacent walls DPC's.
- Insitu concrete Slab: Refer to structural engineers Specification

115A CEMENT:SAND LEVELLING SCREEDSTO GARAGE FLOOR

- Substrate: Beam and Block Flooring.
- Screed construction: Unbonded, as clause 280.
 - Reinforcement for crack control: Steel fabric, as clause 392.
- Thickness:
 - Nominal: 75 mm.
 - Minimum: 70 mm.
- Mix:
 - Proportions (cement:sand): 1:3-4.5.
- In situ crushing resistance (ISCR) category: C (2.5 mm maximum indentation).
 - Mass of test weight: 4 kg.
- Flatness/ Surface regularity class: SR2.
- Finish: Trowelled, as clause 540.
 - To receive: None.
- Other requirements: None, unless laid to a fall towards garage door.

130A PROPRIETARY QUICK DRYING LEVELLING SCREEDSTO GROUND FLOOR OF HOUSES & APARTMENTS

- Substrate: Beam and block flooring OR Insitu Concrete slab.
- Screed manufacturer: Flowcrete UK Ltd.
 - Product reference: Isocrete Standard K screed.
- Screed construction: Floating, as clause 290A.
 - Reinforcement for crack control: Steel fabric, as clause 392.
- Thickness:
 - Nominal: -.
 - Minimum: 75 mm.
- Mix:
 - Cement: As clause 302.
 - Aggregates: Standards as clause 305
 - Coarse aggregate single size: Not required. (For screed thickness in excess of 75 mm, a proportion of the sand is replaced by coarse aggregate, 5 mm (or 6 mm) single size).
 - Proportions: To manufacturer's recommendations.
- In situ crushing resistance (ISCR) category: A/B (3/4 mm maximum indentation).
 - Mass of test weight: 4 kg.
- Flatness/ Surface regularity class: SR2 generally.
- Finish: Trowelled, as clause 540.
 - To receive: Vinyl or carpet tiles - Refer to Finishes Schedule.
- Other requirements: Where a thin floor finish is to be installed, overlay with a suitable thin section smoothing underlayment..

130B PROPRIETARY QUICK DRYING LEVELLING SCREEDSTO UPPER FLOOR OF BLOCKS A, B, C & D

- Substrate: Hollowcore pre-cast planks OR Insitu Concrete slab.
- Screed manufacturer: Flowcrete UK Ltd.
 - Product reference: Isocrete Standard K screed.
- Screed construction: Floating, as clause 295.
 - Reinforcement for crack control: Steel fabric, as clause 392.
- Thickness:
 - Nominal: -.
 - Minimum: 75 mm.
- Mix:
 - Cement: As clause 302.
 - Aggregates: Standards as clause 305
 - Coarse aggregate single size: Not required. (For screed thickness in excess of 75 mm, a proportion of the sand is replaced by coarse aggregate, 5 mm (or 6 mm) single size).
 - Proportions: To manufacturer's recommendations.
- In situ crushing resistance (ISCR) category: A/B (3/4 mm maximum indentation).
 - Mass of test weight: 4 kg.
- Flatness/ Surface regularity class: SR2 generally.
- Finish: Trowelled, as clause 540.
 - To receive: Vinyl or carpet tiles - Refer to Finishes Schedule.
- Other requirements: Where a thin floor finish is to be installed, overlay with a suitable thin section smoothing underlayment..

GENERALLY/ PREPARATION

205 DESIGN LIFE OF SCREEDS

- Duration: 30 years .
 - Subject to reasonable wear and tear.
- Location: All Locations .
- Condition of use: Subject to correct loading and traffic usage throughout duration.

210 SUITABILITY OF SUBSTRATES

- General:
 - Suitable for specified levels and flatness/ regularity of finished surfaces. Consider permissible minimum and maximum thicknesses of screeds.
 - Sound and free from significant cracks and gaps.
- Concrete strength: In accordance with BS 8204-1, Table 2.
- Cleanliness: Remove plaster, debris and dirt.
- Moisture content: To suit screed type. New concrete slabs to receive fully or partially bonded construction must be dried out by exposure to the air for minimum six weeks.

220 PROPRIETARY LEVELLING/ WEARING SCREEDS

- General: Materials, mix proportions, mixing methods, minimum/ maximum thicknesses and workmanship must be in accordance with recommendations of screed manufacturer.
- Standard: In accordance with BS 8204-3.

221 FLOWCRETE PROPRIETARY SCREEDS

- Are to be laid in accordance with the instructions of Flowcrete UK Ltd, The Flooring Technology Centre, Booth Lane, Moston, Sandbach, Cheshire, CW11 3QF (tel 01270 753 753) by one of their Approved Licensees

230 CONTROL SAMPLES

- General: Complete areas of finished work and obtain approval of appearance before proceeding.
- Screed type: As clause 130.
 - Location/ Size: To be agreed.

250 CONDUITS UNDER FLOATING SCREEDS

- Haunching: Before laying insulation for floating screeds, haunch up in 1:4 cement:sand on both sides of conduits.

251 CONDUITS CAST INTO OR UNDER SCREEDS

- Reinforcement: Overlay with reinforcement selected from:
 - 500 mm wide strip of steel fabric to BS 4483, reference D49, or
 - Welded mesh manufactured in rolls from mild steel wire minimum 1.5 mm diameter to BS 1052, mesh size 50 x 50 mm.
- Placing reinforcement: Mid depth between top of conduit and the screed surface.
 - Width of reinforcement (minimum): 300 mm.
- Screed cover over conduit (minimum): 25 mm.

255 PIPE DUCTS/ TRUNKING

- Preformed access ducts: Before laying screed, fix securely to substrates and level accurately in relation to finished floor surface.

263A FULLY BONDED CONSTRUCTION With Isocrete M-Bond Extra

- Removing mortar matrix: Shortly before laying screed, expose coarse aggregate over entire area of hardened substrate.
- Texture of surface: Suitable to accept screed and achieve a full bond over complete area.
- Bonding coat: Brush in or roller apply the bonding agent, Isocrete M-Bond Extra DPM/bonder.
- Lay screed/topping while M-Bond Extra is still wet to ensure a good bond.

280 UNBONDED CONSTRUCTION

- Separation: Lay screed over a suitable sheet dpm or a separating layer.
 - Type: Polyethylene dpm, as section J40/120.
- Installation of separating layer: Lay on clean substrate. Turn up for full depth of screed at abutments with walls, columns, etc. Lap 100 mm at joints.

290A FLOATING CONSTRUCTION GROUND FLOOR

- Insulation:
 - Type: 75mm Jablite Jabfloor 70 or 100 to Block D and Houses AND 60mm to Blocks A, B and C .
 - Installation: Lay with tight butt joints. Continue up at perimeter abutments for full depth of screed.
- Separating layer:
 - Type: Polyethylene sheet minimum 125 micrometres thick (500 gauge), Visqueen (VCL).
 - Installation: Lay over insulation and turn up at perimeter abutments. Lap 100 mm at joints.
- Note: Where a DPM is required this should be laid under the insulation

295 FLOATING CONSTRUCTION (THIN SHEET IMPACT SOUND INSULATION)

- Substrate: Remove projections that may puncture the insulation.
- Insulation:
 - Type: Acoustic Insulation Collecta YELOfon HD10+.
 - Installation: Lay on substrate. Turn up for full depth of screed at perimeter abutments. Lap 100 mm at joints and seal with tape.
 - Perimeter: Maintain isolation of screed.

BATCHING/ MIXING

302A CEMENTS

- Cement types: In accordance with BS 8204-1, clause 5.1.3.
- Cement to be bagged and preblended with Isocrete K-Additive and Isocrete PP Fibres. Refer to Clauses 332 and 392.

305 AGGREGATES

- Sand: To BS EN 13139.
 - Grading limits: In accordance with BS 8204-1, Table B.1.
- Coarse aggregates for fine concrete levelling screeds:
 - Standard: To BS EN 12620.
 - Designation: 4/10.
- Lightweight aggregates: In accordance with BS 8204-1, Annex A.

310 BATCHING WITH DENSE AGGREGATES

- Mix proportions: Specified by weight.
- Batching: Select from:
 - Batch by weight.
 - Batch by volume: Permitted on the basis of previously established weight:volume relationships of the particular materials. Use accurate gauge boxes. Allow for bulking of damp sand.

330 MIXING

- Water content: Minimum necessary to achieve full compaction, low enough to prevent excessive water being brought to surface during compaction.
- Mixing: Mix materials thoroughly to uniform consistency. Mixes other than no-fines must be mixed in a suitable forced action mechanical mixer. Do not use a free fall drum type mixer.
- Consistency: Use while sufficiently plastic for full compaction.
- Ready-mixed retarded screed mortar: Use within working time and site temperatures recommended by manufacturer. Do not retemper.

332 SCREED ADDITIVE

- The screed is to include a proprietary additive complying with the following:
- The screed system is to have a current Agrément certificate from the British Board of Agrément.
- The additive is to be bagged and preblended with cement and Isocrete PP Fibres.

335 IN SITU CRUSHING RESISTANCE (ISCR)

- Standards and category: In accordance with BS 8204-1, table 4.
 - Testing of bonded and unbonded screeds: To Annex D.
 - Testing of floating levelling screeds: To Annex E.

340 ADVERSE WEATHER

- Screeds surface temperature: Maintain above 5°C for a minimum of four days after laying.
- Hot weather: Prevent premature setting or drying out.

LAYING

345 LEVEL OF SCREED SURFACES

- Permissible deviation: (allowing for thickness of coverings) ± 5 mm from datum.

355 FLATNESS/ SURFACE REGULARITY OF FLOOR SCREEDS

- Standard: In accordance with BS 8204-1, Table 5.
- Test: In accordance with BS 8204-1, Annex C.
- Sudden irregularities: Not permitted.

365 FLATNESS/SURFACE REGULARITY OF ROOF SCREEDS

- Sudden irregularities: Not permitted.
- Deviation of surface: Measure from underside of a 2 m straightedge (between points of contact), placed anywhere on surface.
 - Permissible deviation (maximum): 6 mm.

375A COMPACTION OF SCREEDS

- General: Compact thoroughly over entire area.
- Screeds over 75 mm thick: Lay in two layers of approximately equal thickness. Roughen surface of compacted lower layer then immediately lay upper layer.

382 STAIR SCREEDS

- Construction: Fully bonded to treads, risers and landings.
- Risers: Form using fine finish formwork.
- Wearing screed surfaces: Make good with compatible cement:sand mix. Wood float. When hardened remove laitance.

392A GENERAL REINFORCEMENT

- Type: Isocrete PP Fibres
- Installation: To be included in the cement / K-Additive blend at a dosage of 75 grammes per 25 kg

395A STRIP REINFORCEMENT

- Location: Across daywork joints and over the support beams/walls of a precast concrete floor. .
- Type: To BS 4483, Table 1:
 - Type D49 .
 - Width of strips: 500 mm .
- Installation: Place between the two layers of screed and centre over joints. Lap ends minimum 100mm.

405 JOINTS IN LEVELLING SCREEDS GENERALLY

- Laying screeds: Lay continuously using 'wet screeds' between strips or bays. Minimize defined joints.
- Daywork joints: Form with vertical edge.

406 BAY JOINTS IN LEVELLING SCREEDS

- Screed type: As clause 130.
- Bay sizes:
 - Area (maximum): as manufacturers recommendations.
 - Length (maximum): as manufacturers recommendations.
- Location of bay joints: Coordinate with those required for substrate slab and floor covering.

440 CRACK INDUCING GROOVES IN LEVELLING SCREEDS

- Groove depth: At least half the depth of screed.
- Cutting grooves: Straight, vertical and accurately positioned. Select from the following:
 - Trowel cut as screed is laid.
 - Saw cut sufficiently early after laying to prevent random cracking.

FINISHING/CURING

510 FINISHING GENERALLY

- Timing: Carry out all finishing operations at optimum times in relation to setting and hardening of screed material.
- Prohibited treatments to screed surfaces:
 - Wetting to assist surface working.
 - Sprinkling cement.

540 TROWELLED FINISH TO LEVELLING SCREEDS

- Floating: To an even texture with no ridges or steps.
- Trowelling: To a uniform, smooth but not polished surface, free from trowel marks and other blemishes, and suitable to receive specified flooring material.

650 CURING

- General: Prevent premature drying. Immediately after laying, protect surface from wind, draughts and strong sunlight. As soon as screed has set sufficiently, closely cover with polyethylene sheeting.
- Curing period (minimum): Keep polyethylene sheeting in position for: 5 days.
- Drying after curing: Allow screeds to dry gradually. Do not subject screeds to artificial drying conditions that will cause cracking or other shrinkage related problems.

710 FOLLOWING TRADES

- Adequately protect screeds/toppings from damage and contamination by subsequent building operations.

DRAFT

M20

Plastered/ Rendered/ Roughcast coatings

M20 Plastered/ Rendered/ Roughcast coatings

TYPES OF COATING

- 280 GYPSUM PLASTER SKIM COAT ON PLASTERBOARD
- Plasterboard: 12.5 mm Plasterboard and Celotex Insulated plasterboard .
 - Preparation: Bonding agent recommended by plaster manufacturer .
 - Plaster: Board finish/ finish plaster to BS EN 13279-1.
 - Manufacturer: British Gypsum .
 - Product reference: Thistle Multit Finish .
 - Thickness: 2.5mm .
 - Finish: Smooth.
 - Accessories: Beads and stops as required .
- 295A PROPRIETARY PARGE COAT TO TO PARTY WALLS
- Manufacturer: British Gypsum.
 - Web: www.british-gypsum.com.
 - Email: bgtechnical.enquiries@bpb.com.
 - Product reference: Gyproc SoundCoat Plus
 - Application: Apply a continuous parge coat, minimum 6 mm thick, to the entire surface of the background, ensuring all cracked or unfilled joints are covered. Allow to set before completing the plasterboard installation

GENERAL

- 421 SCAFFOLDING
- General: Prevent putlog holes and other breaks in coatings.

MATERIALS AND MARKING OF MORTAR

- 430 READY-TO-USE CEMENT GAUGED RENDER MORTARS
- Time and temperature limitations: Use within limits prescribed by mortar manufacturer
 - Retempering: Restore workability with water only within prescribed time limits.
- 438 CEMENTS FOR MORTARS
- Cement: To BS EN 197-1.
 - Types: Portland cement, CEM I.
Portland slag cement, CEM II.
Portland fly ash cement, CEM II.
 - Strength class: 32.5, 42.5 or 52.5.
 - White cement: To BS EN 197-1.
 - Type: Portland cement, CEM1.
 - Strength class: 52.5.
 - Sulfate resisting Portland cement: To BS EN 197-1.
 - Strength class: 42.5.
 - Masonry cement: To BS EN 998-1 and Kitemarked.
- 440 SAND FOR CEMENT GAUGED MORTARS
- Standard: To BS EN 13139.
 - Grading: 0/2 or 0/4 (CP or MP); Category 2 fines.
 - Colour and texture: Consistent. Obtain from one source.

443 LIME FOR CEMENT GAUGED MORTARS

- Standard: To BS EN 459-1.
 - Type: CL 90S.

445 PIGMENT FOR COLOURED MORTARS

- Standard: To BS EN 12878.

449 ADMIXTURES FOR CEMENT GAUGED MORTARS

- Suitable admixtures: Select from:
 - Air entraining (plasticizing) admixtures: To BS EN 934-2 and compatible with other mortar constituents.
 - Other admixtures: Submit proposals.
- Prohibited admixtures: Calcium chloride and any admixture containing calcium chloride.

450 CHLORIDE CONTENT OF MORTARS

- Chloride content (maximum): 0.1% by dry mass.

495 MIXING

- Render mortars (site prepared):
 - Batching: By volume. Use clean and accurate gauge boxes or buckets.
 - Mix proportions: Based on damp sand. Adjust for dry sand.
 - Lime:sand: Mix thoroughly. Allow to stand, without drying out, for at least 16 hours before using.
- Mixes: Of uniform consistence and free from lumps. Do not retemper or reconstitute mixes.
- Contamination: Prevent intermixing with other materials.

497 COLD WEATHER

- General: Do not use frozen materials or apply coatings on frozen or frost bound substrates.
- External work: Avoid when air temperature is at or below 5°C and falling or below 3°C and rising. Maintain temperature of work above freezing until coatings have fully hardened.
- Internal work: Take precautions to enable internal coating work to proceed without detriment when air temperature is below 3°C.

PREPARING SUBSTRATES

510 SUITABILITY OF SUBSTRATES

- Soundness: Free from loose areas and significant cracks and gaps.
- Cutting, chasing, making good, fixing of conduits and services outlets and the like: Completed.
- Tolerances: Permitting specified flatness/ regularity of finished coatings.
- Cleanliness: Free from dirt, dust, efflorescence and mould, and other contaminants incompatible with coatings.

531 ROUGHENING FOR KEY

- Substrates: Roughen thoroughly and evenly.
 - Depth of surface removal: Minimum necessary to provide an effective key.

541 BONDING AGENT APPLICATION

- General: Apply evenly to substrate to achieve effective bond of plaster/ render coat. Protect adjacent joinery and other surfaces.

BACKINGS/ BEADS/ JOINTS

- 600 ADDITIONAL FRAMING SUPPORTS FOR BACKINGS
- Framing: Accurately position and securely fix to give full support to fixtures, fittings and service outlets.
 - Support board edges and perimeters: As recommended by board manufacturer to suit type and performance of board.
- 610 FIXING PLASTERBOARD BACKINGS TO TIMBER
- Fixings, accessories and installation methods: As recommended by board manufacturer.
 - Fixing: At the following centres (maximum):
 - Nails: 150 mm.
 - Screws to partitions/ walls: 300 mm. Reduce to 200 mm at external angles.
 - Screws to ceilings: 230 mm.
 - Position of nails/ screws from edges of boards (minimum):
 - Bound edges: 10 mm.
 - Cut/ unbound edges: 13 mm.
 - Position of nails/ screws from edges of supports (minimum): 6 mm.
 - Nail/ screw heads: Set below surface. Do not break paper or gypsum core.
- 612 JOINTS IN PLASTERBOARD BACKINGS
- Ceilings:
 - Bound edges: At right angles to supports and with ends staggered in adjacent rows.
 - Two layer boarding: Stagger joints between layers.
 - Partitions/ walls:
 - Vertical joints: Centre on studs. Stagger joints on opposite sides of studs.
 - Two layer boarding: Stagger joints between layers.
 - Horizontal joints:
 - Two layer boarding: Stagger joints between layers by at least 600 mm. Support edges of outer layer.
 - Joint widths (maximum): 3 mm.
- 630 BEADS/ STOPS FOR INTERNAL USE
- Standard: In accordance with BS EN 13914-2, Table 2.
 - Material: Galvanized steel to BS EN 13658-1.
- 646 CRACK CONTROL AT JUNCTIONS BETWEEN DISSIMILAR SOLID SUBSTRATES
- Locations: Where defined movement joints are not required. Where dissimilar solid substrate materials are in same plane and rigidly bonded or tied together.
 - Crack control materials:
 - Isolating layer: Building paper to BS 1521.
 - Metal lathing: Internally: Galvanized steel plain expanded metal with spacers.
 - Installation: Fix metal lathing over isolating layer. Stagger fixings along both edges of lathing.
 - Width of installation over single junctions:
 - Isolating layer: 150 mm.
 - Lathing: 300 mm.
 - Width of installation across face of dissimilar substrate material (column, beam, etc. with face width not greater than 450 mm):
 - Isolating layer: 25 mm (minimum) beyond junctions with adjacent substrate.
 - Lathing: 100 mm (minimum) beyond edges of isolating layer.
- 659 PLASTERBOARD JOINTS
- Joints and angles (except where coincident with metal beads). Reinforce with continuous lengths of jointing tape.

673 PLASTERING OVER CONDUITS/ SERVICE CHASES

- General: Prevent cracking over conduits and other services.
- Services chased into substrate: Isolate from coating by covering with galvanized metal lathing, fixed at staggered centres along both edges.

INTERNAL PLASTERING

710 APPLICATION GENERALLY

- Application of coatings: Firmly and in one continuous operation between angles and joints. Achieve good adhesion.
- Appearance of finished surfaces: Even and consistent. Free from rippling, hollows, ridges, cracks and crazing.
 - Accuracy: Finish to a true plane, to correct line and level, with angles and corners to a right angle unless specified otherwise, and with walls and reveals plumb and square.
- Drying out: Prevent excessively rapid or localized drying out.

715 FLATNESS/ SURFACE REGULARITY

- Sudden irregularities: Not permitted.
- Deviation of plaster surface: Measure from underside of a straight edge placed anywhere on surface.
 - Permissible deviation (maximum) for plaster not less than 13 mm thick: 3 mm in any consecutive length of 1800 mm.

718 JUNCTION OF NEW PLASTERWORK WITH EXISTING

- New plasterwork: Finish flush with original face of existing plasterwork to form a seamless junction.

720 DUBBING OUT

- General: Correct substrate inaccuracies.
- New smooth dense concrete and similar surfaces: Dubbing out prohibited unless total plaster thickness is within range recommended by plaster manufacturer.
- Thickness of any one coat (maximum): 10 mm.
- Mix: As undercoat.
- Application: Achieve firm bond. Allow each coat to set sufficiently before the next is applied. Cross scratch surface of each coat.

725 UNDERCOATS GENERALLY

- General: Rule to an even surface. Cross scratch to provide a key for the next coat.
- Undercoats on metal lathing: Work well into interstices to obtain maximum key.
- Undercoats gauged with Portland cement: Do not apply next coat until drying shrinkage is substantially complete.

742 THIN COAT PLASTER

- Preparation for plasters less than 2 mm thick: Fill holes, scratches and voids with finishing plaster.

777 SMOOTH FINISH

- Appearance: A tight, matt, smooth surface with no hollows, abrupt changes of level or trowel marks. Avoid water brush, excessive trowelling and over polishing.

778 WOOD FLOAT FINISH

- Appearance: An even overall texture. Finish with a dry wood float as soon as wet sheen has disappeared.

DRAFT

M40

Stone/ concrete/ quarry/ ceramic tiling/ mosaic

M40 Stone/ concrete/ quarry/ ceramic tiling/ mosaic

To be read with Preliminaries/ General conditions.

TYPES OF TILING/ MOSAIC

110A TILING TO TO WALLS AND FLOORS

- Refer to WYG Residential Denmise and Fittings Schedule appended to this document

GENERAL

210 SUITABILITY OF BACKGROUNDS/ BASES

- Background/ base tolerances: To permit specified flatness/ regularity of finished surfaces given the permissible minimum and maximum thickness of bedding.
- New background drying times (minimum):
 - Concrete walls: 6 weeks.
 - Brick/ block walls: 6 weeks.
 - Rendering: 2 weeks.
 - Gypsum plaster: 4 weeks.
- New base drying times (minimum):
 - Concrete slabs: 6 weeks.
 - Cement:sand screeds: 3 weeks.

215 FALLS IN THE BASES

- General: Give notice if falls are inadequate.

250 SAMPLES

- General: Submit representative samples of the following:
Each type of tile.

260 CONTROL SAMPLES

- General: Complete sample areas, being part of finished work, in locations as follows: TBC.
 - Approval of appearance: Obtain before proceeding.
- Floor covering slip resistance testing: As clause 900.

PREPARATION

370 NEW IN SITU CONCRETE

- Backgrounds/ bases to be tiled: Remove mould oil, surface retarders and other materials incompatible with bedding.

380 NEW PLASTER

- Plaster: Dry, solidly bedded, free from dust and friable matter.
- Plaster primer: Apply if recommended by adhesive manufacturer.

390 PLASTERBOARD BACKGROUNDS

- Boards: Dry, securely fixed and rigid with no protruding fixings and face to receive decorative finish exposed.

400 BACKGROUNDSGENERALLY

- Boards: Dry, securely fixed and rigid with no protruding fixings.
- Surfaces to be tiled: Seal or prime if recommended by adhesive manufacturer.

- 420 RAKING OUT FOR KEY
- Soft joints in existing masonry: Rake out to a depth of 13 mm (minimum).
- 430 SPATTERDASHING FOR KEY
- Spatterdash mix:
 - Cement: Portland to BS EN 197-1 type CEM I/42.5
 - Sand: Clean coarse.
 - Proportions (cement:sand): 1-1.5:2.
 - Admixture: BBA certified SBR bonding agent.
 - Consistency: Thick slurry with no segregation.
 - Application/finish: Thrown.
 - Thickness: 3-5 mm.
 - Curing/ drying: Keep damp until hardened. Dry out to provide securely bonded finish.
- 438 PREPARING CONCRETE BASES FOR FULLY BONDED BEDDING
- Surface cement:sand matrix: Remove to expose coarse aggregate.
 - Surface preparation: Suitable to achieve a full bond with bedding. Select from:
 - Keep well wetted for several hours. Remove free water then brush in a slurry bonding coat.
Slurry: SBR polymer modified cement.
 - Prepare, prime as necessary and apply a bonding agent.
Bonding agent: BBA certified SBR bonding agent.
- 444 PREPARING CONCRETE BASES FOR PARTIALLY BONDED BEDDING
- Surface finish: Brushed, with no laitance.
 - Surface preparation: Suitable to achieve a full bond with bedding. Select from:
 - Keep well wetted for several hours. Remove free water then brush in a slurry bonding coat.
Slurry: SBR polymer modified cement.
 - Prepare, prime as necessary and apply a bonding agent.
Bonding agent: BBA certified SBR bonding agent.
- 450 PREPARING CONCRETE BASES FOR UNBONDED BEDDING - WITHOUT SEPARATING LAYER
- Surface finish: Smooth.
 - Surface preparation: Dampen lightly before laying mortar bed.
- 460 SMOOTHING UNDERLAYMENT
- Type: Recommended by adhesive manufacturer.
 - Condition: Allow to dry before tiling.

FIXING

510 FIXING GENERALLY

- Colour/ shade: Unintended variations within tiles for use in each area/ room are not permitted.
 - Variegated tiles: Mix thoroughly.
- Adhesive: Compatible with background/ base. Prime if recommended by adhesive manufacturer.
- Use of admixtures with cementitious adhesives: Only admixtures approved by adhesive manufacturer.
- Cut tiles: Neat and accurate.
- Fixing: Provide adhesion over entire background/ base and tile backs.
- Final appearance: Before bedding material sets, make adjustments necessary to give true, regular appearance to tiles and joints when viewed under final lighting conditions.
- Surplus bedding material: Clean from joints and face of tiles without disturbing tiles.

530 SETTING OUT

- Joints: True to line, continuous and without steps.
 - Joints on walls: Horizontal, vertical and aligned round corners.
 - Joints in floors: Parallel to the main axis of the space or specified features.
- Cut tiles: Minimize number, maximize size and locate unobtrusively.
- Joints in adjoining floors and walls: Align.
- Joints in adjoining floors and skirtings: Align.
- Movement joints: Where locations are not indicated, submit proposals.
- Setting out of tbc : Drawing references: tbc.
- Setting out of tbc : Submit proposals.

540 LEVEL OF FLOOR TILING

- Permissible deviation in level from datum tbc.

550 FLATNESS/ REGULARITY OF TILING/ MOSAICS

- Sudden irregularities: Not permitted.
- Deviation of surface: Measure from underside of a 2 m straightedge with 3 mm thick feet placed anywhere on surface. The straightedge should not be obstructed by the tiles and no gap should be greater than 6 mm, i.e. a tolerance of ± 3 mm.

560 LEVEL OF TILING ACROSS JOINTS

- Deviation (maximum) between tile surfaces either side of any type of joint:
 - 1 mm for joints less than 6 mm wide.
 - 2 mm for joints 6 mm or greater in width.

570 MORTAR BEDDING

- Bedding mix:
 - Cement: Portland to BS EN 197-1, type CEM I/42.5.
 - Sand for walls: Fine aggregate to BS EN 13139.
Grading designation: 0/2 (CP or MP) category 2 fines.
 - Sand for floors: Fine aggregate to BS EN 13139.
Grading designation: 0/4 (MP) category 1 fines and between 20%-66% passing a 0.5 sieve.
- Batching: Select from:
 - Batch by weight.
 - Batch by volume: Permitted on the basis of previously established weight:volume relationships of the particular materials. Use accurate gauge boxes. Allow for bulking of damp sand.
- Mixing: Mix materials thoroughly to uniform consistence. Use a suitable forced action mechanical mixer. Do not use a free fall type mixer.
- Application: At normal temperatures use within two hours. Do not use after initial set. Do not retemper.

578 CRACK CONTROL REINFORCEMENT

- Type to BS 4483: D49.
- Installation: Place centrally in depth of bed. Lap not less than 100 mm and securely tie together with steel wire.
- Corners: Avoid a four layer build at corners.

580 POROUS TILES

- Tiles to be bedded in cement:sand: Soak in clean water for at least 30 minutes. Fix as soon as surface water has drained.

590 COVED TILE SKIRTINGS

- Sequence: Bed solid to wall before laying floor tiles.
- Bedding: Cement based adhesive.

600 SIT-ON TILE SKIRTINGS

- Sequence: Bed solid to wall after laying floor tiles.
- Bedding: Cement based adhesive.

650 ADHESIVE BED - NOTCHED TROWEL METHOD (WALLS)

- Application: By 3 mm floated coat of adhesive to dry background in areas of approximately 1 m². Comb surface.
- Tiling: Press tiles firmly onto float coat.

651 ADHESIVE BED - NOTCHED TROWEL AND BUTTERING METHOD (WALLS)

- Application: By floated coat of adhesive to dry background in areas of about 1 m². Comb surface.
- Tiling: Apply thin even coat of adhesive to backs of dry tiles. Fill any ribbed, deep keyed or button profiles. Press tiles firmly onto float coat.
- Finished adhesive thickness: 3 mm or within the range allowed by the adhesive manufacturer.

652 ADHESIVE BED - BUTTERING METHOD (WALLS)

- Tiling: Apply even coat of adhesive to backs of dry tiles. Fill any ribbed, deep keyed or button profiles.
- Finished adhesive thickness: 3 mm or within the range allowed by the adhesive manufacturer.

- 660 ADHESIVE BED - NOTCHED TROWEL METHOD FOR MESH BACKED MOSAIC (WALLS)
- Application: By 3 mm floated coat of adhesive to dry background. Comb surface.
 - Placing mosaic sheets: Hang in horizontal rows, working downwards. Stagger vertical joints. Prevent slippage of sheets. Lightly beat mosaics into adhesive.
 - Width, plane and alignment of joints between sheets: To match joints between mosaic tiles.
- 661 ADHESIVE BED - NOTCHED TROWEL METHOD FOR PAPER FACED MOSAIC (WALLS)
- Application: By 3 mm floated coat of adhesive to dry background. Comb surface.
 - Preparing mosaic sheets: Pre-grout. Remove surplus before fixing.
 - Placing mosaic sheets: Hang in horizontal rows, working downwards. Stagger vertical joints.
 - Width, plane and alignment of joints between sheets: To match joints between mosaic tiles.
 - Paper face: Before adhesive hardens completely, remove paper face. Complete grouting. Wash off glue from face of mosaic.
- 690 CEMENT:SAND MORTAR BEDDING (WALLS)
- Preparation: Dampen background.
 - Application: By floated coat to background: 1:3-4 cement:sand mortar bedding.
 - Thickness (maximum): 10 mm.
 - Finish: Equivalent to wood float. Before tiling allow to stiffen slightly.
 - Tiling: Without delay, apply 2 mm thick coat of 1:1 cement:fine sand mortar to backs of tiles, filling keys. Press tiles firmly onto float coat. Tap firmly into position
- 710 ADHESIVE BED - NOTCHED TROWEL AND BUTTERING METHOD (FLOORS)
- Application: Floated coat of adhesive to dry base and comb surface.
 - Tiling: Apply coat of adhesive to backs of dry tiles. Fill any ribbed, deep keyed or button profiles. Press tiles firmly onto float coat.
 - Finished adhesive thickness: Within range allowed by manufacturer.
- 711 ADHESIVE BED - BUTTERING METHOD (FLOORS)
- Tiling: Apply even coat of adhesive to backs of dry tiles. Fill any ribbed, deep keyed or button profiles.
 - Finished adhesive thickness: Within the range allowed by the adhesive manufacturer.
- 720 CEMENT:SAND MORTAR BEDDING (FLOORS)
- Mortar bedding mix: 1:3-4 cement:sand.
 - Consistency: Stiff plastic.
 - Laying: Lay suitably small working areas of screeded bed. Compact thoroughly to level.
 - Finished bed thickness:
 - For tiles up to 10 mm thick: 10-15 mm.
 - For tiles greater than 10 mm thick: 15-20 mm.
 - Tiling: Within two hours and before bedding sets, evenly coat backs of tiles with adhesive. Press tiles firmly into position.
 - Finished adhesive thickness: Within range recommended by manufacturer.
- 885 COLOURED GROUT
- Staining of tiles: Not permitted
 - Evaluating risk of staining: Apply grout to a few tiles in a small trial area. If discoloration occurs apply a protective sealer to tiles and repeat trial.

DRAFT

M50

Rubber/ plastics/ cork/ lino/ carpet tiling/ sheeting

M50 Rubber/ plastics/ cork/ lino/ carpet tiling/ sheeting

TYPES OF COVERING

110A FLOOR FINISHES FOR ALL HOUSES AND APARTMENTS

- Refer to WYG Residential Demise and Fittings Schedule

GENERAL REQUIREMENTS

210 WORKMANSHIP GENERALLY

- Base condition after preparation: Rigid, dry, sound, smooth and free from grease, dirt and other contaminants.
- Finished coverings: Accurately fitted, tightly jointed, securely bonded, smooth and free from air bubbles, rippling, adhesive marks and stains.

220 SAMPLES

- Covering samples: Before placing orders, submit representative sample of each type.

230 CONTROL SAMPLES

- General: Complete areas of finished work in approved locations as follows, and obtain approval of appearance before proceeding: tbc .

270 EXTRA MATERIAL

- Provision of extra material: At completion, hand to Employer extra material of each type of covering to extent of tbc.

330 COMMENCEMENT

- Required condition of works prior to laying materials:
 - Building is weathertight and well dried out.
 - Wet trades have finished work.
 - Paintwork is finished and dry.
 - Conflicting overhead work is complete.
 - Floor service outlets, duct covers and other fixtures around which materials are to be cut are fixed.
- Notification: Submit not less than 48 hours before commencing laying.

340 CONDITIONING

- Prior to laying: Condition materials by unpacking and separating in spaces where they are to be laid. Maintain resilient flooring rolls in an upright position. Unroll carpet and keep flat on a supporting surface.
- Conditioning time and temperature (minimum): As recommended by manufacturer with time extended by a factor of two for materials stored or transported at a temperature of less than 10°C immediately prior to laying.

350 ENVIRONMENT

- Temperature and humidity: Before, during and after laying, maintain approximately at levels which will prevail after building is occupied.
- Ventilation: Before during and after laying, maintain adequate provision.

360 FLOORS WITH UNDERFLOOR HEATING

- Commencement of laying: Not before a period of 48 hours after heating has been turned off.
- Post laying start up of heating system: Slowly return heating to its operative temperature not less than 48 hours after completing laying.

PREPARING BASES

410 NEW BASES

- Suitability of bases and conditions within any area: Commencement of laying of coverings will be taken as acceptance of suitability.

430 NEW WET LAID BASES

- Base drying aids: Not used for at least four days prior to moisture content testing.
- Base moisture content test: Carry out in accordance with BS 5325, Annexe A or BS 8203, Annexe A.
 - Locations for readings: In all corners, along edges, and at various points over area being tested.
- Commencement of laying coverings: Not until all readings show 75% relative humidity or less.

440 SUBSTRATES TO RECEIVE THIN COVERINGS

- Trowelled finishes: Uniform, smooth surface free from trowel marks and other blemishes. Abrade suitably to receive specified floor covering material.

460 SMOOTHING/ LEVELLING UNDERLAYMENT COMPOUND

- Type: Latex cement.
- Manufacturer: Contractor's choice.
 - Product reference: Submit proposals.

520 TIMBER BOARDING/ STRIP FLOORING

- Substrate: Boards/ strips securely fixed and acceptably level with no protruding fasteners. Plane, sand or apply smoothing underlayment compound to give a smooth, even surface.

530 PARTICLEBOARD FLOORING

- Substrate: Boards securely fixed, level and free from surface sealers and contaminants.
 - Gaps between boards: Not more than 1 mm.
 - Priming: As required by covering adhesive manufacturer.
- Equilibrium moisture content at time of laying covering: As in service conditions.

535 PARTICLEBOARD FLOATING FLOORING

- Substrate: Boards securely fixed and free from surface sealers and contaminants.
- Level: Sand to remove areas raised more than 1 mm.
 - Gaps between boards: Not more than 1 mm.
 - Priming: As required by covering adhesive manufacturer.
- Equilibrium moisture content at time of laying covering: As in service conditions.

LAYING COVERINGS

610 SETTING OUT TILES

- Method: Set out from centre of area/ room, so that wherever possible:
 - Tiles along opposite edges are of equal size.
 - Edge tiles are more than 50% of full tile width.

- 620 COLOUR CONSISTENCY
- Finished work in any one area/ room: Free from banding or patchiness.
- 640 ADHESIVE FIXING GENERALLY
- Adhesive type: As specified, as recommended by covering/ underlay, manufacturer or as approved.
 - Primer: Type and usage as recommended by adhesive manufacturer.
 - Application: As necessary to achieve good bond.
 - Finished surface: Free from trowel ridges, high spots caused by particles on the substrate, and other irregularities.
- 650 SEAMS
- Patterns: Matched.
 - Joints: Tight without gaps.
- 670 BORDERS/ AND FEATURE STRIPS IN SHEET MATERIAL
- Curl: Not acceptable.
 - Corners: Mitre joints.
- 700 LOOSE LAID CARPET TILES
- Areas of adhered tiles: Secure using double sided tape or peelable adhesive.
 - Joints: Butted.
 - Perimeter joints: Accurately cut to match abutment and prevent movement.
- 720 DOORWAYS
- Joint location: On centre line of door leaf.
- 740 EDGINGS AND COVER STRIPS
- Manufacturer: Contractor's choice .
 - Product reference: Submit proposals .
 - Material/ finish: tbc .
 - Fixing: Secure with edge of covering gripped. Use matching fasteners where exposed to view
- 750 STAIR NOSINGS AND TRIMS
- Manufacturer: tgc.
 - Product reference: tbc.
 - Material/ finish: tbc.
 - Fixing: Secure, level and with mitred joints. Adjusted to suit thickness of covering with continuous packing strips of hardboard or plywood. Nosings and packing strips bedded in gap-filling adhesive recommended by nosing manufacturer.
 - Screw fixing with matching plugs: Required.
- 760 STAIR COVERINGS - SEPARATE RISERS AND TREADS
- Fixing: Accurately cut. Fit risers before treads.

COMPLETION

- 880 WASTE
- Spare covering material: Retain suitable material for patching. On completion submit pieces for selection. Hand over selected pieces to Employer.

DRAFT

M60
Painting/clear finishing

M60 Painting/clear finishing

COATING SYSTEMS

PAINT SPECIFICATION

Refer to WYG 'Residential Demise and Fittings Schedule' and Careys 'Outline Specification'

Any variations to be highlighted.

ALL ITEMS TO BE FULLY REVIEWED BEFORE SPECIFICATION CONFIRMED

110A EMULSION PAINT TO INTERNAL PLASTERED SURFACES

- Manufacturer: Dulux Trade, brand of AkzoNobel.
 - Web: www.duluxtrade.co.uk.
 - Email: john.ashford@akzonobel.com.
 - Product reference: Diamond Matt
 - System code: D251 - New plaster/ cement/ calcium silicate boards, etc..
 - Colour: Brilliant White and Gardenia
- Coats: 1 mist coat and 2 full coats emulsion paint to plasterboard surfaces

150A EGG SHELL/ SATIN PAINT TO INTERNAL EXPOSED SOFTWOOD

- Manufacturer: Dulux Trade, brand of AkzoNobel.
 - Web: www.duluxtrade.co.uk.
 - Email: john.ashford@akzonobel.com.
 - Product reference: Diamond Matt
 - System code: D251 - New plaster/ cement/ calcium silicate boards, etc..
 - Colour: Brilliant White
- Coats: 1 undercoat and 1 finishing coat

GENERALLY

215 HANDLING AND STORAGE

- Coating materials: Deliver in sealed containers, labelled clearly with brand name, type of material and manufacturer's batch number.
- Materials from more than one batch: Store separately. Allocate to distinct parts or areas of the work.

220 COMPATIBILITY

- Coating materials selected by contractor:
 - Recommended by their manufacturers for the particular surface and conditions of exposure.
 - Compatible with each other.
 - Compatible with and not inhibiting performance of preservative/fire retardant pretreatments.

280 PROTECTION

- 'Wet paint' signs and barriers: Provide where necessary to protect other operatives and general public, and to prevent damage to freshly applied coatings.

PREPARATION

400 PREPARATION GENERALLY

- Standard: In accordance with BS 6150.
- Refer to any pre-existing CDM Health and Safety File.
- Refer to CDM Construction Phase Plan where applicable.
- Suspected existing hazardous materials: Prepare risk assessments and method statements covering operations, disposal of waste, containment and reoccupation, and obtain approval before commencing work.
- Preparation materials: Types recommended by their manufacturers and the coating manufacturer for the situation and surfaces being prepared.
- Substrates: Sufficiently dry in depth to suit coating.
- Efflorescence salts: Remove.
- Dirt, grease and oil: Remove. Give notice if contamination of surfaces/ substrates has occurred.
- Surface irregularities: Remove.
- Joints, cracks, holes and other depressions: Fill flush with surface, to provide smooth finish.
- Dust, particles and residues from preparation: Remove and dispose of safely.
- Water based stoppers and fillers:
 - Apply before priming unless recommended otherwise by manufacturer.
 - If applied after priming: Patch prime.
- Oil based stoppers and fillers: Apply after priming.
- Doors, opening windows and other moving parts:
 - Ease, if necessary, before coating.
 - Prime resulting bare areas.

425 IRONMONGERY

- Removal: Before commencing work: Remove ironmongery from surfaces to be coated.
- Hinges: Do not remove.
- Replacement: Refurbishment as necessary; refit when coating is dry.

471 PREPRIMED WOOD

- Areas of defective primer: Take back to bare wood and reprime.

481 UNCOATED WOOD

- General: Provide smooth, even finish with arrises and moulding edges lightly rounded or eased.
- Heads of fasteners: Countersink sufficient to hold stoppers/fillers.
- Resinous areas and knots: Apply two coats of knotting.

500 PREPRIMED STEEL

- Areas of defective primer, corrosion and loose scale: Take back to bare metal. Reprime as soon as possible.

511 GALVANIZED, SHERARDIZED AND ELECTROPLATED STEEL

- White rust: Remove.
- Pretreatment: Apply one of the following:
 - Mordant solution to blacken whole surface.
 - Etching primer recommended by coating system manufacturer.

521 UNCOATED STEEL - MANUAL CLEANING

- Oil and grease: Remove.
- Corrosion, loose scale, welding slag and spatter: Remove.
- Residual rust: Treat with a proprietary removal solution.
- Primer: Apply as soon as possible.

560 UNCOATED CONCRETE

- Release agents: Remove.

570 UNCOATED MASONRY/ RENDERING

- Loose and flaking material: remove.

580 UNCOATED PLASTER

- Nibs, trowel marks and plaster splashes: Scrape off.
- Overtrowelled 'polished' areas: Key lightly.

590 UNCOATED PLASTERBOARD

- Depressions around fixings: Fill with stoppers/ fillers

601 UNCOATED PLASTERBOARD - TO RECEIVE TEXTURED COATING

- Joints: Fill, tape and feather out with materials recommended by textured coating manufacturer.

640 EXTERNAL POINTING TO EXISTING FRAMES

- Defective sealant pointing: Remove.
- Joint depth: Approximately half joint width; adjust with backing strip if necessary.
- Sealant:
 - Manufacturer: Contractor's choice.
 - Product reference: Submit proposals.
 - Preparation and application: As section Z22.

APPLICATION

711 COATING GENERALLY

- Application standard: In accordance with BS 6150, clause 9.
- Conditions: Maintain suitable temperature, humidity and air quality during application and drying.
- Surfaces: Clean and dry at time of application.
- Thinning and intermixing of coatings: Not permitted unless recommended by manufacturer.
- Overpainting: Do not paint over intumescent strips or silicone mastics.
- Priming coats:
 - Thickness: To suit surface porosity.
 - Application: As soon as possible on same day as preparation is completed.
- Finish:
 - Even, smooth and of uniform colour.
 - Free from brush marks, sags, runs and other defects.
 - Cut in neatly.
- Doors, opening windows and other moving parts: Ease before coating and between coats.

720 PRIMING JOINERY

- Preservative treated timber: Retreat cut surfaces with two flood coats of a suitable preservative before priming.
- End grain: Coat liberally allow to soak in, and recoat.

- 730 WORKSHOP COATING OF CONCEALED JOINERY SURFACES
- General: Apply coatings to all surfaces of components.
- 770 EXTERNAL DOORS
- Bottom edges: Prime and coat before hanging doors.
- 780 BEAD GLAZING TO COATED WOOD
- Before glazing: Apply first two coats to rebates and beads.
- 790 LINSEED OIL PUTTY GLAZING
- Setting: Allow putty to set for seven days.
 - Sealing:
 - Within a further 14 days, seal with a solvent-borne primer.
 - Fully protect putty with coating system as soon as it is sufficiently hard.
Extend finishing coats on to glass up to sight line.
- 800 GLAZING
- Etched, sand blasted and ground glass: Treat or mask edges before coating to protect from contamination by oily constituents of coating materials.
- 810 WATER REPELLENT
- Application: Liberally flood surface, giving complete and even coverage.

DRAFT

N
Furniture/Equipment

DRAFT

N11

Domestic kitchen fittings, furnishings and equipment

N11 Domestic kitchen fittings, furnishings and equipment

PRODUCTS

PRODUCTS

Refer to WYG 'Residential Demise and Fittings Schedule' and Careys 'Outline Specification'

Any variations to be highlighted.

ALL ITEMS TO BE FULLY REVIEWED BEFORE SPECIFICATION CONFIRMED

390 SEALANT

- Standard: To BS EN ISO 11600, class F20 HM.
- Type: One part silicone.
 - Manufacturer: Contractor's choice.
 - Product reference: Submit proposals.
- Colour: TBC.

EXECUTION

610 MOISTURE CONTENT OF WOOD AND WOOD BASED BOARDS

- Control and monitoring:
 - Method statement: Submit.

620 INSTALLATION GENERALLY

- Fixings and adhesives: As section Z20.
- Services: As Engineering Services specification.

630 INSTALLING UNITS AND WORKTOPS

- General: Well fitting, stable and secure.

640 INSTALLING APPLIANCES

- Connections: Provide to electric, gas, and hot and cold water services.

650 INSTALLING SINKS, TAPS AND WASTES

- Water supply: To BS EN 806-2 and -4.
- Taps:
 - Fixing: Secure, watertight seal with the appliance.
 - Positioning: Hot tap to left of cold tap as viewed by the user of the appliance.
- Wastes:
 - Bedding: Waterproof jointing compound.
 - Fixing: With resilient washer between appliance and backnut.

660 SEALANT BEDDING AND POINTING

- Application: As section Z22.
- Bedding: Sink to top of worktop.
- Pointing: Between units and splash backs.

670 INSTALLING TRIMS AND MOULDINGS

- Lengths: Un-jointed between angles or ends of runs.
- Angle joints: Mitred.

COMPLETION

910 GENERAL

- Doors and drawers: Accurately aligned, not binding. Adjusted to ensure smooth operation.
- Ironmongery: Checked, adjusted and lubricated to ensure correct functioning.

920 APPLIANCE COMMISSIONING

- Appliance operation, functions and controls: Verify.
- Documentation: Submit guarantees, instruction manuals, etc.

DRAFT

N13

Sanitary appliances and fittings

N13 Sanitary appliances and fittings**PRODUCTS**

Refer to WYG 'Residential Demise and Fittings Schedule' and Careys 'Outline Specification'

Any variations to be highlighted.

ALL ITEMS TO BE FULLY REVIEWED BEFORE SPECIFICATION CONFIRMED

EXECUTION**610 INSTALLATION GENERALLY**

- Assembly and fixing: Surfaces designed to falls to drain as intended.
- Fasteners: Nonferrous or stainless steel.
- Supply and discharge pipework: Fix before appliances.
- Fixing: Fix appliances securely to structure. Do not support on pipework.
- Jointing and bedding compounds: Recommended by manufacturers of appliances, accessories and pipes being jointed or bedded.
- Appliances: Do not use. Do not stand on appliances.
- On completion: Components and accessories working correctly with no leaks.
- Labels and stickers: Remove.

613 COMPATIBILITY OF COMPONENTS

- General: Each sanitary assembly must consist of functionally compatible components, preferably obtained from a single manufacturer.
 - Exceptions: Wastes and traps.

620 NOGGINGS AND BEARERS

- Noggings, bearers, etc. to support sanitary appliances and fittings: Position accurately. Fix securely.

630 TILED BACKGROUNDS OTHER THAN SPLASHBACKS

- Timing: Complete before fixing appliances.
- Fixing appliances: Do not overstress tiles.

650 INSTALLING WC PANS

- Floor mounted pans: Screw fix and fit cover caps over screw heads. Do not use mortar or other beddings.
- Seat and cover: Stable when raised.

670 INSTALLING CISTERNS

- Cistern operating components: Obtain from cistern manufacturer.
- Inlet and flushing valves: Match to pressure of water supply.
- Internal overflows: Into pan, to give visible warning of discharge.
- External overflows: Fix pipes to falls and locate to give visible warning of discharge. Agree location where not shown on drawings.

710 INSTALLING TAPS

- Fixing: Secure against twisting.
- Seal with appliance: Watertight.
- Positioning: Hot tap to left of cold tap as viewed by user of appliance.

720 INSTALLING WASTES AND OVERFLOWS

- Bedding: Waterproof jointing compound.
- Fixing: With resilient washer between appliance and backnut.

755 SEALANT BEDDING AND POINTING

- Bedding: Bed sinks to top of worktops. .
- Pointing: tbc .

DRAFT

N15

Internal fire and safety signage systems

N15 Internal fire and safety signage systems

GENERAL

110 FIRE AND SAFETY SIGNAGE SYSTEMS FOR COMMON STAIR

- System manufacturer: Contractor's choice.
 - System reference: Submit proposals.
- Locations and layout: Submit proposals.
 - Language: English.
- Material: Aluminium plate.
 - Other properties: Fixing holes.

SYSTEM PERFORMANCE

210 GENERAL REQUIREMENTS

- Signage and way guiding system design: TBC.
 - For fire escape and evacuation signage: In accordance with: BS 5499-4 or BS ISO 16069.
 - For way guiding systems: In accordance with BS ISO 16069.
 - For safety signs other than escape route signage: In accordance with: BS 5499-10.
- Comply with the requirements of: Fire Strategy Report.

290 SIGNAGE SAMPLES

- Sign type: Fire escape.
 - Action: Submit labelled samples.
 - Conformity: Retain samples on site for the duration of the contract or until instructed to remove.
 - Delivered products: To conform with labelled samples.

PRODUCTS

320 ALUMINIUM PLATE FOR ESCAPE ROUTE SIGNS

- Manufacturer: Contractor's choice.
 - Product reference: Submit proposals.
- Component thickness: 2 mm .
- Finish: Manufacturer's standard.
- Perimeters: Manufacturer's standard.
- Supports/ Fixings: Screw fixed.
- Accessories: Not required.

EXECUTION

610 FIXING SIGNS GENERALLY

- Installation: Secure, plumb and level.
- Fasteners and adhesives: As section Z20.
- Strength of fasteners: Sufficient to support live and dead loads.
- Fixings showing on surface of sign: Must not detract from the message being displayed.

COMPLETION

910 DOCUMENTATION

- Submit:
 - Manufacturer's maintenance instructions.
 - Guarantees, warranties, test certificates, record schedules and logbooks.

DRAFT

N17

Portable fire fighting systems

N17 Portable fire fighting systems

GENERAL

- 110 PORTABLE FIRE EXTINGUISHER SYSTEM FOR COMMON STAIR AT EACH FLOOR
 - Type: Water .
 - Capacity: 6 L.
 - Supports: Fixing bracket.
- 150 FIRE BLANKET SYSTEM FOR KITCHENS
 - Type: Light duty.
 - Size: 1 x 1 m.
 - Supports: Submit proposals.

SYSTEM PERFORMANCE

- 220 COLOUR CODING
 - Portable fire extinguishers: Colour code in accordance with BS 7863.

PRODUCTS

- 340 WATER EXTINGUISHERS
 - Standard: To BS EN 3-10.
 - Manufacturer: Submit proposals.
 - Product reference: Submit proposals.
- 410 FIRE BLANKETS - LIGHT DUTY
 - Standard: To BS EN 1869.
 - Manufacturer: Submit proposals.
 - Product reference: Submit proposals.

EXECUTION

- 610 INSTALLING PORTABLE FIRE EXTINGUISHERS
 - Mounting height above finished floor level: Submit proposals.
- 650 INSTALLING FIRE BLANKETS
 - Mounting height above finished floor level: Submit proposals.

COMPLETION

- 910 CLEANING
 - Protective wrappings: Remove.
 - Cleaning: Clean off and wipe down container finishes.
- 920 TESTING
 - Test standard: To BS 5603-0.
 - Test times: At completion.
 - Notice for testing (minimum): 5 days.

930 TRAINING

- Training: Submit instruction manuals or supply other appropriate resources to train the users of the building in the safe and appropriate use of the fire extinguishers and fire blankets.
- Fire brigade: Submit contact details.

940 MAINTENANCE

- Servicing: Arrange the first annual service of the portable fire fighting systems.
- Maintenance standard: To BS 5603-0.

DRAFT

P

Building fabric sundries

DRAFT

P10

Sundry insulation/ proofing work

P10 Sundry insulation/ proofing work

SUNDRY INSULATION/ PROOFING WORK

To be read with Preliminaries/ General conditions.

TYPES OF INSULATION

135A INSULATION LAID BETWEEN ROOF TRUSSES JOISTS TO VENTILATED COLD ROOF **To achieve a U value of 0.11 W/m²K in conjunction with clause 135B**

- Manufacturer: ROCKWOOL Ltd.
 - Web: www.rockwool.co.uk.
 - Email: info@rockwool.com.
 - Product reference: Rockwool rolls.
 - Product reference: Rockwolol Roll.
- Thickness: 100 mm.
- Installation requirements:
 - Installation standard: To BS 5803-5.
 - Insulation widths: Widest practical.
 - Laid direction: At right angles to ties/ joists.
 - Joints: Butted, no gaps.
 - Insulation: Fitted neatly around rafter ends and extended over wall plates.
 - Service holes: Sealed, and debris removed before laying insulation.
 - Eaves ventilation: Unobstructed.
 - Electric cables overlaid by insulation: Sized accordingly.
 - Water cistern platforms: Not applicable.

135B INSULATION LAID OVER ROOF TRUSSES/JOISTS TO VENTILATED COLD ROOF

- **To achieve a U value of 0.11 W/m²K in conjunction with clause 135A**
- Manufacturer: ROCKWOOL Ltd.
 - Web: www.rockwool.co.uk.
 - Email: info@rockwool.com.
 - Product reference: Rockwool rolls.
 - Product reference: Rockwolol Roll.
- Thickness: 300 mm
- Installation requirements:
 - Installation standard: To BS 5803-5.
 - Insulation widths: Widest practical.
 - Laid direction: At right angles to ties/ joists.
 - Joints: Butted, no gaps.
 - Insulation: Fitted neatly around rafter ends and extended over wall plates.
 - Service holes: Sealed, and debris removed before laying insulation.
 - Eaves ventilation: Unobstructed.
 - Electric cables overlaid by insulation: Sized accordingly.
 - Water cistern platforms: Not applicable.

165 INSULATION TO EAVES

- Manufacturer: Contractor's choice.
 - Product reference: Submit proposals.
- Material: Rock wool quilt to BS EN 13162.
- Recycled content: Contractor's choice.
- Thickness: To achieve a U value of 0.11 W/m²K.
- Installation: Continuous with roof insulation, with no gaps.

170 INSULATION TO LOFT ACCESS HATCHES

- Manufacturer: _____.
- Product reference: _____.
- Material: _____.
- Recycled content: _____.
- Thickness: _____.
- Laying: Cut to fit, with no gaps, and fix securely.
- Edges of hatch: Sealed with an approved compressible draught excluder.

212 INSULATION FIXED ACROSS INNER FACE OF STUD WALLING

- Manufacturer: _____.
- Product reference: _____.
- Material: _____.
- Thickness: _____.
- Number of layers: _____.
- Installation requirements:
 - Fixing: Secure to inner face of studs, with vapour control facing (if provided) on warm side.
 - Fasteners: _____.
 - Fixing centres: _____.
 - Joints: Butt or overlap, as recommended by insulation manufacturer, with no gaps. Seal with vapour impermeable adhesive tape.

220 SOUND INSULATION WITHIN TIMBER FRAMED SEPARATING WALL

- Location: In cavity between frames and blockwork walls.
- Manufacturer: Saint-Gobain Isover..
- Product reference: RD Party wall Rolls.
- Material: Glass wool to BS EN 13162.
- Recycled content: Contractor's choice.
- Density (minimum): 12 kg/m³.
- Thickness: 75 mm and 100mm to fill cavity to party walls.
- Width: Widest practical.
- Installation requirements:
 - Joints: Butted, no gaps.
 - Head fixing: Staples or large head nails.

220A SOUND INSULATION WITHIN TIMBER FRAME / METAL STUD WALL TO PARTITIONS BETWEEN BEDROOMS, OR ANY ROOM CONTAINING A WC

Acoustic Insulation

In accordance with requirement E2(b) of Approved Document E, partitions between bedrooms, or any room containing a WC and any other room are to have a sound insulation rating of 40dB Rw or better.

This can be achieved by incorporating suitable sound-absorbent mineral wool batts (minimum density of 10kg/m³)

Isover 65mm APR 1200 in the stud cavity.

NB: This is not required where the wall contains a door or where the wall separates an en-suite from the associated bedroom.

- Location: In cavity between frames and blockwork walls.
- Manufacturer: Saint-Gobain Isover..
 - Product reference: Isover APR 1200.
- Material: Glass wool to BS EN 13162.
- Recycled content: Contractor's choice.
- Density (minimum): 10 kg/m³.
- Thickness: 65 mm.
- Width: Widest practical.
- Installation requirements:
 - Joints: Butted, no gaps.
 - Head fixing: Staples or large head nails.

310 VAPOUR CONTROL LAYER _____

- Manufacturer: _____.
 - Product reference: _____.
- Material: _____.
- Minimum vapour resistance: _____.
- Installation requirements:
 - Setting out: Joints minimized.
 - Method of fixing: _____.
 - Joints: At supports only, lapped 150 mm minimum.
 - Openings: Membrane fixed to reveals.
 - Joints and edges: Sealed with double sided tape with vapour resistivity not less than the vapour control layer.
- Penetrations: Sealed.
- Other requirements: _____.

410A SMOKE AND FIRE BARRIER

- Manufacturer: Knauf Insulation Ltd.
 - Web: www.knaufinsulation.co.uk.
 - Email: info@knaufinsulation.com.
 - Product reference: Rocksilks® Smoke and Fire Barrier
- Width: 900 mm.

430 WIRED MINERAL WOOL SMALL CAVITY BARRIERS

- Material: Wire reinforced mineral wool minimum 50 mm thick.
- Fire resistance rating: To BS 476-20, 30/15 integrity/ insulation.
- Installation requirements:
 - Fasteners: Staples at maximum 150 mm centres. Fold cavity barrier if necessary to ensure a tight fit.
 - Joints and intersections: Butted, no gaps.

432A FIREPROOF INSULATION CAVITY BARRIER TO EXTERNAL WALLS

- Manufacturer: ROCKWOOL Ltd.
 - Web: www.rockwool.co.uk.
 - Email: info@rockwool.com.
 - Product reference: ROCKWOOL SP 60 Firestop Slab.
- Dimensions: 90 mm x 650 mm x 1000 mm.
- Facing: Aluminium faced foil both sides.
- Reaction to fire class: A1 to BS EN 13501-1.
- Accessories: SP Firestop fixing brackets.

DRAFT

P12

Fire stopping systems

P12 Fire stopping systems

GENERAL

- 110 FIRE STOPPING SYSTEM TO INDIVIDUAL SERVICE PENETRATIONS
 - Penetration seal/ Gap filler: Submit proposals.
- 140 FIRE STOPPING SYSTEM TO MULTIPLE SERVICES PENETRATIONS GENERALLY
 - Fire resistance: 30 / 60 minutes subject to location.
 - Board barrier:
 - Material: Submit proposals.
 - Thickness: to meet fire rating.
 - Number of layers: submit proposals.
 - Framing: Submit proposals.
 - Finish: Submit Proposals.
 - Capping sealant: Submit proposals.
 - Colour: Submit proposals.
- 150 LOOSE FIRE STOPPING GENERALLY
 - Fire resistance: 30 / 60 minutes subject to location.
 - Penetration seal: Submit proposals.
 - Size: Contractor's choice .
- 160 LINEAR GAP SEALING GENERALLY
 - Fire resistance: 30 / 60 minutes subject to location.
 - Gap width or height (nominal): 20 mm.
 - Gap filler: Submit proposals.
 - Capping sealant: Submit proposals.
 - Colour: Submit proposals.

SYSTEM PERFORMANCE

- 210 DESIGN
 - Design: Complete the design of the fire stopping system.
 - Proposals: Submit drawings, technical information, calculations and manufacturers' literature.
- 240 FIRE PERFORMANCE GENERALLY
 - Resistance to fire: To BS 476-20 and -22, 30 minutes integrity and insulation UNLESS noted otherwise.
 - Reaction to fire: In accordance with Building Regulations, Class 0.
 - Smoke resistance:
 - Air leakage rate (maximum): 3 m³/m²·hr.
- 260 DESIGN LIFE
 - Effective design life: 30 years.

PRODUCTS

- 305 PRODUCT CERTIFICATION
 - Certification: For products specified generically, submit evidence of compliance with the specification.
 - Acceptable evidence: Agrément certificate.

370 PIPE COLLAR - CONCEALED INTUMESCENT

- Manufacturer: Contractor's choice.
- Product reference: Submit proposals.

EXECUTION

620 WORKMANSHIP GENERALLY

- Gaps: Seal gaps between building elements and services, to provide fire resistance and resist the passage of smoke.
- Adjacent surfaces: Prevent overrun of sealant or mortar on to finished surfaces.

640 INSTALLING BOARDING

- Position of boarding: Across face of opening.
- Framing: Provide framing to all edges of boarding.
- Bedding: Bed boarding on fire resisting silicone.
- Multiple board layers: Stagger joints between layers.
 - Joints: Seal with board adhesive.
- Fixing: Contractor's choice.
- Other requirements: Finish joint sealant flush with boards.

660 APPLYING INTUMESCENT FOAM

- New joints: Remove builder's debris, mortar droppings, grease, and other contaminants.
- Old joints: Clean and remove existing sealant from each joint.
- Priming: Lightly moisten substrate with water.
- Application: Fill joint to approximately half its depth, and allow foam to expand to face of joint.
- Trimming: Trim excess foam to give a neat, flush appearance.

670 APPLYING INTUMESCENT MORTAR

- Sequence: Install mortar after services are permanently installed.
- Loose dust and combustible materials: Remove from the opening.
- Shuttering: Install suitable shuttering panels to the faces of the opening.
- Temperature: Do not apply mortar when it could be damaged by frost.
- Powder:water ratio: As manufacturers recommendations.
- Mortar cure: Do not disturb mortar before final set has taken place.
- Shuttering: Remove after mortar has cured.

710 INSTALLING MINERAL WOOL BATTS

- Installing batts: Fit tight into void between the penetrating services and the surrounding construction to form a solid barrier.
 - Brackets: Impale batts on proprietary pressed steel brackets at 500 mm maximum centres and not greater than 250 mm from ends of batts..
 - Bracket fixing: As manufacturers recommendations.
- Face of batts: Flush with the surface of wall, floor or soffit.
- Joints between batts: Close butt joints, seal with fire resisting sealant.
- Gaps between services and barrier: Seal with fire resisting sealant.

730 FIXING PIPE COLLARS

- Collar fixing: Contractor's choice .
- Gap around collar: Seal with gap filler and sealant.
- Length of wraps: Project 50 mm from each side of the element.

740 INSERTING SEALANT BACKING MATERIAL

- Preparation: Removed debris from service penetration.
- Installation: Build-in joint filler as the work proceeds.

745 APPLYING SEALANTS GENERALLY

- Application: As section Z22.

750 APPLYING CAPPING SEALANT

- Preparation: De-grease using cleaner recommended by sealant manufacturer.
- Priming: Primer recommended by sealant manufacturer.
- Depth of sealant: 10 mm.
- Temperature: Do not apply water based sealants when they could be damaged by frost.

COMPLETION

910 CLEANING

- Masking tapes: Remove.
- Cleaning: Clean off splashes and droppings. Wipe down finishes.

920 INSPECTION

- Notice for inspection (minimum): 5 working days.

DRAFT

P20

Unframed isolated trims/ skirtings/ sundry items

P20 Unframed isolated trims/ skirtings/ sundry items

To be read with Preliminaries/ General conditions

170 PROPRIETARY INTUMESCENT DOORS SEALS

- Manufacturer: Intumescent Systems Ltd
Envirograf House, Barfrestone, DOVER
Kent, CT15 7JG, United Kingdom
+44 (0) 1 304 842 555. .
- Product reference: Envirograph Product 100. .
- Size: 5 X 10mm .
- Finish/ Colour: white .
- Accessories: None .
- Fixing: Rebated into edge of doors leaf .
Jointing: Butt joints .

170A PROPRIETARY INTUMESCENT DOOR BRUSH SMOKE SEALS

- Manufacturer: Intumescent Systems Ltd
Envirograf House, Barfrestone, DOVER
Kent, CT15 7JG, United Kingdom
+44 (0) 1 304 842 555. .
- Product reference: Envirograph Product 100. .
- Size: 5 X 10mm .
- Finish/ Colour: white .
- Accessories: None .
- Fixing: Rebated into edge of doors leaf .
Jointing: Butt joints .

172 PROPRIETARY ACOUSTIC DOOR SEALS

- Manufacturer: Pemco Ltd
P.O. Box No. 515, Bury St. Edmunds IP29 4WP UK
Tel/Fax : 01284 735005. .
- Product reference: S773 Silicone Seal .
- Product Description: Flexible finger seal fitted to door frame to seal gaps
Size: 13 X 12mm .
- Finish/ Colour: tbc .
- Accessories: None .
- Fixing: Applied to door frame at junction with planted stop .
Jointing: Butt joints .

173 PROPRIETARY ACOUSTIC DOOR SEALS

- Manufacturer: Pemco Ltd
P.O. Box No. 515, Bury St. Edmunds IP29 4WP UK
Tel/Fax : 01284 735005. .
- Product reference: 234 Recessed Door Shoe .
- Product Description: Concealed Acoustic door seal to bottom edge of door
Size: 13 X 30mm .
- Finish/ Colour: tbc .
- Accessories: None .
- Fixing: Rebated into bottom edge of door so as to be fully concealed .
Jointing: Not applicable .

200 MEDIUM DENSITY FIBREBOARD WINDOW SILL BOARDS

- Manufacturer: Contractor's choice.
 - Product reference: Submit proposals.
- Standard: To BS EN 622-5.
 - Type: MDF.
 - Formaldehyde class: To BS EN 622-1, Class E1.
- Fire rating: To BS 476-7, Class 1.
- Thickness: 25 mm.
 - Edges: Quarter rounded.
- Finish: Prepared and primed as section M60 .
- Recycled content: 60% (minimum) to BS EN ISO 14021.
- Support/ Fixing: Fix to softwood grounds with lost head nails at 600 mm centres..

200A MEDIUM DENSITY FIBREBOARD SKIRTING BOARDS

- Manufacturer: Contractor's choice.
 - Product reference: Submit proposals.
- Standard: To BS EN 622-5.
 - Type: MDF.
 - Formaldehyde class: To BS EN 622-1, Class E1.
- Fire rating: To BS 476-7, Class 1.
- Thickness: 22 mm.
- Height: 150mm
- Edges: Square.
- Finish: Prepared and primed as section M60 .
- Recycled content: 60% (minimum) to BS EN ISO 14021.
- Support/ Fixing: Fix to softwood grounds with lost head nails at 600 mm centres..

EXECUTION

510 INSTALLATION GENERALLY

- Joinery workmanship: As section Z10.
- Metal workmanship: As section Z11.
- Methods of fixing and fasteners: As section Z20 where not specified.
- Straight runs: To be in one piece, or in long lengths with as few joints as possible.
- Running joints: Location and method of forming to be agreed where not detailed.
- Joints at angles: Mitre, unless shown otherwise.
- Position and level: To be agreed where not detailed.

DRAFT

P31

Holes, chases, covers and supports for services

P31 Holes, chases, covers and supports for services

EXECUTION

- 620 HOLES, RECESSES AND CHASES IN IN SITU CONCRETE
 - Cast in: Holes larger than 10 mm diameter, recesses and chases.
 - Cutting and drilling:
 - Permitted for holes not larger than 10 mm diameter.
 - Not permitted for holes larger than 10 mm diameter except as indicated on drawings.
- 630 HOLES, RECESSES AND CHASES IN PRECAST CONCRETE
 - Cutting and drilling: Not permitted except as indicated on drawings.
- 640 HOLES IN STRUCTURAL STEELWORK
 - Cutting and drilling: Not permitted except as indicated on drawings.
- 650 HOLES, RECESSES AND CHASES IN MASONRY
 - Locations: To maintain integrity of strength, stability and sound resistance of construction.
 - Sizes: Minimum needed to accommodate services.
 - Holes (maximum): 300 mm².
 - Walls of hollow or cellular blocks: Do not chase.
 - Walls of other materials:
 - Vertical chases: No deeper than one third of single leaf thickness, excluding finishes.
 - Horizontal or raking chases: No longer than 1 m. No deeper than one sixth of the single leaf thickness, excluding finishes.
 - Chases and recesses: Do not set back to back. Offset by a clear distance at least equal to the wall thickness.
 - Cutting: Do not cut until mortar is fully set. Cut carefully and neatly. Avoid spalling, cracking and other damage to surrounding structure.
- 660 PREFORMED HOLES IN MASONRY
 - Width of holes without bridging over (maximum): 460 mm or one block.
 - Holes requiring bridging: Submit proposals.
- 690 INSTALLING PIPE SLEEVES
 - Sleeves: Fit to pipes passing through building fabric.
 - Material: Match pipeline.
 - Size: One or two sizes larger than pipe to allow clearance.
 - Finish: Install sleeves flush with building finish. In areas where floors are washed down, install protruding 100 mm above floor finish.
 - Masking plates: Fit at visible penetrations, including through false ceilings of occupied rooms.
- 740 INSTALLING METER CABINETS
 - Fixing: As drawing ???.
 - Keys: Hand over to Employer at completion.

DRAFT

R
Disposal systems

DRAFT

R10

Rainwater drainage systems

R10 Rainwater drainage systems

GENERAL

110A GRAVITY RAINWATER DRAINAGE SYSTEM GENERALLY

- Rainwater outlets: Proprietary and Submit proposals.
- Gutters: PVC-U 125mm half round gutter.
- - Colour: Anthracite Grey RAL 7016 to match windows
- Pipework: PVC-U, external round profile RWP.
- - Colour: Anthracite Grey RAL 7016 to match windows
- Below ground drainage: As civil specification.
- Disposal: To rainwater storage tanks and To surface water drainage.
- Controls: Not applicable.
- Accessories: Sealant for gutters.
- For extent of gutter - refer to drawings
- For number and location of Rainwater pipes - refer to drawings.

115A DOWNPIPES TO CONCEAL GAS PIPES GENERALLY GENERALLY

- Allow for 1 No per dwelling, locations to be confirmed.
- PPipes to match those specified in 110A

140 DRAINAGE CHANNELS TO GARAGE ENTRANCE DOORS GENERALLY

- Refer to DLA Drawings 1132 - ????
- Allow 1 No per dwelling
- Manufacturer: Contractors Choice
- Product: Proprietary linear drainage channel with galvanised steel grating suitable for vehicle loadings.
- Samples: Submit for approval.

145 THRESHOLD DRAIN TO DWELLING ENTRANCE DOORS GENERALLY

- Refer to DLA Drawings 1132 - ????
- Allow 1 No per dwelling
- Manufacturer: Hepworth
- Product: Hepworth threshold drain or equivalent approved
- Samples: Submit for approval.

SYSTEM PERFORMANCE

210 DESIGN

- Design: Complete the design of the rainwater drainage system.
- Standard:
 - To BS EN 12056-3, clauses 3–7, Annex A and National Annexes.
 - To BS EN 12056-5, clauses 3, 4, 6 and 11.
- Proposals: Submit drawings, technical information, calculations and manufacturers' literature.

221 COLLECTION AND DISTRIBUTION OF RAINWATER

- General: Complete, and without leakage or noise nuisance.

PRODUCTS

- 340 GRP GUTTERS
- Manufacturer: Contractor's choice.
 - Product reference: Submit proposals.
 - Profile: Half-round.
 - Construction: Solid single skin.
 - Nominal size: 125 mm.
 - Finish/ Colour: GRP to match windows RAL 7016.
 - Brackets: Galvanized steel fascia type.
 - Fixings: Stainless steel screws.
 - Size: 40 x 4 mm.
 - Accessories: Stop ends.

- 360 SEALANT FOR GUTTERS
- Type: As recommended by manufacturer.

EXECUTION

- 600 PREPARATION
- Work to be completed before commencing work specified in this section:
 - Below ground drainage. Alternatively, make temporary arrangements for dispersal of rainwater without damage or disfigurement of the building fabric and surroundings.
 - Painting of surfaces which will be concealed or inaccessible.
- 605 INSTALLATION GENERALLY
- Electrolytic corrosion: Avoid contact between dissimilar metals where corrosion may occur.
 - Plastics and galvanized steel pipes: Do not bend.
 - Allowance for thermal and building movement: Provide and maintain clearance as fixing and jointing proceeds.
 - Protection:
 - Fit purpose made temporary caps to prevent ingress of debris.
 - Fit access covers, cleaning eyes and blanking plates as the work proceeds.
- 610 FIXING AND JOINTING GUTTERS
- Joints: Watertight.
 - Brackets: Securely fixed.
 - Fixings: Plugged and screwed into masonry.
 - Fixing centres: to be confirmed.
 - Additional brackets: Where necessary to maintain support and stability, provide at joints in gutters and near angles and outlets.
 - Roofing underlay: Dressed into gutter.
- 615 SETTING OUT EAVES GUTTERS - TO FALLS
- Setting out: To true line and even gradient to prevent ponding or backfall. Position high points of gutters as close as practical to the roof and low points not more than 50 mm below the roof.
 - Outlets: Align with connections to below ground drainage.
- 616 SETTING OUT EAVES GUTTERS - LEVEL
- Setting out: Level and as close as practical to the roof.
 - Outlets: Aligned with connections to below ground drainage.

630 INSTALLING RAINWATER OUTLETS

- Fixing: Secure. Fix before connecting pipework.
 - Method: Grout into preformed holes.
- Junctions between outlets and pipework: Accommodate movement in structure and pipework.

635 FIXING PIPEWORK

- Pipework: Fix securely, plumb and/ or true to line.
- Branches and low gradient sections: Fix with uniform and adequate falls to drain efficiently.
- Externally socketed pipes and fittings: Fix with sockets facing upstream.
- Additional supports: Provide as necessary to support junctions and changes in direction.
- Vertical pipes:
 - Provide a loadbearing support at least at every storey level.
 - Tighten fixings as work proceeds so that every storey is self supporting.
 - Wedge joints in unsealed metal pipes to prevent rattling.
- Wall and floor penetrations: Isolate pipework from structure.
 - Pipe sleeves: As section P31.
 - Masking plates: Fix at penetrations if visible in the finished work.
- Expansion joint pipe sockets: Fix rigidly to buildings. Elsewhere, provide brackets and fixings that allow pipes to slide.

640 FIXING VERTICAL PIPEWORK

- Bracket fixings: Plugged and screwed into masonry.
- Distance between bracket fixing centres (maximum): 1200 mm.

650 JOINTING PIPEWORK AND GUTTERS

- General: Joint with materials and fittings that will make effective and durable connections.
- Jointing differing pipework and gutter systems: Use adaptors intended for the purpose.
- Cut ends of pipes and gutters: Clean and square. Remove burrs and swarf. Chamfer pipe ends before inserting into ring seal sockets.
- Jointing or mating surfaces: Clean and, where necessary, lubricate immediately before assembly.
- Junctions: Form with fittings intended for the purpose.
- Jointing material: Strike off flush. Do not allow it to project into bore of pipes and fittings.
- Surplus flux, solvent jointing materials and cement: Remove.

660 JOINTING EXTERNAL PIPEWORK

- Jointing: Solvent welded with lubricated ring seal joints at no more than 1.8 m spacing.

665 INSTALLING PREFABRICATED GALVANIZED STEEL PIPEWORK

- Installation: Do not cut pipes to length or otherwise damage coatings.

685 IDENTIFICATION OF INTERNAL RAINWATER PIPEWORK

- Standard: In accordance with Water Regulations Advisory Scheme (WRAS) Information and guidance note 9-02-05 and BS 8515.

690 ELECTRICAL CONTINUITY - PIPEWORK

- Joints in metal pipes with flexible couplings: Clips (or suitable standard pipe couplings) supplied for earth bonding by pipework manufacturer to ensure electrical continuity.

695 ELECTRICAL CONTINUITY - GUTTERS

- Joints in metal gutters: Purpose made links supplied by the gutter manufacturer to ensure electrical continuity.

700 ACCESS FOR TESTING AND MAINTENANCE

- General: Install pipework and gutters with adequate clearance to permit testing, cleaning and maintenance, including painting where necessary.
- Access fittings and rodding eyes: Position so that they are not obstructed.

COMPLETION

910 GUTTER TEST

- Preparation: Temporarily block all outlets.
- Testing: Fill gutters to overflow level and after 5 minutes closely inspect for leakage.

915 MAINTENANCE INSTRUCTIONS

- General: At completion, submit printed instructions recommending procedures for maintenance of the rainwater installation, including full details of recommended inspection, cleaning and repair procedures.

920 IMMEDIATELY BEFORE HANDOVER

- Construction rubbish, debris, swarf, temporary caps and fine dust which may enter the rainwater system: Remove. Do not sweep or flush into the rainwater system.
- Access covers, rodding eyes, outlet gratings and the like: Secure complete with fixings.

DRAFT

Z

Building fabric reference specification

DRAFT

Z11

Purpose made metalwork

Z11 Purpose made metalwork**310 MATERIALS GENERALLY**

- Grades of metals, section dimensions and properties: To appropriate British Standards. When not specified, select grades and sections appropriate for the purpose.
- Prefinished metal: May be used if methods of fabrication do not damage or alter appearance of finish, and finish is adequately protected.
- Fasteners: To appropriate British Standards and, unless specified otherwise, of same metal as component being fastened, with matching coating or finish.

320 STEEL LONG AND FLAT PRODUCTS

- Hot rolled structural steels (excluding structural hollow sections and tubes): To BS EN 10025-1.
- Fine grain steels, including special steels: To BS EN 10025-3 and -4.
- Steels with improved atmospheric corrosion resistance: To BS EN 10025-5.

330 STEEL PLATE, SHEET AND STRIP

- Plates and wide flats, high yield strength steel: To BS EN 10025-6.

340 HOT ROLLED STEEL PLATE, SHEET AND STRIP

- Flat products, high yield strength for cold forming: To BS EN 10149-1, -2 and -3.
- Carbon steel sheet and strip for cold forming: To BS EN 10111.
- Narrow strip, formable steel and steel for general engineering purposes: To BS 1449-1.8 and BS 1449-1.14.

350 COLD ROLLED STEEL PLATE, SHEET AND STRIP

- Steel sections: To BS EN 10162.
- Flat products, high yield strength micro-alloyed steels for cold forming: To BS EN 10268.
- Carbon steel flat products for cold forming: To BS EN 10130 and BS EN 10131.
- Uncoated carbon steel narrow strip for cold forming: To BS EN 10139 and BS EN 10140.
- Narrow strip steel for general engineering purposes: To BS EN 10132-1, -2, and -3.
- Carbon steel flat products for vitreous enamelling: To BS EN 10209.

360 COATED STEEL FLAT PRODUCTS

- Hot dip zinc coated carbon steel sheet and strip for cold forming: To BS EN 10346 and BS EN 10143.
- Hot dip zinc coated structural steel sheet and strip: To BS EN 10143 and BS EN 10346.
- Hot dip zinc-aluminium (za) coated sheet and strip: To BS EN 10346.
- Hot dip aluminium-zinc (az) coated sheet and strip: To BS EN 10346.
- Organic coated flat products: To BS EN 10169.

370 STEEL STRUCTURAL HOLLOW SECTIONS (SHS)

- Non alloy and fine grain steels, hot finished: To BS EN 10210-1 and -2.
- Non-alloy and fine grain steels, cold formed welded: To BS EN 10219-2.
- Weather resistant steels, hot finished: To BS 7668.

380 OTHER STEEL SECTIONS

- Equal flange tees: To BS EN 10055.
- Equal and unequal angles: To BS EN 10056-1 and -2.
- Wire, carbon steel for general engineering purposes: To BS 1052.
- Wire and wire products, general: To BS EN 10218-2.
- Tubes:
 - Seamless circular: To BS EN 10297-1.
 - Seamless cold drawn: To BS EN 10305-1.
 - Welded and cold sized square and rectangular: To BS EN 10305-5.
 - Welded circular: To BS EN 10296-1.
 - Welded cold drawn: To BS EN 10305-2.
 - Welded cold sized: To BS EN 10305-3.

400 STAINLESS STEEL PRODUCTS

- Chemical composition and physical properties: To BS EN 10088-1.
- Sheet, strip and plate: To BS EN 10088-2.
- Semi-finished products bars, rods and sections: To BS EN 10088-3.
- Wire: To BS EN 1088-3.
- Tubes:
 - Welded circular: To BS EN 10296-2.
 - Seamless circular: To BS EN 10297-2.

410 ALUMINIUM ALLOY PRODUCTS

- Designations:
 - Designation system, chemical composition and forms: To BS EN 573-1, -2, -3 and -5.
 - Temper designations: To BS EN 515.
- Sheet, strip and plate: To BS EN 485-1 to -4.
- Cold drawn rods, bars and tubes: To BS EN 754-1 and -2.
- Extruded rods, bars, tubes and profiles: To BS EN 755-1 and -2.
- Drawn wire: To BS EN 1301-1, -2 and -3.
- Rivet, bolt and screw stock: To BS 1473.
- Structural sections: To BS 1161.

420 COPPER ALLOY PRODUCTS

- Sheet, strip, plate and circles for general purposes: To BS EN 1652.
- Sheet and strip for building purposes: To BS EN 1172.
- Rods: To BS EN 12163.
- Profiles and rectangular bars: To BS EN 12167.
- Wire: To BS EN 12166.
- Tubes: To BS EN 12449.

FABRICATION

515 FABRICATION GENERALLY

- Contact between dissimilar metals in components: Avoid.
- Finished components: Rigid and free from distortion, cracks, burrs and sharp arrises.
 - Moving parts: Free moving without binding.
- Corner junctions of identical sections: Mitre.

520 COLD FORMED WORK

- Profiles: Accurate, with straight arrises.

525 ADHESIVE BONDING

- Preparation of surfaces of metals to receive adhesives:
 - Degrease.
 - Abrade mechanically or chemically etch.
 - Prime: To suit adhesive.
- Adhesive bond: Form under pressure.

527 WELDING MILD STEEL .

- Welding procedures:
 - Method and standard: Metal arc welding to BS EN 1011-1 and -2..
 - Welding Procedure Specification (WPS): Not required.
- Preparation:
 - Joint preparation: Clean thoroughly.
 - Surfaces of materials that will be self-finished and visible in the completed work: protect from weld splatter.
- Jointing:
 - Joints: Fully bond parent and filler metal throughout with no inclusions, holes, porosity or cracks.
 - Dissimilar metals: Welding not permitted.
 - Strength requirements: Welds to achieve design loads.
 - Heat straightening: Obtain approval.
 - Complex assemblies: Agree priority for welding members to minimize distortion caused by subsequent welds.
 - Tack welds: Use only for temporary attachment.
 - Jigs: Provide to support and restrain members during welding.
 - Filler plates: Obtain approval.
 - Lap joints: Minimum 5 x metal thickness or 25 mm, whichever is greater.
 - Weld terminations: Clean and sound.

FINISHING

710 FINISHING WELDED AND BRAZED JOINTS VISIBLE IN COMPLETE WORK

- Standard: To BS EN ISO 8501-3.
 - Preparation grade: P1.
- Butt joints: Smooth, and flush with adjacent surfaces.
- Fillet joints: Neat.
- Grinding: Grind smooth where indicated on drawings.

745 PREPARATION FOR APPLICATION OF COATINGS

- General: Complete fabrication, and drill fixing holes before applying coatings.
- Paint, grease, flux, rust, burrs and sharp arrises: Remove.

750 LIQUID ORGANIC COATING FOR ALUMINIUM ALLOY COMPONENTS

- Standard: To BS 4842.

780 GALVANIZING

- Standard: To BS EN ISO 1461.
- Preparation:
 - Vent and drain holes: Provide in accordance with BS EN ISO 14713-1 and -2. Seal after sections have been drained and cooled.
 - Components subjected to cold working stresses: Heat treat to relieve stresses before galvanizing.
 - Welding slag: Remove.
 - Component cleaning: To BS EN ISO 8501-3.
- Grade: Engineer to advise.

COMPLETION

910 DOCUMENTATION

- Submit:
 - Manufacturer's maintenance instructions.
 - Guarantees, warranties, test certificates, record schedules and log books.

920 COMPLETION

- Protection: Remove.
- Cleaning and maintenance: Carry out in accordance with procedures detailed in fabricators' guarantees.

DRAFT

Z12

Preservative/ fire retardant treatment

Z12 Preservative/ fire retardant treatment

110 TREATMENT APPLICATION

- Timing: After cutting and machining timber, and before assembling components.
- Processor: Licensed by manufacturer of specified treatment solution.
 - Operatives: WPA certified.
- Certification: For each batch of timber provide a certificate of assurance that treatment has been carried out as specified.

120 COMMODITY SPECIFICATIONS

- Standard: In accordance with the Wood Protection Association (WPA) publication 'Industrial wood preservation specification and practice'.

130 PRESERVATIVE TREATMENT SOLUTION STRENGTHS/ TREATMENT CYCLES

- General: Select to achieve specified service life and to suit treatability of specified wood species.

150 WATER-BASED ORGANIC PRESERVATIVE TREATMENT

- Solution:
 - Manufacturer: Contractor's choice.
 - Product reference: Submit proposals.
 - Application: High pressure impregnation.
- Moisture content of wood:
 - At time of treatment: Not more than 28%.
 - After treatment: Timber to be surface dry before use.

610 MAKING GOOD TO PRESERVATIVE TREATMENT ON-SITE

- Preservative solution: Compatible with off-site treatment.
- Application: In accordance with preservative manufacturer's recommendations.

620 MAKING GOOD TO FIRE RETARDANT TREATMENT ON-SITE

- Fire retardant: Compatible with off-site treatment.
- Application: In accordance with fire retardant manufacturer's recommendations.

DRAFT

Z20

Fixings and adhesives

Z20 Fixings and adhesives

PRODUCTS

- 310 FASTENERS GENERALLY
 - Materials: To have:
 - Bimetallic corrosion resistance appropriate to items being fixed.
 - Atmospheric corrosion resistance appropriate to fixing location.
 - Appearance: Submit samples on request.
- 320 PACKINGS
 - Materials: Noncompressible, corrosion proof.
 - Area of packings: Sufficient to transfer loads.
- 330 NAILED TIMBER FASTENERS
 - Nails:
 - Steel: To BS 1202-1 or BS EN 10230-1.
 - Copper: To BS EN 1202-2.
 - Aluminium: To BS 1202-3.
- 340 MASONRY FIXINGS
 - Light duty: Plugs and screws.
 - Heavy duty: Expansion anchors or chemical anchors.
- 350 PLUGS
 - Type: Proprietary types to suit substrate, loads to be supported and conditions expected in use.
- 360 ANCHORS
 - Types:
 - Expansion: For use in substrate strong enough to resist forces generated by expansion of anchor.
 - Adhesive or chemical:
 - For use in substrate where expansion of anchor would fracture substrate.
 - For use in irregular substrate where expansion anchors cannot transfer load on anchor.
 - Cavity: For use where the anchor is retained by toggles of the plug locking onto the inside face of the cavity.
- 370 WOOD SCREWS
 - Type:
 - Wood screws (traditional pattern).
 - Standard: To BS 1210.
 - Wood screws.
 - Pattern: Parallel, fully threaded shank or twin thread types.
 - Washers and screw cups: Where required are to be of same material as screw.
- 380 MISCELLANEOUS SCREWS
 - Type: To suit the fixing requirement of the components and substrate.
 - Pattern: Self-tapping, metallic drive screws, or power driven screws.
 - Washers and screw cups: Where required to be of same material as screw.

390 ADHESIVES GENERALLY

- Standards:
 - Hot-setting phenolic and aminoplastic: To BS 1203.
 - Thermosetting wood adhesives: To BS EN 12765.
 - Thermoplastic adhesives: To BS EN 204.

410 POWDER ACTUATED FIXING SYSTEMS

- Types of fastener, accessories and consumables: As recommended by tool manufacturer.

EXECUTION

610 FIXING GENERALLY

- Integrity of supported components: Select types, sizes, quantities and spacings of fixings, fasteners and packings to retain supported components without distortion or loss of support.
- Components, substrates, fixings and fasteners of dissimilar metals: Isolate with washers/sleeves to avoid bimetallic corrosion.
- Appearance: Fixings to be in straight lines at regular centres.

620 FIXING THROUGH FINISHES

- Penetration of fasteners and plugs into substrate: To achieve a secure fixing.

630 FIXING PACKINGS

- Function: To take up tolerances and prevent distortion of materials and components.
- Limits: Do not use packings beyond thicknesses recommended by fixings and fasteners manufacturer.
- Locations: Not within zones to be filled with sealant.

640 FIXING CRAMPS

- Cramp positions: Maximum 150 mm from each end of frame sections and at 600 mm maximum centres.
- Fasteners: Fix cramps to frames with screws of same material as cramps.
- Fixings in masonry work: Fully bed in mortar.

650 NAILED TIMBER FIXING

- Penetration: Drive fully in without splitting or crushing timber.
- Surfaces visible in completed work: Punch nail heads below wrot surfaces.
- Nailed timber joints: Two nails per joint (minimum), opposed skew driven.

660 SCREW FIXING

- Finished level of countersunk screw heads:
 - Exposed: Flush with timber surface.
 - Concealed (holes filled or stopped): Sink minimum 2 mm below surface.

670 PELLETTED COUNTERSUNK SCREW FIXING

- Finished level of countersunk screw heads: Minimum 6 mm below timber surface.
- Pellets: Cut from matching timber, match grain and glue in to full depth of hole.
- Finished level of pellets: Flush with surface.

680 PLUGGED COUNTERSUNK SCREW FIXING

- Finished level of countersunk screw heads: Minimum 6 mm below timber surface.
- Plugs: Glue in to full depth of hole.
- Finished level of plugs: Projecting above surface.

690 USING POWDER ACTUATED FIXING SYSTEMS

- Powder actuated fixing tools: To BS 4078-2 and Kitemark certified.
- Operatives: Trained and certified as competent by tool manufacturer.

700 APPLYING ADHESIVES

- Surfaces: Clean. Adjust regularity and texture to suit bonding and gap filling characteristics of adhesive.
- Support and clamping during setting: Provide as necessary. Do not mark surfaces of or distort components being fixed.
- Finished adhesive joints: Fully bonded. Free of surplus adhesive.

DRAFT

**Z21
Mortars**

Z21 Mortars

CEMENT GAUGED MORTARS

- 110 CEMENT GAUGED MORTAR MIXES
 - Specification: Proportions and additional requirements for mortar materials are specified elsewhere.
- 120 SAND FOR SITE MADE CEMENT GAUGED MASONRY MORTARS
 - Standard: To BS EN 13139.
 - Grading: 0/2 (FP or MP).
 - Fines content where the proportion of sand in a mortar mix is specified as a range (e.g. 1:1: 5-6):
Lower proportion of sand: Use category 3 fines.
Higher proportion of sand: Use category 2 fines.
 - Sand for facework mortar: Maintain consistent colour and texture. Obtain from one source.
- 131 READY-MIXED LIME:SAND FOR CEMENT GAUGED MASONRY MORTARS
 - Standard: To BS EN 998-2.
 - Lime: Nonhydraulic to BS EN 459-1.
 - Type: CL 90S.
 - Pigments for coloured mortars: To BS EN 12878.
- 135 SITE MADE LIME:SAND FOR CEMENT GAUGED MASONRY MORTARS
 - Permitted use: Where a special colour is not required and in lieu of factory made ready-mixed material.
 - Lime: Nonhydraulic to BS EN 459-1.
 - Type: CL 90S.
 - Mixing: Thoroughly mix lime with sand, in the dry state. Add water and mix again. Allow to stand, without drying out, for at least 16 hours before using.
- 160 CEMENTS FOR MORTARS
 - Cement: To BS EN 197-1 and CE marked.
 - Types: Portland cement, CEM I.
Portland limestone cement, CEM II/A-L or CEM II/A-LL.
Portland slag cement, CEM II/B-S.
Portland fly ash cement, CEM II/B-V.
Strength class: 32.5, 42.5 or 52.5.
 - White cement: To BS EN 197-1 and CE marked.
 - Type: Portland cement, CEM I.
 - Strength class: 52.5.
 - Sulfate resisting Portland cement:
 - Type: To BS EN 197-1 Sulfate resisting Portland cement, CEM I/SR and CE marked.
To BS EN 197-1 fly ash cement, CEM II/B-V and CE marked.
 - Strength class: 32.5, 42.5 or 52.5.
 - Masonry cement: To BS EN 413-1 and CE marked.
Class: MC 12.5.

- 180 ADMIXTURES FOR SITE MADE CEMENT GAUGED MORTARS
 - Air entraining (plasticizing) admixtures: To BS EN 934-3 and compatible with other mortar constituents.
 - Other admixtures: Submit proposals.
 - Prohibited admixtures: Calcium chloride, ethylene glycol and any admixture containing calcium chloride.

- 190 RETARDED READY TO USE CEMENT GAUGED MORTAR
 - Standard: To BS EN 998-2.
 - Lime for cement:lime:sand mortars: Nonhydraulic to BS EN 459-1.
 - Type: CL 90S.
 - Pigments for coloured mortars: To BS EN 12878.
 - Time and temperature limitations: Use within limits prescribed by mortar manufacturer.
Retempering: Restore workability with water only within prescribed time limits.

- 200 STORAGE OF CEMENT GAUGED MORTAR MATERIALS
 - Sands and aggregates: Keep different types/ grades in separate stockpiles on hard, clean, free-draining bases.
 - Factory made ready-mixed lime:sand/ ready to use retarded mortars: Keep in covered containers to prevent drying out or wetting.
 - Bagged cement/ hydrated lime: Store off the ground in dry conditions.

- 210 MAKING CEMENT GAUGED MORTARS
 - Batching: By volume. Use clean and accurate gauge boxes or buckets.
 - Mix proportions: Based on dry sand. Allow for bulking of damp sand.
 - Mixing: Mix materials thoroughly to uniform consistency, free from lumps.
 - Mortars containing air entraining admixtures: Mix mechanically. Do not overmix.
 - Working time (maximum): Two hours at normal temperatures.
 - Contamination: Prevent intermixing with other materials.

- LIME:SAND MORTARS**

- 310 LIME:SAND MORTAR MIXES
 - Specification: Proportions and additional requirements for mortar materials are specified elsewhere.

- 320 SAND FOR LIME:SAND MASONRY MORTARS
 - Type: Sharp, well graded.
 - Quality, sampling and testing: To BS EN 13139.
 - Grading/ Source: As specified elsewhere in relevant mortar mix items.

- 330 READY PREPARED LIME PUTTY
 - Type: Slaked directly from CL 90 quicklime to BS 890, using an excess of water.
 - Maturation: In pits/ containers that allow excess water to drain away.
 - Density of matured lime putty: 1.3 - 1.4 kg/litre.
 - Maturation period before use (minimum): Seek instructions.

- 350 STORAGE OF LIME:SAND MORTAR MATERIALS
 - Sands and aggregates: Keep different types/ grades in separate stockpiles on hard, clean, free-draining bases.
 - Ready prepared nonhydraulic lime putty: Prevent drying out and protect from frost.
 - Nonhydraulic lime:sand mortar: Store on clean bases or in clean containers that allow free drainage. Prevent drying out or wetting and protect from frost.
 - Bagged hydrated hydraulic lime: Store off the ground in dry conditions.

360 MAKING LIME:SAND MORTARS GENERALLY

- Batching: By volume. Use clean and accurate gauge boxes or buckets.
- Mixing: Mix materials thoroughly to uniform consistency, free from lumps.
- Contamination: Prevent intermixing with other materials, including cement.

370 SITE PREPARED NONHYDRAULIC LIME:SAND MORTARS

- Mixing: Mix materials thoroughly by compressing, beating and chopping. Do not add water.
 - Equipment: Roller pan mixer or submit proposals.
- Maturation period before use (maximum): Seek instructions.

DRAFT

**Z22
Sealants**

Z22 Sealants

PRODUCTS

310 JOINTS TO WINDOWS AND DOORS

- Primer, backing strip, bond breaker: Types recommended by sealant manufacturer.

EXECUTION

610 SUITABILITY OF JOINTS

- Presealing checks:
 - Joint dimensions: Within limits specified for the sealant.
 - Substrate quality: Surfaces regular, undamaged and stable.
- Joints not fit to receive sealant: Submit proposals for rectification.

620 PREPARING JOINTS

- Surfaces to which sealant must adhere:
 - Remove temporary coatings, tapes, loosely adhering material, dust, oil, grease, surface water and contaminants that may affect bond.
 - Clean using materials and methods recommended by sealant manufacturer.
- Vulnerable surfaces adjacent to joints: Mask to prevent staining or smearing with primer or sealant.
- Backing strip and/ or bond breaker installation: Insert into joint to correct depth, without stretching or twisting, leaving no gaps.
- Protection: Keep joints clean and protect from damage until sealant is applied.

630 APPLYING SEALANTS

- Substrate: Dry (unless recommended otherwise) and unaffected by frost, ice or snow.
- Environmental conditions: Do not dry or raise temperature of joints by heating.
- Sealant application: Fill joints completely and neatly, ensuring firm adhesion to substrates.
- Sealant profiles:
 - Butt and lap joints: Slightly concave.
 - Fillet joints: Flat or slightly convex.
- Protection: Protect finished joints from contamination or damage until sealant has cured.

DRAFT

Z31
Powder coatings

Z31 Powder coatings

120 POWDER COATING MATERIALS

- Manufacturer: Obtain from one only of the following: Contractor's choice.
- Selected manufacturer: Submit details before commencement of powder coating including:
 - Name and contact details.
 - Details of accreditation schemes.
 Technical data of product including current Agrément certificates.

210 WORKING PROCEDURES

- Comply with the follow following standards.
 - Aluminium components: To BS 6496 or BS EN 12206-1.
 - Steel components: To BS EN 13438.
 - Safety standards: To British Coatings Federation 'Code of safe practice - Application of thermosetting powder coatings by electrostatic spraying'.

220 POWDER COATING APPLICATORS

- Applicator requirements:
 - Approved by powder coating manufacturer.
 - Currently certified to BS EN ISO 9001.
 - Comply with quality procedures, guarantee conditions, standards and tests required by powder coating manufacturer.
 - Applicator to use only one plant.
 - Selected applicator: Submit details before commencement of powder coating including:
 - Name and contact details.
 - Details of accreditation schemes.

225 GUARANTEES

- Powder coating manufacturer and applicator guarantees:
 - Submit sample copies before commencement of powder coating.
 - Submit signed project specific copies on completion of work.

230 CONTROL SAMPLES

- Sequence: Prior to ordering materials for the works, obtain approval of appearance for:
 - Powder coated samples: Of various grades and forms of background metal to be used, showing any colour, texture and gloss variation.
 - Fabrication samples: Showing joint assembly, how powder coating is affected and how any cut metal edges are finished and protected.
- Samples to include the following information:
 - Product reference.
 - Colour.
 - Reference number.
 - Name.
 - Gloss level.

250 COMPONENT DESIGN

- Condition of components to be powder coated:
 - To comply with relevant recommendations of BS 4479-1, -3, and -4.
 - Of suitable size to fit plant capacity.
 - Of suitable thickness to withstand oven curing.

310 PRETREATMENT OF ALUMINIUM COMPONENTS

- Condition of components to be pretreated:
 - Free from corrosion and damage.
 - All welding and jointing completed and finish off as specified.
 - Free from impurities including soil, grease, oil.
 - Suitable for and compatible with the pretreatment process.
- Conversion coating requirements:
 - Chromate system: To BS 6496 or BS EN 12206-1.
 - Chromate-free system: To BS EN 12206-1. Submit details before using.
- Rinsing requirements: Use demineralized water. Drain and dry.

320 PRETREATMENT OF STEEL COMPONENTS

- Condition of components to be pretreated:
 - Free from corrosion and damage.
 - All welding and jointing completed and finish off as specified.
 - Free from impurities including soil, grease, oil.
 - Suitable for and compatible with the pretreatment process.
- Conversion coating requirements: To BS EN 13438.
- Rinsing requirements: Use demineralized water. Drain and dry.

430 EXTENT OF POWDER COATINGS

- Application: To visible component surfaces, and concealed surfaces requiring protection. Coated surfaces will be deemed 'significant surfaces' for relevant BS 6496 or BS EN 13438 performance requirements.

435 APPLICATION OF POWDER COATINGS

- Surfaces to receive powder coatings: Free from dust or powder deposits.
- Powder colours: Obtain from one batch of one manufacturer.
- Commencement of powder coating: To be continuous from pretreatment.
- Jig points: Not visible on coated components.
- Curing: Controlled to attain metal temperatures and hold periods recommended by powder coating manufacturer.
- Stripping and recoating of components: Only acceptable by prior agreement of powder coating manufacturer. Stripping, pretreatment and powder coating are to be in accordance with manufacturer's requirements.
- Overcoating of components: Not acceptable.

440 PERFORMANCE AND APPEARANCE OF POWDER COATINGS

- For aluminium components:
 - Standard: To BS 6496 or BS EN 12206-1.
- For steel components:
 - Standard: To BS EN 13438.
- Visual inspection after powder coating: Significant surface viewing distances to be as specified in the relevant Standard, unless specified otherwise.
- Colour and gloss levels: To conform with approved samples.

450 ALUMINIUM ALLOY FABRICATIONS

- Units may be assembled:
 - Before powder coating.
 - From components powder coated after cutting to size.
 - Where approved, from components powder coated before cutting to size.
- Exposure of uncoated background metal: Not acceptable.
- Assembly sealants: Compatible with powder coatings. Obtain approval of colour if sealants are visible after fabrication.

460 STEEL FABRICATIONS

- Unit assembly: Wherever practical, before powder coating.
- Exposure of uncoated background metal: Not acceptable.
- Assembly sealants: Compatible with powder coatings. Obtain approval of colour if sealants are visible after fabrication.

470 FIXINGS

- Exposed metal fixings: Powder coat together with components, or coat with matching repair paint system applied in accordance with the powder coating manufacturer's recommendations.

480 DAMAGED COMPONENTS - REPAIR/ REPLACEMENT

- Before delivery to site: Check all components for damage to powder coatings. Replace damaged components.
- Site damage: Submit proposals for repair or replacement.

510 PROTECTION

- Powder coated surfaces of components: Protect from damage during handling and installation, or by subsequent site operations.
- Protective coverings: Must be:
 - Resistant to weather conditions.
 - Partially removable to suit building in and access to fixing points.
- Protective tapes in contact with powder coatings: Must be:
 - Low tack, self adhesive and light in colour.
 - Applied and removed in accordance with tape and powder coating manufacturers' recommendations. Do not use solvents to remove residues as these are detrimental to the coating.
- Inspection of protection: Carry out monthly. Promptly repair any deterioration or deficiency.

535 DOCUMENTATION

- Submit the following information for each batch of powder coated components:
 - Supplier.
 - Trade name.
 - Colour.
 - Type of powder.
 - Method of application.
 - Batch and reference number.
 - Statutory requirements.
 - Test certificates.
 - Maintenance instructions.

540 COMPLETION

- Protection: Remove.
- Cleaning and maintenance of powder coatings: Carry out in accordance with procedures detailed in powder coating manufacturer and applicator guarantees.